

Chapter 1

Complex Numbers and Quadratic Equations

Solutions

SECTION - A

Objective Type Questions (One option is correct)

1. If $z = \frac{1+i}{\sqrt{2}}$, then the value of z^{1929} is

(1) $1+i$

(2) -1

(3) $\frac{1+i}{2}$

(4) $\frac{1+i}{\sqrt{2}}$

Sol. Answer (4)

$$z = \frac{1+i}{\sqrt{2}}$$

$$\Rightarrow z^2 = \frac{(1+i)^2}{(\sqrt{2})^2} = \frac{1+i^2+2i}{2} = \frac{2i}{2} = i$$

$$\begin{aligned}\text{Now, } z^{1929} &= (z^2)^{964} \cdot z \\ &= i^{964} \cdot z \\ &= z \\ &= \frac{1+i}{\sqrt{2}}\end{aligned}$$

2. If $z = \left(\frac{1+i}{1-i}\right)$, then z^8 equals

(1) 1

(2) -1

(3) i

(4) 0

Sol. Answer (1)

$$\begin{aligned}z &= \frac{1+i}{1-i} = \frac{1+i}{1-i} \cdot \frac{1+i}{1+i} \\ &= \frac{(1+i)^2}{1-i^2} = \frac{1+i^2+2i}{2} \\ &= i \\ z^8 &= (i)^8 = (i^2)^4 \\ &= 1\end{aligned}$$

3. If the multiplicative inverse of a complex number is $\frac{\sqrt{2}+5i}{17}$, then the complex number is

(1) $\frac{\sqrt{2}-5i}{17}$ (2) $\frac{\sqrt{2}+5i}{29}$ (3) $\frac{17}{27}(\sqrt{2}-5i)$ (4) $\frac{17}{27}(\sqrt{2}+5i)$

Sol. Answer (3)

Let z be the complex number, then

$$z \cdot \frac{(\sqrt{2}+5i)}{17} = 1$$

$$\Rightarrow z = \frac{17}{\sqrt{2}+5i}$$

$$= \frac{17}{\sqrt{2}+5i} \cdot \frac{\sqrt{2}-5i}{\sqrt{2}-5i}$$

$$= \frac{17(\sqrt{2}-5i)}{2-25i^2}$$

$$= \frac{17}{27}(\sqrt{2}-5i)$$

4. The additive inverse of $5+7i$ is

(1) $5-7i$ (2) $-5+7i$ (3) $5+7i$ (4) $-5-7i$

Sol. Answer (4)

Additive inverse of $5+7i$ is $-5-7i$

5. The complex number $\frac{1+2i}{1-i}$ lies in

(1) First quadrant (2) Second quadrant (3) Third quadrant (4) Fourth quadrant

Sol. Answer (2)

$$\frac{1+2i}{1-i} = \frac{1+2i}{1-i} \cdot \frac{1+i}{1+i}$$

$$= \frac{(1+2i)(1+i)}{1-i^2}$$

$$= \frac{1-2+i(2+1)}{2}$$

$$= \frac{-1+3i}{2}$$

$$= -\frac{1}{2} + \frac{3}{2}i$$

$\therefore \left(-\frac{1}{2}, \frac{3}{2}\right)$ lies in the second quadrant.

6. If $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3 = a + ib$, then values of a and b respectively are

(1) 0 and 2

(2) 0 and -2

(3) 2 and 0

(4) 2 and 2

Sol. Answer (2)

$$\frac{1+i}{1-i} = \frac{(1+i) \cdot (1+i)}{(1-i)(1+i)}$$

$$= \frac{(1+i)^2}{1-i^2}$$

$$= \frac{1+i^2+2i}{2}$$

$$= i$$

$$\text{and } \frac{1-i}{1+i} = \frac{(1-i) \cdot (1-i)}{(1+i)(1-i)}$$

$$= \frac{(1-i)^2}{1-i^2}$$

$$= \frac{1+i^2-2i}{2}$$

$$= -i$$

$$\therefore \left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3 = (i)^3 - (-i)^3 = a + ib$$

$$\Rightarrow -i - i = a + ib$$

$$\Rightarrow 0 - 2i = a + ib$$

$$\Rightarrow a = 0 \text{ and } b = -2$$

7. If $x = -2 - \sqrt{3}i$, then value of $2x^4 + 5x^3 + 7x^2 - x + 41$ is

(1) 1

(2) 3

(3) 6

(4) 8

Sol. Answer (3)

$$x = -2 - \sqrt{3}i$$

$$\Rightarrow x + 2 = -\sqrt{3}i$$

$$\Rightarrow (x+2)^2 = (-\sqrt{3}i)^2$$

$$\Rightarrow x^2 + 4 + 4x = 3i^2$$

$$\Rightarrow x^2 + 4x + 7 = 0$$

$$\text{Now, } 2x^4 + 5x^3 + 7x^2 - x + 41$$

$$= 2x^2(x^2 + 4x + 7) - 3x(x^2 + 4x + 7) + 5(x^2 + 4x + 7) + 6$$

$$= 0 - 0 + 0 + 6$$

$$= 6$$

8. $\left[i^{17} + \frac{1}{i^{315}}\right]^9$ is equal to

- (1) $32i$ (2) -512 (3) 512 (4) $512i$

Sol. Answer (4)

$$\begin{aligned}\left[i^{17} + \frac{1}{i^{315}}\right]^9 &= \left[(i^2)^8 \cdot i + \frac{1}{(i^2)^{157} \cdot i}\right]^9 \\ &= \left[i - \frac{1}{i}\right]^9 \\ &= \left[i - \frac{i}{i^2}\right]^9 \\ &= [i + i]^9 \\ &= (2i)^9 \\ &= 512(i^2)^4 \cdot i \\ &= 512i\end{aligned}$$

9. $\frac{(1+i)^3}{2+i}$ is equal to

- (1) $\frac{2}{5} - \frac{6}{5}i$ (2) 0 (3) $-\frac{1}{5} + \frac{6}{5}i$ (4) $-\frac{2}{5} + \frac{6}{5}i$

Sol. Answer (4)

$$\frac{(1+i)^3}{2+i} = \frac{1-i+3i-3}{2+i} = \frac{-2+2i}{2+i} = \frac{(-2+2i)(2-i)}{4+1} = \frac{-2+6i}{5} = \left(-\frac{2}{5}\right) + i\left(\frac{6}{5}\right)$$

10. $\left(\frac{2}{1-i} + \frac{3}{1+i}\right)\left(\frac{2+3i}{4+5i}\right)$ is equal to

- (1) $-\frac{117}{82} - \frac{13}{82}i$ (2) $-\frac{117}{82} + \frac{13}{82}i$ (3) $\frac{117}{82} - \frac{13}{82}i$ (4) $\frac{117}{82} + \frac{13}{82}i$

Sol. Answer (3)

$$\left(\frac{2(1+i)+3(1-i)}{(1+i)(1-i)}\right)\left(\frac{2+3i}{4+5i}\right) = \left(\frac{5-i}{2}\right)\left(\frac{2+3i}{4+5i}\right) = \frac{117}{82} - \frac{13}{82}i$$

11. In the Argand plane, the conjugate of the complex number $3 - 7i$ will lie in

- (1) First quadrant (2) Second quadrant (3) Third quadrant (4) Fourth quadrant

Sol. Answer (1)

$$\overline{(3-7i)} = 3+7i \text{ (1st quadrant)}$$

12. The conjugate of $\frac{(1+2i)^2}{3-i}$ is

(1) $\frac{-13}{10} + \frac{9}{10}i$

(2) $\frac{-13}{10} - \frac{9}{10}i$

(3) $\frac{13}{10} + \frac{9}{10}i$

(4) $\frac{13}{10} - \frac{9}{10}i$

Sol. Answer (2)

$$\frac{1-4+4i}{3-i} = \left(\frac{-3+4i}{3-i} \right) \left(\frac{3+i}{3+i} \right)$$

$$= \frac{-9-4+12i-3i}{10}$$

$$= \frac{-13+9i}{10}$$

$$= \left(\frac{-13}{10} \right) + i \left(\frac{9}{10} \right)$$

Conjugate is $\frac{-13}{10} - i \left(\frac{9}{10} \right)$

13. If $z_1 = 1 + i$ and $z_2 = -3 + 2i$, then $\operatorname{Im} \left(\frac{z_1 z_2}{\bar{z}_1} \right)$ is

(1) 2

(2) -3

(3) 3

(4) -2

Sol. Answer (2)

$$\frac{z_1 z_2}{\bar{z}_1} = \frac{(1+i)(-3+2i)}{(1-i)}$$

$$= \frac{-3-3i+2i-2}{1-i}$$

$$= \frac{-5-i}{1-i} \times \frac{(1+i)}{(1+i)}$$

$$= \frac{(-5+1)+i(-1-5)}{2}$$

$$= \frac{-4-6i}{2}$$

$$= -2 - 3i$$

$$\operatorname{Im} \left(\frac{z_1 z_2}{\bar{z}_1} \right) = -3$$

14. The multiplicative inverse of $(3 + \sqrt{5}i)^2$ is

(1) $\frac{1}{49} - \frac{3\sqrt{5}}{98}i$

(2) $\frac{1}{49} + \frac{3\sqrt{5}}{98}i$

(3) $4 + 6\sqrt{5}i$

(4) $4 - 6\sqrt{5}i$

Sol. Answer (1)

$$\begin{aligned}\text{Multiplicative inverse} &= \frac{1}{(3 + i\sqrt{5})^2} \\ &= \frac{1}{9 - 5 + i6\sqrt{5}} \\ &= \frac{1}{4 + i6\sqrt{5}} \times \frac{4 - i6\sqrt{5}}{4 - i6\sqrt{5}} \\ &= \frac{4 - i6\sqrt{5}}{196} \\ &= \frac{1}{99} - \frac{i3\sqrt{5}}{98}\end{aligned}$$

15. If $z = 3 + i + 9i^2 - 6i^3$, then $(\overline{z^{-1}})$ is

(1) $2 + i$

(2) $-\frac{3}{79} + \frac{4}{79}i$

(3) $1 - i$

(4) $-\frac{6}{85} + \frac{7}{85}i$

Sol. Answer (4)

$$z = 3 + i - 9 + 6i$$

$$z = -6 + 7i$$

$$\begin{aligned}\overline{z^{-1}} &= \frac{1}{-6 - 7i} \\ &= \frac{-6 + 7i}{36 + 49} \\ &= \frac{-6}{85} + i\frac{7}{85}\end{aligned}$$

16. If $z_1 = 3 + i$ and $z_2 = 2 - i$, then $\left| \frac{z_1 + z_2 - 1}{z_1 - z_2 + i} \right|$ is

(1) $\sqrt{\frac{8}{5}}$

(2) $\frac{\sqrt{8}}{5}$

(3) $\frac{8}{5}$

(4) $\frac{8}{\sqrt{5}}$

Sol. Answer (1)

$$z_1 = 3 + i, z_2 = 2 - i$$

$$\left| \frac{3+i+2-i-1}{3+i-2+i+i} \right|$$

$$= \left| \frac{4}{1+3i} \right|$$

$$= \frac{4}{\sqrt{10}}$$

$$= \sqrt{\frac{8}{5}}$$

17. The modulus of $\frac{(2+3i)^2}{2+i}$ is

(1) $\frac{\sqrt{13}}{5}$

(2) $\frac{\sqrt{147}}{5}$

(3) $\frac{13}{\sqrt{5}}$

(4) $\frac{\sqrt{185}}{5}$

Sol. Answer (3)

$$\frac{(2+3i)^2}{2+i} = \frac{(\sqrt{4+9})^2}{\sqrt{2^2+1}} = \frac{13}{\sqrt{5}}$$

18. The value of $(1+i)(1-i^2)(1+i^4)(1-i^5)$ is

(1) $2i$

(2) 8

(3) -8

(4) $8i$

Sol. Answer (2)

$$(1+i)(1-i^2)(1+i^4)(1-i^5)$$

$$4(1+i)(1-i) = 4 \times 2 = 8$$

19. If $z = \frac{1}{(1+i)(1-2i)}$, then $|z|$ is

(1) $\frac{2}{10}$

(2) $\frac{\sqrt{7}}{10}$

(3) $\frac{9}{\sqrt{10}}$

(4) $\frac{1}{\sqrt{10}}$

Sol. Answer (4)

$$z = \frac{1}{(1+i)(1-2i)}$$

$$|z| = \frac{1}{|1+i||1-2i|} = \frac{1}{\sqrt{2} \cdot \sqrt{5}} = \frac{1}{\sqrt{10}}$$

20. The value of $\left| \frac{1}{2+i} - \frac{1}{2-i} \right|$ is

- (1) $-\frac{2}{5}$ (2) $\frac{4}{25}$ (3) $\frac{2}{5}$ (4) 0

Sol. Answer (3)

$$\left| \frac{1}{2+i} - \frac{1}{2-i} \right| = \left| \frac{2-i-2-i}{5} \right| = \frac{2}{5}$$

21. The modulus of $i^{25} + (i+2)^3$ is

- (1) $\sqrt{47}$ (2) $4\sqrt{15}$ (3) $\sqrt{35}$ (4) $2\sqrt{37}$

Sol. Answer (4)

$$i^{25} + (i+2)^3$$

$$i + (-i) + 8 + 12i - 6 = 2 + 12i$$

$$\text{Now, } |2 + 12i| = \sqrt{4 + 144} = \sqrt{148} = 2\sqrt{37}$$

22. The argument of the complex number $(1+i)^4$ is

- (1) 135° (2) 180° (3) 90° (4) 45°

Sol. Answer (2)

$$(1+i)^4 = (\sqrt{2})^4 (e^{i\pi/4})^4 = e^{i\pi}$$

$$\text{Argument} = \pi = 180^\circ$$

23. The polar form of $(i^{41})^3$ is

- (1) $\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$ (2) $\cos \pi + i \sin \pi$ (3) $-\cos \pi - i \sin \pi$ (4) $\cos \left(\frac{-\pi}{2} \right) + i \sin \left(\frac{-\pi}{2} \right)$

Sol. Answer (4)

$$(i^{41})^3 = -i = 1 \left(\cos \left(\frac{-\pi}{2} \right) + i \sin \left(\frac{-\pi}{2} \right) \right)$$

24. If $z = \frac{-4 + 2\sqrt{3}i}{5 + \sqrt{3}i}$, then the value of $\arg(z)$ is

- (1) π (2) $\frac{\pi}{3}$ (3) $\frac{2\pi}{3}$ (4) $\frac{\pi}{4}$

Sol. Answer (3)

$$\begin{aligned}
 z &= \frac{-4 + 2\sqrt{3}i}{5 + \sqrt{3}i} \times \frac{(5 - \sqrt{3}i)}{(5 - \sqrt{3}i)} \\
 &= \frac{-20 + 6 + i(10\sqrt{3} + 4\sqrt{3})}{28} \\
 &= \frac{-14 + i(14\sqrt{3})}{28}
 \end{aligned}$$

$$\begin{aligned}
 \arg(z) &= \pi - \tan^{-1}(\sqrt{3}) \\
 &= \frac{2\pi}{3}
 \end{aligned}$$

25. If $z = \cos \frac{\pi}{4} + i \sin \frac{\pi}{6}$, then

$$(1) |z| = 1, \arg(z) = \frac{\pi}{4}$$

$$(2) |z| = 1, \arg(z) = \frac{\pi}{6}$$

$$(3) |z| = \frac{\sqrt{3}}{2}, \arg(z) = \frac{5\pi}{24}$$

$$(4) |z| = \frac{\sqrt{3}}{2}, \arg(z) = \tan^{-1} \frac{1}{\sqrt{2}}$$

Sol. Answer (4)

$$z = \cos \frac{\pi}{4} + i \sin \frac{\pi}{6} = \frac{1}{\sqrt{2}} + i \frac{1}{2}$$

$$|z| = \sqrt{\frac{1}{2} + \frac{1}{4}} = \frac{\sqrt{3}}{2}, \tan \alpha = \frac{\frac{1}{2}}{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}$$

$$\alpha = \tan^{-1} \left(\frac{1}{\sqrt{2}} \right)$$

26. The square root of $-8i$ is

$$(1) \pm 2(1 - i)$$

$$(2) 2(1 + i)$$

$$(3) \pm (1 - i)$$

$$(4) \pm (1 + i)$$

Sol. Answer (1)

$$\sqrt{0 - 8i} = x + iy$$

$$-8i = x^2 - y^2 + 2ixy$$

$$x^2 - y^2 = 0 \quad xy = -4$$

$$x = \pm y \quad x(\pm x) = -4 \quad x^2 = 4$$

$$x = \pm 2$$

$$y = \pm 2$$

$$\therefore \pm 2(1 - i)$$

27. The square root of $3 + 4i$ is

(1) $\pm (2 - i)$

(2) $\pm (2 + i)$

(3) $\pm (3 + i)$

(4) $\pm (3 - i)$

Sol. Answer (2)

$$\sqrt{3 + 4i} = x + iy$$

$$3 + 4i = x^2 - y^2 + 2ixy$$

$$x^2 - y^2 = 3$$

$$xy = 2$$

.....

$$x + iy = \pm(2 + i)$$

28. If α and β are the roots of $4x^2 + 3x + 7 = 0$, then the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is

(1) $\frac{4}{7}$

(2) $-\frac{3}{7}$

(3) $\frac{3}{7}$

(4) $-\frac{3}{4}$

Sol. Answer (2)

$$4x^2 + 3x + 7 = 0$$

Quadratic equation whose roots are

$$\frac{1}{\alpha}, \frac{1}{\beta} \text{ is } 7x^2 + 3x + 4 = 0$$

$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} = \frac{-3}{7}$$

29. If a and b are the roots of the equation $x^2 + x + 1 = 0$, then $a^2 + b^2$ is equal to

(1) 1

(2) 2

(3) -1

(4) 3

Sol. Answer (3)

$$x^2 + x + 1 = 0$$

$$\Rightarrow a^2 + b^2 = (a + b)^2 - 2ab$$

$$= (-1)^2 - 2$$

$$= -1$$

30. If the difference of the roots of the equation $x^2 - px + q = 0$ is unity,

(1) $p^2 + 4q = 1$

(2) $p^2 - 4q = 1$

(3) $p^2 - 4q^2 = (1 + 2q)^2$

(4) $4p^2 + q^2 = (1 + 2p)^2$

Sol. Answer (2)

$$x^2 - px + q = 0$$

Let α, β be roots

$$(\alpha - \beta) = 1$$

$$(\alpha - \beta)^2 = 1$$

$$(\alpha + \beta)^2 - 4\alpha\beta = 1$$

$$p^2 - 4q = 1$$

31. If α and β are the roots of the equation $x^2 - px + 16 = 0$, such that $\alpha^2 + \beta^2 = 9$, then the value of p is

(1) $\pm\sqrt{6}$

(2) $\pm\sqrt{41}$

(3) ± 8

(4) ± 7

Sol. Answer (2)

$$x^2 - px + 16 = 0$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$9 = p^2 - 32$$

$$p^2 = 41$$

$$p = \pm\sqrt{41}$$

32. The solution of the equation $z(\overline{z - 3i}) = 2(2 + 3i)$ is/are

(1) $2 + i, 3 - 2i$

(2) $2 + 2i, 3i$

(3) $3 + 2i, 2i$

(4) $2, 2 + 3i$

Sol. Answer (4)

$$\Rightarrow z(\overline{z - 3i}) = 2(2 + 3i)$$

$$z(\overline{z} + 3i) = 2(2 + 3i)$$

$$|z|^2 + 3iz = 4 + 6i$$

$$(x^2 + y^2) + 3i(x + iy) = 4 + 6i$$

$$(x^2 + y^2 - 3y) + 3ix = 4 + 6i$$

$$\Rightarrow 3x = 6 \text{ and } x^2 + y^2 - 3y = 4$$

$$\Rightarrow x = 2 \text{ and } 4 + y^2 - 3y = 4$$

$$\Rightarrow y = 0, 3$$

$$z = x + iy$$

$$= 2 + i \cdot 0 \text{ and } 2 + 3i$$

$$= 2, 2 + 3i$$

33. If $f(x) = x^4 - 8x^3 + 4x^2 + 4x + 39$ and $f(3 + 2i) = a + ib$, then $a : b$ equals

(1) $\frac{1}{8}$

(2) $-\frac{1}{4}$

(3) $\frac{1}{4}$

(4) $-\frac{1}{8}$

Sol. Answer (4)

$$f(x) = x^4 - 8x^3 + 4x^2 + 4x + 39$$

$$x = 3 + 2i \Rightarrow (x - 3)^2 = -4$$

$$x^2 - 6x + 13 = 0$$

$$\text{Now } x^4 - 8x^3 + 4x^2 + 4x + 39 = (x^2 - 6x + 13)(x^2 - 2x - 21) + (-96x + 312)$$

$$\text{Now } f(3 + 2i) = -96(3 + 2i) + 312$$

$$= -288 - 192i + 312$$

$$= 24 - 192i$$

$$= a + ib$$

$$\Rightarrow a = 24, b = -192.$$

$$\text{Required ratio} = \frac{24}{-192} = \frac{-1}{8}$$

34. If centre of a regular hexagon is at origin and one of the vertex on Argand diagram is $1 + 2i$, then its perimeter is

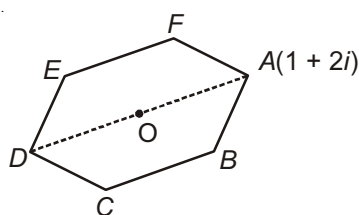
(1) $6\sqrt{5}$

(2) $4\sqrt{5}$

(3) $6\sqrt{2}$

(4) $2\sqrt{5}$

Sol. Answer (1)



In regular hexagon $OA = AB = BC = CD = ED = EF = FA$

Length of perimeter $= 6 \times |OA|$

$$= 6 \times \sqrt{1+4}$$

$$= 6\sqrt{5}$$

35. The sum of principal arguments of complex numbers $1 + i, -1 + i\sqrt{3}, -\sqrt{3} - i, \sqrt{3} - i, i, -3i, 2, -1$ is

(1) $\frac{11\pi}{12}$

(2) $\frac{13\pi}{12}$

(3) $\frac{12\pi}{13}$

(4) $\frac{\pi}{15}$

Sol. Answer (1)

$$\arg(1 + i) = \frac{\pi}{4}$$

$$\arg(-1 + i\sqrt{3}) = \frac{2\pi}{3}$$

$$\arg(-\sqrt{3} - i) = \frac{-5\pi}{6}$$

$$\arg(\sqrt{3} - i) = \frac{-\pi}{6}$$

$$\arg(i) = \frac{\pi}{2}$$

$$\arg(-3i) = \frac{-\pi}{2}$$

$$\arg(2) = 0$$

$$\arg(-1) = \pi$$

$$\text{Required sum} = \frac{11\pi}{12}$$

36. If $a = \cos\theta + i\sin\theta$, $b = \cos\phi + i\sin\phi$, $c = \cos\Psi + i\sin\Psi$ and $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = 2$, then

$\sin(\theta - \phi) + \sin(\phi - \psi) + \sin(\psi - \theta)$ equals

(1) 3

(2) $-\frac{3}{2}$

(3) 0

(4) $\frac{3}{2}$

Sol. Answer (3)

From question

$$\cos(\theta - \phi) + i\sin(\theta - \phi) + \cos(\phi - \psi) + i\sin(\phi - \psi) + \cos(\psi - \theta) + i\sin(\psi - \theta) = 2 + 0i$$

$$\Rightarrow \sin(\theta - \phi) + \sin(\phi - \psi) + \sin(\psi - \theta) = 0$$

37. The value of $(i + \sqrt{3})^{100} + (i - \sqrt{3})^{100} + 2^{100} =$

(1) 1

(2) -1

(3) 0

(4) 2

Sol. Answer (3)

$$(i + \sqrt{3})^{100} + (i - \sqrt{3})^{100} + 2^{100}$$

$$= (-i)^{100} \left(\frac{-1 - i\sqrt{3}}{2} \right)^{100} \times 2^{100} + (-i)^{100} \left(\frac{-1 + i\sqrt{3}}{2} \right)^{100} \cdot 2^{100} + 2^{100}$$

$$= 2^{100} ((\omega^2)^{100} + (\omega)^{100} + 1)$$

$$= 2^{100} (\omega^2 + \omega + 1) \text{ \{where } \omega \text{ is complex cube root of unity\}}$$

$$= 0$$

38. Which of the following is not true?

- (1) The number whose conjugate is $\frac{1}{1-i}$ is $\frac{1}{1+i}$
- (2) If $\sin x + i \cos 2x$ and $\cos x - i \sin 2x$ are conjugate to each other then number of values of x is zero
- (3) If $x + 1 + iy$ and $2 + 3i$ are conjugate of each other then the value of $x + y$ is -2
- (4) $2 + i > 3 + i$

Sol. Answer (4)

If z_1 and z_2 are two complex number if $\text{Im}(z_1) \neq 0$ and $\text{Im}(z_2) \neq 0$ then $z_1 > z_2$ or $z_1 < z_2$ does not hold.

39. The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ are the vertices of a triangle which is

- (1) Of area zero (2) Right angled isosceles (3) Equilateral (4) Obtuse angle isosceles

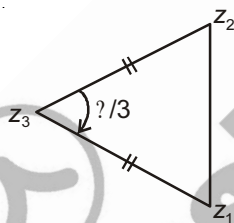
Sol. Answer (3)

$$(z_1 - z_3) = (z_2 - z_3) \left(\frac{1}{2} - i \frac{\sqrt{3}}{2} \right)$$

$$(z_1 - z_3) = (z_2 - z_3) e^{-i\pi/3}$$

Now using concept of rotation.

$\Rightarrow z_1, z_2, z_3$ are vertices of equilateral triangle.



40. Let $a = i^i$ and consider the following statements

$$S_1: a = e^{-\frac{\pi}{2}}$$

$$S_2: \text{The value of } \sin(\ln a) = -1$$

$$S_3: \text{Im}(a) + \arg(a) = 0$$

Now identify the correct combination of the true statements.

- (1) S_1, S_2 only (2) S_1, S_3 only (3) S_1, S_2, S_3 (4) S_1 only

Sol. Answer (3)

$$a = (i)^i$$

$$\log_e a = i \log_e i$$

$$\log_e a = i \log_e \left(e^{\frac{\pi}{2}} \right)$$

$$\log_e a = -\frac{\pi}{2}$$

$$a = e^{-\frac{\pi}{2}}$$

Therefore $\sin(\ln a) = \sin\left(-\frac{\pi}{2}\right) = -1$

$\operatorname{Im}(a) + \arg(a) = 0 + 0 = 0$

Therefore S_1, S_2, S_3 are correct.

41. If $z^2 + z + 1 = 0$ then the value of $\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + \dots + \left(z^{21} + \frac{1}{z^{21}}\right)^2$ is equal to

(1) 21

(2) 42

(3) 0

(4) 11

Sol. Answer (2)

if $z^2 + z + 1 = 0$

$\Rightarrow (z - \omega)(z - \omega^2) = 0$

$\Rightarrow z = \omega, \omega^2$

if $z = \omega$, then $\frac{1}{z} = \omega^2$

To find the value of $\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + \dots + \left(z^{21} + \frac{1}{z^{21}}\right)^2$

Now, $z + \frac{1}{z} = \omega + \frac{1}{\omega} = -1$, $z^2 + \frac{1}{z^2} = \omega^2 + \frac{1}{\omega^2} = -1$, $z^3 + \frac{1}{z^3} = 2$

$z^4 + \frac{1}{z^4} = \omega^4 + \frac{1}{\omega^4} = \omega + \frac{1}{\omega} = -1$, $z^5 + \frac{1}{z^5} = \omega^2 + \frac{1}{\omega^2} = -1$ and

$z^6 + \frac{1}{z^6} = 2$ and so on

Therefore,

$\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + \dots + \left(z^{21} + \frac{1}{z^{21}}\right)^2$

$= \{(-1)^2 + (-1)^2 + (2)^2\} + \{(-1)^2 + (-1)^2 + (2)^2\} \times \dots 7 \text{ times}$

$= (1 + 1 + 4) + (1 + 1 + 4) \times \dots 7 \text{ times}$

$= 6 + 6 \times \dots 7 \text{ times}$

$= 6 \times 7 = 42$

42. If $1, \alpha_1, \alpha_2, \dots, \alpha_{3n}$ be the roots of equation $x^{3n+1} - 1 = 0$, and ω be an imaginary cube root of unity,

then $\frac{(\omega^2 - \alpha_1)(\omega^2 - \alpha_2) \dots (\omega^2 - \alpha_{3n})}{(\omega - \alpha_1)(\omega - \alpha_2) \dots (\omega - \alpha_{3n})}$

(1) ω

(2) $-\omega$

(3) 1

(4) ω^2

Sol. Answer (3)

We have

$$x^{3n+1} - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{3n})$$

$$\text{Thus, } \frac{(\omega^2 - \alpha_1)(\omega^2 - \alpha_2) \dots (\omega^2 - \alpha_{3n})}{(\omega - \alpha_1)(\omega - \alpha_2) \dots (\omega - \alpha_{3n})}$$

$$= \frac{1}{(\omega + 1)} \cdot \frac{(\omega^2 - 1)(\omega^2 - \alpha_1)(\omega^2 - \alpha_2) \dots (\omega^2 - \alpha_{3n})}{(\omega - 1)(\omega - \alpha_1)(\omega - \alpha_2) \dots (\omega - \alpha_{3n})}$$

$$= \frac{1}{\omega + 1} \cdot \frac{(\omega^2)^{3n+1} - 1}{\omega^{3n+1} - 1} = \frac{1}{\omega + 1} \cdot \frac{(\omega^2)^{3n} \cdot \omega^2 - 1}{\omega^{3n} \cdot \omega - 1}$$

$$\frac{1}{\omega + 1} \cdot \frac{\omega^2 - 1}{\omega - 1} \quad [\because \omega^{3n} = \omega^{6n} = 1]$$

$$= \frac{\omega^2 - 1}{\omega^2 - 1} = 1$$

43. If z_1, z_2 and z_3, z_4 are two pairs of conjugate complex numbers, then $\arg\left(\frac{z_1}{z_3}\right) + \arg\left(\frac{z_2}{z_4}\right)$ is

- (1) 0 (2) $\frac{\pi}{2}$ (3) $\frac{3\pi}{4}$ (4) π

Sol. Answer (1)We have $z_2 = \bar{z}_1$ and $z_4 = \bar{z}_3$ Therefore, $z_1 z_2 = z_1 \bar{z}_1 = |z_1|^2$ and $z_3 z_4 = z_3 \bar{z}_3 = |z_3|^2$

$$\text{Now } \arg\left(\frac{z_1}{z_3}\right) + \arg\left(\frac{z_2}{z_4}\right) = \arg\left(\frac{z_1}{z_3} \times \frac{z_2}{z_4}\right)$$

$$= \arg\left(\frac{|z_1|^2}{|z_3|^2}\right) = 0 \quad (\because \text{Argument of a positive real number is } 0)$$

44. If $|z - 4 + 3i| \leq 2$, then the least and the greatest values of $|z|$ are

- (1) 3, 7 (2) 4, 7 (3) 3, 9 (4) 4, 5

Sol. Answer (1)

$$|z - (4 - 3i)| \leq 2$$

$$\|z| - |4 - 3i|\| \leq 2$$

$$\Rightarrow -2 \leq |z| - 5 \leq 2$$

$$\Rightarrow 3 \leq |z| \leq 7$$

$$|z|_{\min} = 3, |z|_{\max} = 7$$

45. If $|z_1| = 2$, $|z_2| = 3$, $|z_3| = 4$ and $|2z_1 + 3z_2 + 4z_3| = 4$, then the expression $|8z_2z_3 + 27z_3z_1 + 64z_1z_2|$ equals

(1) 72

(2) 24

(3) 96

(4) 92

Sol. Answer (3)

$$|8z_2z_3 + 27z_3z_1 + 64z_1z_2| = |z_1| |z_2| |z_3| \left| \frac{8}{z_1} + \frac{27}{z_2} + \frac{64}{z_3} \right|$$

$$= (2)(3)(4) \left| \frac{8\bar{z}_1}{|z_1|^2} + \frac{27\bar{z}_2}{|z_2|^2} + \frac{64\bar{z}_3}{|z_3|^2} \right| = (2)(3)(4) \left| \frac{8\bar{z}_1}{4} + \frac{27\bar{z}_2}{9} + \frac{64\bar{z}_3}{16} \right|$$

$$= 24 |2\bar{z}_1 + 3\bar{z}_2 + 4\bar{z}_3| = 24 |\overline{2z_1 + 3z_2 + 4z_3}| = 24 |2z_1 + 3z_2 + 4z_3|$$

$$= 24(4) = 96$$

46. If $z_1 = \cos\theta + i\sin\theta$ and $1, z_1, z_1^2, z_1^3, \dots, z_1^{n-1}$ are vertices of a regular polygon such that $\frac{I_m(z_1^2)}{R_e(z_1)} = \frac{\sqrt{5}-1}{2}$,

then the value of n is

(1) 20

(2) 10

(3) 18

(4) 15

Sol. Answer (1)

$z_1 = \cos\theta + i\sin\theta$ and $1, z_1, z_1^2, z_1^3, \dots, z_1^{n-1}$ are vertices of a regular polygon,

$$\theta = \frac{2\pi}{n}, z_1 = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}, z_1^2 = e^{i4\frac{\pi}{n}} = \cos\frac{4\pi}{n} + i\sin\frac{4\pi}{n}$$

$$\text{Now, } \frac{I_m(z_1^2)}{R_e(z_1)} = \frac{\sqrt{5}-1}{2}$$

$$\Rightarrow \frac{\sin\frac{4\pi}{n}}{\cos\left(\frac{2\pi}{n}\right)} = \frac{\sqrt{5}-1}{2} \Rightarrow \frac{2\sin\left(\frac{2\pi}{n}\right)\cos\left(\frac{2\pi}{n}\right)}{\cos\left(\frac{2\pi}{n}\right)} = \frac{\sqrt{5}-1}{2}$$

$$\Rightarrow \sin\left(\frac{2\pi}{n}\right) = \frac{\sqrt{5}-1}{4} = \sin 18^\circ = \sin \frac{\pi}{10} \Rightarrow \frac{2\pi}{n} = \frac{\pi}{10}$$

$$\Rightarrow n = 20$$

47. The area of a triangle whose vertices are represented by complex numbers 0, z and $ze^{i\alpha}$ ($0 < \alpha < \pi$) equals

- (1) $\frac{1}{2}|z|^2 \cos \alpha$ (2) $\frac{1}{2}|z|^2 \sin \alpha$ (3) $\frac{1}{2}|z|^2 \sin \alpha \cos \alpha$ (4) $\frac{1}{2}|z|^2$

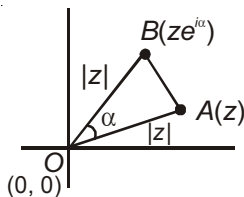
Sol. Answer (2)

To find area of a Δ whose vertices are represented by complex number 0, z and $ze^{i\alpha}$ ($0 < \alpha < \pi$)

$$\text{Area of } \Delta = \frac{1}{2}bc \sin A$$

$$= \frac{1}{2}|z||z| \sin \alpha$$

$$= \frac{1}{2}|z|^2 \sin \alpha$$



48. The maximum value of $|z|$ when z satisfies the condition $\left|z + \frac{2}{z}\right| = 2$ is

- (1) $\sqrt{3} - 1$ (2) $\sqrt{3} + 1$ (3) $\sqrt{3}$ (4) $\sqrt{2} + \sqrt{3}$

Sol. Answer (2)

$$\left|z + \frac{2}{z}\right| = 2$$

$$\left|z - \frac{2}{z}\right| \leq 2$$

$$\left|r - \frac{2}{r}\right| \leq 2$$

$$-2 \leq \frac{r^2 - 2}{r} \leq 2$$

$$\text{when } 0 \leq \frac{r^2 - 2}{r} \leq 2$$

$$r^2 - 2r - 2 \leq 0$$

$$1 - \sqrt{3} \leq r \leq 1 + \sqrt{3} \quad \dots(i)$$

$$r_{\max} \equiv 1 + \sqrt{3}$$

49. The value of $\sqrt{i}$ is

- (1) $\frac{1+i}{\sqrt{2}}, \frac{-1-i}{\sqrt{2}}$ (2) $\frac{1+i}{\sqrt{2}}, \frac{1-i}{\sqrt{2}}$ (3) $\frac{-1+i}{\sqrt{2}}, \frac{1-i}{\sqrt{2}}$ (4) $\frac{-1+i}{\sqrt{2}}, \frac{-1+i}{\sqrt{2}}$

Sol. Answer (1)Let the value of $\sqrt{i} = x + iy$

$$i = (x + iy)^2$$

$$i = x^2 - y^2 + 2xy$$

On comparing real and imaginary part

$$x^2 - y^2 = 0 \text{ and } 2xy = 1$$

$$x = \pm y \text{ and } xy = \frac{1}{2}$$

$$\text{Therefore } \sqrt{i} = \frac{1+i}{\sqrt{2}}, \frac{-1-i}{\sqrt{2}}$$

50. The roots of the equation

 $(x-a)(x-b)+(x-b)(x-c)+(x-a)(x-c) = 0$ are equal, then which of the following is not true?

(1) $a + b\omega + c\omega^2 = 0$

(2) $a + b\omega^2 + c\omega = 0$

(3) $a^2 + b^2 + c^2 - ab - bc - ca = 0$

(4) $a + b + c = 0$

Sol. Answer (4)

Roots of the equations

$$(x-a)(x-b) + (x-b)(x-c) + (x-a)(x-c) = 0$$

$$\text{i.e., } \{x^2 - (a+b)x + ab\} + \{x^2 - (b+c)x + bc\} + \{x^2 - (a+c)x + ac\} = 0$$

$$\text{i.e., } 3x^2 - 2(a+b+c)x + (ab+bc+ca) = 0$$

have equal roots, Therefore

$$B^2 - 4AC = 0$$

$$4(a+b+c)^2 - 4 \times 3(ab+bc+ca) = 0$$

$$4[a^2 + b^2 + c^2 + 2ab + 2bc + 2ca] - 12ab - 12bc - 12ca = 0$$

$$4(a^2 + b^2 + c^2) - 4ab - 4bc - 4ca = 0$$

$$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = 0$$

51. If $\log_{\sqrt{3}} \left(\frac{|z|^2 - |z| + 1}{2 + |z|} \right) < 2$, then the locus of z is

(1) $|z| = 5$

(2) $|z| < 5$

(3) $|z| > 5$

(4) $|z| = 0$

Sol. Answer (2)

$$\log_{\sqrt{3}} \left(\frac{|z|^2 - |z| + 1}{2 + |z|} \right) < 2$$

$$\Rightarrow \frac{z^2 - |z| + 1}{2 + |z|} < (\sqrt{3})^2$$

$$\Rightarrow |z|^2 - |z| + 1 < 6 + 3|z|$$

$$\Rightarrow |z|^2 - 4|z| - 5 < 0$$

$$\Rightarrow (|z| + 1)(|z| - 5) < 0$$

$$\text{but } |z| + 1 > 0$$

$$\Rightarrow |z| - 5 < 0$$

$$\Rightarrow |z| < 5$$

52. If $\arg z = \frac{\pi}{4}$, then

(1) $\operatorname{Re}(z^2) = 9\operatorname{Im}(z^2)$

(2) $\operatorname{Im}(z^2) = 0$

(3) $\operatorname{Re}(z^2) = 0$

(4) $\operatorname{Re}(z) = 0$

Sol. Answer (3)

$$\arg z = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1} \left| \frac{y}{x} \right| = \frac{\pi}{4}$$

$$\Rightarrow \left| \frac{y}{x} \right| = 1$$

$$\Rightarrow |y| = |x| \Rightarrow x^2 - y^2 = 0$$

$$\therefore \operatorname{Re}(z^2) = 0$$

53. If $z^2 + z|z| + |z|^2 = 0$, then locus of z is

(1) Circle

(2) Straight line

(3) A pair of straight line

(4) None of these

Sol. Answer (3)

$$\text{Let } z = x + iy$$

Given equation is,

$$z^2 + z|z| + |z|^2 = 0$$

$$\Rightarrow (x + iy)^2 + (x + iy) + \sqrt{x^2 + y^2} + (x^2 + y^2) = 0$$

$$\Rightarrow x^2 - y^2 + 2ixy + x\sqrt{x^2 + y^2} + iy\sqrt{x^2 + y^2} + x^2 + y^2 = 0$$

$$\Rightarrow 2x^2 + x\sqrt{x^2 + y^2} + i(2xy + y\sqrt{x^2 + y^2}) = 0$$

$$\text{Now, } 2x^2 + x\sqrt{x^2 + y^2} = 0$$

$$\Rightarrow x(2x + \sqrt{x^2 + y^2}) = 0$$

$$\Rightarrow x = 0 \text{ or } x^2 + y^2 = 4x^2$$

$$\Rightarrow 3x^2 - y^2 = 0$$

Alternative

$$\frac{z^2}{|z|^2} + \frac{z}{|z|} + 1 = 0$$

$$\Rightarrow \left(\frac{z}{|z|}\right)^2 + \left(\frac{z}{|z|}\right) + 1 = 0$$

$$\therefore \frac{z}{|z|} = \omega, \omega^2$$

$$\Rightarrow z = \omega|z|, z = \omega^2|z|$$

54. The least value of p for which the two curves $\arg z = \frac{\pi}{6}$ and $|z - 2\sqrt{3}i| = p$ intersect is

(1) $p = \sqrt{3}$

(2) $p = 3$

(3) $p = \frac{1}{\sqrt{3}}$

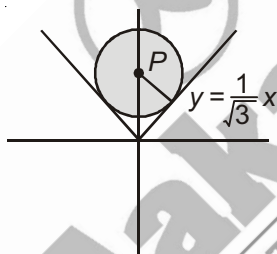
(4) $p = \frac{1}{3}$

Sol. Answer (2)

$$\sqrt{3}y - x = 0$$

$$\text{Now, } \left| \frac{\sqrt{3} \cdot 2\sqrt{3} - 0}{\sqrt{(\sqrt{3})^2 + 1^2}} \right| = p$$

$$\Rightarrow p = \left| \frac{6}{2} \right| = 3$$



55. If $\operatorname{Re}\left(\frac{z+4}{2z-1}\right) = \frac{1}{2}$, then z is represented by a point lying on

(1) A circle

(2) An ellipse

(3) A straight line

(4) No real locus

Sol. Answer (3)

$$\operatorname{Re}\left(\frac{z+4}{2z-1}\right) = \frac{1}{2}$$

$$\Rightarrow \left(\frac{z+4}{2z-1}\right) + \overline{\left(\frac{z+4}{2z-1}\right)} = 1$$

$$\Rightarrow \frac{z+4}{2z-1} + \frac{\bar{z}+4}{2\bar{z}-1} = 1$$

$$\Rightarrow \frac{2z\bar{z} - z + 8\bar{z} - 4 + 2z\bar{z} + 8z - \bar{z} - 4}{(2z-1)(2\bar{z}-1)} = 1$$

$$\Rightarrow 4z\bar{z} + 7z + 7\bar{z} - 8 = 4z\bar{z} - 2z - 2\bar{z} + 1$$

$$\Rightarrow 9z + 9\bar{z} - 9 = 0$$

$$\Rightarrow \boxed{z + \bar{z} = 1}$$

Hence point z lies on a straight line.

56. If $f(x)$ and $g(x)$ are two polynomials such that the polynomial $h(x) = xf(x^3) + x^2g(x^6)$ is divisible by $x^2 + x + 1$, then

- (1) $f(1) + g(1) = 1$ (2) $f(1) = -g(1)$ (3) $f(1) = g(1) \neq 0$ (4) $f(1) = \pm g(1)$

Sol. Answer (2)

$\because h(x) = xf(x^3) + x^2g(x^6)$ is divisible by $x^2 + x + 1$,

So, when $h(x)$ will be divided by $x - \omega$ and $x - \omega^2$ remainder will be 0.

$$h(\omega) = \omega f(1) + \omega^2 g(1) = 0 \quad \dots(i)$$

$$h(\omega^2) = \omega^2 f(1) + \omega g(1) = 0 \quad \dots(ii)$$

Now, adding (i) & (ii),

$$(\omega + \omega^2)f(1) + (\omega + \omega^2)g(1) = 0$$

$$\Rightarrow -f(1) - g(1) = 0 \quad \Rightarrow f(1) = -g(1)$$

57. The value of

$(x + \omega + \omega^2)(x + \omega^2 + \omega^4)(x + \omega^4 + \omega^8) \dots$ till $2n$ factors

- (1) $(x - 1)^{2n}$ (2) $(x - 1)^{2n+1}$ (3) $(x - 1)^{2n-1}$ (4) $(x - 1)^{2n+2}$

Sol. Answer (1)

The given expression is $(x - 1)(x - 1) \dots (x - 1) \dots$ till $2n$ factors.

$$= (x - 1)^{2n}$$

58. If $z = \frac{\sqrt{3} - i}{2}$, then $(i^{101} + z^{101})^{103}$ equals

- (1) iz (2) z
(3) $\bar{z}$ (4) ωz , (ω is complex cube root of unity)

Sol. Answer (2)

$$z^{101} + i^{101} = \left[\cos\left(\frac{-\pi}{6}\right) + i \sin\left(\frac{-\pi}{6}\right) \right]^{101} + i^{101}$$

$$= \cos\left(\frac{-101\pi}{6}\right) + i \sin\left(\frac{-101\pi}{6}\right) + i^{101}$$

$$= \cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6} + i^{101}$$

$$= -\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$$

$$= -\frac{\sqrt{3}}{2} - \frac{i}{2} + i$$

$$= -\frac{\sqrt{3}}{2} + \frac{i}{2}$$

Now,

$$(i^{101} + z^{101})^{103} = \left(-\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^{103}$$

$$= \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)^{103}$$

$$= \cos \frac{515\pi}{6} + i \sin \frac{515\pi}{6}$$

$$= \cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}$$

$$= \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}$$

$$= \frac{\sqrt{3}}{2} - \frac{i}{2}$$

59. The region of the complex plane for which $\left|\frac{z-a}{z+a}\right| = 1$ is (a is real)

(1) x -axis

(2) y -axis

(3) Straight line $x = a$

(4) The straight line $y = a$

Sol. Answer (2)

$$\left|\frac{z-a}{z+a}\right| = 1$$

$|z-a| = |z+a|$, let $z = x + iy$ then,

$$(x-a)^2 + (y)^2 = (x+a)^2 + y^2$$

$$\Rightarrow x^2 - 2ax + a^2 + y^2 = x^2 + a^2 + 2ax + y^2$$

$$\Rightarrow 4ax = 0$$

$$\Rightarrow x = 0$$

is y -axis

60. The imaginary part of $\frac{2z+1}{iz+1}$ is -2 ; then the locus of the point representing z in the Argand plane is
- (1) A circle (2) A straight line (3) A parabola (4) An ellipse

Sol. Answer (2)

Let $z = x + iy$

$$z = \frac{2x + 2iy + 1}{i(x + iy) + 1}$$

$$\frac{2x + 1 + i2y}{(1 - y) + ix} \times \frac{(1 - y) - ix}{(1 - y) - ix} = \frac{(2x + 1)(1 - y) + 2xy + i((2y)(1 - y) - x(2x + 1))}{(1 - y)^2 + x^2}$$

from given condition

$$\frac{(2y)(1 - y) - x(2x + 1)}{(1 - y)^2 + x^2} = -2$$

$$2y - 2y^2 - 2x^2 - x = -2(x^2 + y^2 + 1 - 2y) = 2y - x = -2 + 4y$$

$$\Rightarrow 2y + x - 2 = 0, \text{ i.e., a straight line}$$

61. If z is a complex number satisfying $|2008z - 1| = 2008|z - 2|$, then locus z is
- (1) y - axis (2) x - axis (3) Circle (4) A line parallel to y -axis

Sol. Answer (4)

$z = x + iy$

$$|2008z - 1| = 2008|z - 2|$$

$$\Rightarrow \left| z - \frac{1}{2008} \right| = |z - 2|$$

Put $z = x + iy$

$$\left(x - \frac{1}{2008} \right)^2 + (y)^2 = (x - 2)^2 + y^2$$

$$\Rightarrow \left(\frac{1}{2008} \right)^2 - 2 \times x \times \frac{1}{2008} = 4 - 4x$$

$$\Rightarrow 4x - \frac{x}{1004} = 4 - \left(\frac{1}{2008} \right)^2$$

$$x = \frac{4 - \left(\frac{1}{2008} \right)^2}{4 - \frac{1}{1004}}$$

a line parallel to y -axis.

62. The locus of the point z satisfying the condition $\arg \frac{z-1}{z+1} = \frac{\pi}{3}$ is

- (1) A straight line (2) Circle (3) A parabola (4) Ellipse

Sol. Answer (2)

$$\arg \left(\frac{z-1}{z+1} \right) = \angle APB = \frac{\pi}{3}$$

$\Rightarrow z$ lies on a circle

Alternatively

put $z = x + iy$

$$\arg \left(\frac{z-1}{z+1} \right) = \frac{\pi}{3}$$

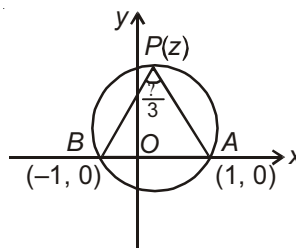
$$\Rightarrow \arg \left[\frac{(x-1)+iy}{(x+1)+iy} \right] = \frac{\pi}{3} \Rightarrow \arg \left[\frac{(x-1)+iy}{(x+1)+iy} \times \frac{(x+1)-iy}{(x+1)-iy} \right] = \frac{\pi}{3}$$

$$\Rightarrow \arg \left[\frac{(x^2+y^2-1)+i(2y)}{(x+1)^2+y^2} \right] = \frac{\pi}{3} \Rightarrow \tan^{-1} \left[\frac{2y}{x^2+y^2-1} \right] = \frac{\pi}{3}$$

$$\Rightarrow \frac{2y}{x^2+y^2-1} = \sqrt{3} \Rightarrow \sqrt{3}(x^2+y^2) - \sqrt{3} - 2y = 0$$

$$\Rightarrow x^2 + y^2 - \frac{2}{\sqrt{3}}y - 1 = 0$$

$$\Rightarrow \text{Which represents a circle having centre at } \left(0, \frac{1}{\sqrt{3}} \right) \text{ and radius } \sqrt{\frac{1}{3} + 1} = \frac{2}{\sqrt{3}}$$



63. The locus of $z = i + 2 \exp \left(i \left(\theta + \frac{\pi}{4} \right) \right)$, (where θ is parameter) is

- (1) A circle (2) An ellipse (3) A parabola (4) A hyperbola

Sol. Answer (1)

$$z = i + 2 \exp \left(i \left(\theta + \frac{\pi}{4} \right) \right), \text{ where } \theta \text{ is parameter put } z = x + iy$$

$$x + iy = i + 2 \left[\cos \left(\theta + \frac{\pi}{4} \right) + i \sin \left(\theta + \frac{\pi}{4} \right) \right]$$

equating real and imaginary parts we get

$$x = 2 \cos \left(\theta + \frac{\pi}{4} \right) \dots\dots(1)$$

$$y = 1 + 2 \sin \left(\theta + \frac{\pi}{4} \right)$$

$$\text{or } y - 1 = 2 \sin \left(\theta + \frac{\pi}{4} \right) \dots\dots(2)$$

squaring and adding (1) and (2), we get

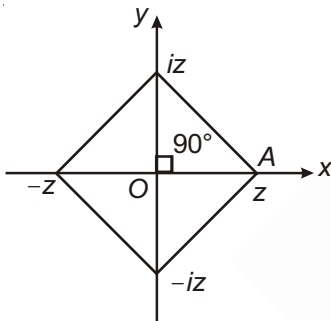
$$x^2 + (y - 1)^2 = 4$$

which represents a circle with centre (0, 1) and radius 2.

64. If one vertex and centre of a square are z and origin then which of the following cannot be the vertex of the square?

- (1) iz (2) $-z$ (3) $-iz$ (4) $2z$

Sol. Answer (4)



On rotating OA by 90° angle we can find other vertices.

65. If z_1, z_2, z_3 represent the vertices of an equilateral triangle such that $|z_1| = |z_2| = |z_3|$, then

- (1) $z_1 + z_2 = z_3$ (2) $z_1 + z_2 + z_3 = 0$ (3) $z_1 z_2 = \frac{1}{z_3}$ (4) $z_1 - z_2 = z_3 - z_2$

Sol. Answer (2)

z_1, z_2, z_3 are the vertices of an equilateral triangle such that $|z_1| = |z_2| = |z_3|$

$$\text{or } |z_1 - 0| = |z_2 - 0| = |z_3 - 0|$$

$\Rightarrow$ origin is the circumcentre of the $\Delta \Rightarrow$ origin is the centroid of the equilateral Δ

$$\Rightarrow \frac{z_1 + z_2 + z_3}{3} = 0 \quad \Rightarrow \quad z_1 + z_2 + z_3 = 0$$

66. If $|z - 2 - 3i| + |z + 2 - 6i| = 4$, $i = \sqrt{-1}$, then locus of z is

- (1) An ellipse (2) A point
(3) Segment joining the points $(2+3i)$ and $(-2+6i)$ (4) Empty

Sol. Answer (4)

$$|z - (2 + 3i)| + |z - (-2 + 6i)| = 4$$

$$\text{Let } z_1 = 2 + 3i, \quad z_2 = -2 + 6i$$

$$|z_1 - z_2| = |4 - 3i| = 5 > k$$

$$\Rightarrow |z - z_1| + |z - z_2| = 2a, \text{ where } k < |z_1 - z_2|$$

$\Rightarrow$ This does not represent any curve

$\Rightarrow$ Locus of z is an empty set.

Alternatively : If we put $z = x + iy$, then we got an equation in x and y which does not have any solution.

67. If z_1, z_2, z_3 and u, v, w are complex numbers representing the vertices of two triangles such that $z_3 = (1 - t)z_1 + tz_2$ and $w = (1 - t)u + tv$, where t is a complex number, then the two triangles

- (1) Have the same area (2) Are similar (3) Are congruent (4) Are equilateral

Sol. Answer (2)

z_1, z_2, z_3 and u, v, w are complex numbers representing the vertices of two triangles such that

$$z_3 = (1 - t)z_1 + tz_2 \text{ and } w = (1 - t)u + tv, \quad t \in \mathbb{C}$$

$$z_3 = z_1 - tz_1 + tz_2 \text{ and } w - u = -tu + tv$$

$$z_3 - z_1 = t(z_2 - z_1) \text{ and } w - u = t(v - u)$$

$$\Rightarrow t = \frac{z_3 - z_1}{z_2 - z_1} \quad \dots(1)$$

$$\text{and } t = \frac{w - u}{v - u} \quad \dots(2)$$

$\Rightarrow$ From (1) & (2)

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{w - u}{v - u}$$

$$\Rightarrow \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \arg\left(\frac{w - u}{v - u}\right) \quad \dots(3)$$

$$\Rightarrow \arg\left(\frac{z_3 - z_1}{z_2 - z_1} - 1\right) = \arg\left(\frac{w - u}{v - u} - 1\right) \Rightarrow \arg\left(\frac{z_3 - z_2}{z_2 - z_1}\right) = \arg\left(\frac{w - v}{v - u}\right)$$

$$\Rightarrow \arg\left(\frac{z_3 - z_2}{z_1 - z_2}\right) = \arg\left(\frac{w - v}{u - v}\right) \quad \dots(4)$$

From (3) & (4) we conclude that two triangles are similar.

68. For two complex numbers z_1 and z_2 , we have $\left|\frac{z_1 - z_2}{1 - \bar{z}_1 z_2}\right| = 1$, then

(1) Both z_1 and z_2 lie on circle $|z| = 1$

$$(2) \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{3}$$

(3) At least one of z_1 and z_2 lies on the circle $|z| = 1$

$$(4) |z_1| = 2|z_2|$$

Sol. Answer (1)

$$\left| \frac{z_1 - z_2}{1 - \bar{z}_1 z_2} \right| = 1$$

$$\Rightarrow |z_1 - z_2| = |1 - \bar{z}_1 z_2| \Rightarrow |z_1 - z_2|^2 = |1 - \bar{z}_1 z_2|^2$$

$$\Rightarrow |z_1|^2 + |z_2|^2 - z_1 \bar{z}_2 - \bar{z}_1 z_2 = 1 + |\bar{z}_1|^2 |z_2|^2 - \bar{z}_1 z_2 - z_1 \bar{z}_2$$

$$\Rightarrow |z_1|^2 |z_2|^2 - |z_1|^2 - |z_2|^2 + 1 = 0 \Rightarrow (|z_1|^2 - 1)(|z_2|^2 - 1) = 0$$

$$\Rightarrow |z_1| = 1, |z_2| = 1 \Rightarrow \text{Both } z_1 \text{ and } z_2 \text{ lie on the circle } |z| = 1$$

69. Let α and β are the roots of the equation $x^2 + x + 1 = 0$ then

(1) $\alpha^2 + \beta^2 = 4$

(2) $(\alpha - \beta)^2 = 3$

(3) $\alpha^3 + \beta^3 = 2$

(4) $\alpha^4 + \beta^4 = 1$

Sol. Answer (3)

$$x^2 + x + 1 = 0$$

$$\alpha + \beta = -1$$

$$\alpha\beta = -1$$

(1) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$$= (-1)^2 - 2(1)$$

$$= 1 - 2 = -1$$

(2) $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$$= (-1)^2 - 4 \times 1 = -3$$

(3) $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta)$

$$= (\alpha + \beta)((\alpha + \beta)^2 - 2\alpha\beta - \alpha\beta)$$

$$= (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)$$

$$= (-1)((-1)^2 - 3 \times 1)$$

$$= (-1)(1 - 3) = 2$$

Alternative

$$x^2 + x + 1 = 0$$

$$x = \omega, \omega^2 \text{ (complex root of unity)}$$

$$\therefore \omega^3 + (\omega^2)^3 = 2$$

(4) $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$

$$= (-1)^2 - 2 \cdot 1$$

$$= 1 - 2$$

$$= -1$$

70. If the ratio of the roots of $lx^2 - nx + n = 0$ is $p : q$, then

$$(1) \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{n}{l}} = 0 \quad (2) \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} - \sqrt{\frac{n}{l}} = 0 \quad (3) \sqrt{\frac{q}{p}} + \sqrt{\frac{p}{q}} + \sqrt{\frac{l}{n}} = 1 \quad (4) \sqrt{\frac{q}{p}} + \sqrt{\frac{p}{q}} + \sqrt{\frac{l}{n}} = 0$$

Sol. Answer (2)

$$\frac{p}{q} = \frac{\alpha}{\beta}, \quad \alpha + \beta = +\frac{n}{l}, \quad \alpha\beta = \frac{n}{l}$$

$$\text{Now, } \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} = \sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}}$$

$$= \frac{\alpha + \beta}{\sqrt{\alpha\beta}} = \frac{+\frac{n}{l}}{\sqrt{\frac{n}{l}}} = +\sqrt{\frac{n}{l}}$$

$$\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} = +\sqrt{\frac{n}{l}} \Rightarrow \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} - \sqrt{\frac{n}{l}} = 0$$

71. For the equation $|x^2| + |x| - 6 = 0$, the roots are

- (1) Real and equal (2) Real with sum 0 (3) Real with sum 1 (4) Real with product 0

Sol. Answer (2)

$$|x^2| + |x| - 6 = 0 \Rightarrow |x|^2 + |x| - 6 = 0$$

$$\Rightarrow (|x| + 3)(|x| - 2) = 0 \Rightarrow |x| = -3, |x| = 2$$

$$\Rightarrow x = \pm 2$$

Two roots are real, with sum 0.

72. If $a + b + c = 0$ and a, b, c are rational, then the roots of the equation $(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$ are

- (1) Rational (2) Irrational (3) Imaginary (4) Equal

Sol. Answer (1)

$$(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$$

$$\text{Put } x = 1, \quad b + c - a + c + a - b + a + b - c = a + b + c = 0$$

 $\therefore 1$ is the root of the equation. $\therefore$ Roots are rational.73. If $\sec\alpha, \tan\alpha$ are roots of $ax^2 + bx + c = 0$, then

- (1) $a^4 - b^4 + 4ab^2c = 0$ (2) $a^4 + b^4 - 4ab^2c = 0$ (3) $a^2 - b^2 = 4ac$ (4) $a^2 + b^2 = ac$

Sol. Answer (1)

We know

$$\sec^2\alpha - \tan^2\alpha = 1$$

$$(\sec\alpha - \tan\alpha)(\sec\alpha + \tan\alpha) = 1$$

$$\left(\frac{\sqrt{b^2 - 4ac}}{a} \right) \left(-\frac{b}{a} \right) = -1$$

Squaring both side

$$(b^2 - 4ac)b^2 = a^4$$

$$a^4 - b^4 + 4ab^2c = 0$$

74. If x is real, then the expression $\frac{x^2 + 34x - 71}{x^2 + 2x - 7}$

(1) Lies between 4 and 7

(2) Lies between 5 and 9

(3) Has no value between 4 and 7

(4) Has no value between 5 and 9

Sol. Answer (4)

$$\text{Let } y = \frac{x^2 + 34x - 71}{x^2 + 2x - 7}$$

$$\Rightarrow x^2(y - 1) + x(2y - 34) + 71 - 7y = 0$$

For real x , discriminant should be ≥ 0

$$\Rightarrow (2y - 34)^2 - 4(y - 1)(71 - 7y) \geq 0 \Rightarrow 4(y - 17)^2 - 4(y - 1)(71 - 7y) \geq 0$$

$$\Rightarrow (y - 17)^2 - (-7y^2 + 78y - 71) \geq 0 \Rightarrow 8y^2 - 112y + 360 \geq 0$$

$$\Rightarrow y^2 - 14y + 45 \geq 0 \Rightarrow (y - 9)(y - 5) \geq 0$$

$$\Rightarrow y \geq 9 \text{ or } y \leq 5$$



75. If α, β are roots of $ax^2 + bx + c = 0$, then the equation $ax^2 - bx(x - 1) + c(x - 1)^2 = 0$ has roots

(1) $\frac{\alpha}{1 - \alpha}, \frac{\beta}{1 - \beta}$

(2) $\frac{1 - \alpha}{\alpha}, \frac{1 - \beta}{\beta}$

(3) $\frac{\alpha}{1 + \alpha}, \frac{\beta}{1 + \beta}$

(4) $\frac{1 + \alpha}{\alpha}, \frac{1 + \beta}{\beta}$

Sol. Answer (3)

$$ax^2 + bx + c = 0,$$

$$\text{Given equation is } ax^2 - bx(x - 1) + c(x - 1)^2 = 0$$

$$a\left(\frac{-x}{x - 1}\right)^2 + b\left(\frac{-x}{x - 1}\right) + 1 = 0$$

Now,

$$\text{Replacing } x \text{ by } \alpha = -\frac{x}{x - 1}$$

$$\frac{ax^2}{(x-1)^2} - \frac{bx}{x-1} + c = 0 \Rightarrow ax^2 - bx(x-1) + c(x-1)^2 = 0$$

$$\alpha = \frac{-x}{x-1} \Rightarrow x = \frac{\alpha}{1+\alpha} \text{ is the root of the above equation.}$$

76. Let α, β be the roots of $ax^2 + bx + c = 0$, γ, δ be the roots of $px^2 + qx + r = 0$ and D_1 and D_2 be their respective discriminant. If $\alpha, \beta, \gamma, \delta$ are in A.P., then the ratio $D_1 : D_2$ is equal to

(1) $\frac{a^2}{b^2}$

(2) $\frac{a^2}{p^2}$

(3) $\frac{b^2}{q^2}$

(4) $\frac{c^2}{r^2}$

Sol. Answer (2)

$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}, D_1 = b^2 - 4ac$$

$$\gamma + \delta = -\frac{q}{p}, \gamma\delta = \frac{r}{p}, D_2 = q^2 - 4rp$$

Let common difference of A.P. be k .

$$k = |\alpha - \beta| = |\gamma - \delta|$$

$$\Rightarrow \frac{\sqrt{b^2 - 4ac}}{a} = \frac{\sqrt{q^2 - 4pr}}{p} \Rightarrow \frac{\sqrt{b^2 - 4ac}}{\sqrt{q^2 - 4pr}} = \frac{a}{p} \Rightarrow \frac{\sqrt{D_1}}{\sqrt{D_2}} = \frac{a}{p}$$

$$\Rightarrow \frac{D_1}{D_2} = \frac{a^2}{p^2}$$

77. The equation

$$\frac{a(x-b)(x-c)}{(a-b)(a-c)} + \frac{b(x-c)(x-a)}{(b-c)(b-a)} + \frac{c(x-a)(x-b)}{(c-a)(c-b)} = x$$

is satisfied by

(1) No value of x

(2) Exactly two values of x

(3) Exactly three values of x

(4) All values of x

Sol. Answer (4)

$$\frac{a(x-b)(x-c)}{(a-b)(a-c)} + \frac{b(x-c)(x-a)}{(b-c)(b-a)} + \frac{c(x-a)(x-b)}{(c-a)(c-b)} = x$$

is satisfied by $x = a, x = b, x = c$.

A quadratic equation is satisfied by more than two values of x . So it is an identity. Hence it is satisfied by all values of x .

78. Consider the equation $ax^2 + bx + c = 0$, where $a \neq 0$, $a, b, c \in R$ then

- (1) If one root is $\alpha + \sqrt{\beta}$, then other root is $\alpha - \sqrt{\beta}$
- (2) If $a = 1$ and b and c are integers, then root will be integer
- (3) If one root is $\alpha + i\beta$, then other root will be $\alpha - i\beta$
- (4) If roots are of opposite sign, then $b \neq 0$

Sol. Answer (3)

- (1) Root will be of the form $\alpha \pm \sqrt{\beta}$ if a, b, c are rational.
- (2) There is no information about $b^2 - 4ac$
Hence statement is false.
- (3) As a, b, c are real and one root is $\alpha + i\beta$ then other root will be $\alpha - i\beta$.
- (4) If roots are of opposite sign then $\alpha\beta < 0 \Rightarrow \frac{c}{a} < 0$

79. If the equation $(k^2 - 3k + 2)x^2 + (k^2 - 5k + 4)x + (k^2 - 6k + 5) = 0$ is an identity then the value of k is

- (1) 1
- (2) 2
- (3) 3
- (4) 4

Sol. Answer (1)

For an identity

$$(k^2 - 3k + 2) = 0$$

$$\Rightarrow (k - 1)(k - 2) = 0$$

$$\Rightarrow k = 1, k = 2$$

$$k^2 - 5k + 4 = 0 \Rightarrow (k - 1)(k - 4) = 0$$

$$k = 1, 4$$

$$k^2 - 6k + 5 = 0 \Rightarrow (k - 5)(k - 1) = 0$$

$$k = 1, 5$$

Common value of $k = 1$.

80. The value of k if

- (1) The roots of $5x^2 + 13x + k = 0$ are reciprocal to each other is 5
- (2) The roots of $x^2 + x + k = 0$ are consecutive integer is 1
- (3) The roots of $x^2 - 6x + k = 0$ are in the ratio 2 : 1 is 7
- (4) The roots of the equation $x^2 + kx - 1 = 0$ are real, equal in magnitude but opposite in sign is 1

Sol. Answer (1)

- (1) Let roots are α, β

$$\alpha + \beta = -\frac{13}{5} \quad \dots(i)$$

$$\alpha\beta = \frac{k}{5} \quad \dots(ii)$$

For reciprocal roots $\alpha\beta = 1 \Rightarrow k = 5$.

(2) If roots are cosecutive integer then

$$|\alpha - \beta| = 1$$

$$\Rightarrow |\alpha - \beta|^2 = 1$$

$$(\alpha + \beta)^2 - 4\alpha\beta = 1$$

$$\Rightarrow 1 - 4k = 1$$

$$\Rightarrow k = 0$$

(3) Let roots are $2\alpha, \alpha$

$$\Rightarrow 2\alpha + \alpha = 6 \quad \dots(i)$$

$$2\alpha \cdot \alpha = k \quad \dots(ii)$$

By (i), (ii)

$$\alpha = 2, k = 8$$

(4) In this case

$$\alpha + \beta = 0$$

$$\Rightarrow k = 0$$

81. If the difference of the roots of the equation $x^2 + ax + b = 0$ is equal to the difference of the roots of the equation $x^2 + bx + a = 0$, then

$$(1) a + b = 4$$

$$(2) a + b = -4$$

$$(3) a - b = 4$$

$$(4) a - b = -4$$

Sol. Answer (2)

$$x^2 + ax + b = 0 \Rightarrow \alpha + \beta = -a, \alpha\beta = b, |\alpha - \beta| = \sqrt{a^2 - 4b}$$

$$x^2 + bx + a = 0 \Rightarrow \gamma + \delta = -b, \gamma\delta = a, |\gamma - \delta| = \sqrt{b^2 - 4a}$$

$$\text{Now, } |\alpha - \beta| = |\gamma - \delta| \Rightarrow \sqrt{a^2 - 4b} = \sqrt{b^2 - 4a}$$

$$\Rightarrow a^2 - 4b = b^2 - 4a \Rightarrow a^2 - b^2 = 4(b - a)$$

$$\Rightarrow (a - b)(a + b + 4) = 0 \Rightarrow a + b = -4 \quad (a \neq b)$$

82. If the equations $px^2 + 2qx + r = 0$ and $px^2 + 2rx + q = 0$ ($q \neq r$) have a common root, then $p + 4q + 4r$ equals

$$(1) 0$$

$$(2) 1$$

$$(3) 2$$

$$(4) -2$$

Sol. Answer (1)

$$\text{Let } \alpha \text{ be common root, } p\alpha^2 + 2q\alpha + r = 0 \quad \dots(1)$$

$$\text{and } p\alpha^2 + 2r\alpha + q = 0 \quad \dots(2)$$

$$\text{Now } (1) - (2) \Rightarrow 2\alpha(q - r) + r - q = 0 \Rightarrow \alpha = \frac{1}{2}$$

Common root is $\alpha = \frac{1}{2}$, substituting in (1)

$$p\left(\frac{1}{2}\right)^2 + 2q\left(\frac{1}{2}\right) + r = 0 \Rightarrow 4r + 4q + p = 0$$

83. If the equations $ax^2 + bx + c = 0$ and $x^2 + x + 1 = 0$ has one common root then $a : b : c$ is equal to

- (1) $1 : 1 : 1$ (2) $1 : 2 : 3$ (3) $2 : 3 : 1$ (4) $3 : 2 : 1$

Sol. Answer (1)

$$x^2 + x + 1 = 0 \quad \dots (i)$$

$$\text{Discriminant} = b^2 - 4ac = 1 - 4 \times 1 \times 1 = -3$$

Hence the roots of $x^2 + x + 1 = 0$ are not real.

So roots will be in pair.

Also the roots of $ax^2 + bx + c = 0$ will be non-real.

Clearly both roots of the equations are common.

$$\Rightarrow \frac{a}{1} = \frac{b}{1} = \frac{c}{1}$$

$$\Rightarrow a : b : c = 1 : 1 : 1$$

84. If 1, 2, 3 are the roots of the equation $x^3 + ax^2 + bx + c = 0$, then

- (1) $a = 1, b = 2, c = 3$ (2) $a = -6, b = 11, c = -6$
 (3) $a = 6, b = 11, c = 6$ (4) $a = 6, b = 6, c = 6$

Sol. Answer (2)

If 1, 2, 3 are roots of equation then

$$x^3 + ax^2 + bx + c = 0$$

$$\Rightarrow 1 + 2 + 3 = -a \Rightarrow a = -6$$

$$1 \cdot 2 + 2 \cdot 3 + 1 \cdot 3 = b \Rightarrow b = 11$$

$$1 \cdot 2 \cdot 3 = -c \Rightarrow c = -6$$

85. Consider that $f(x) = ax^2 + bx + c$, $D = b^2 - 4ac$, then which of the following is not true?

- (1) If $a > 0$, then minimum value of $f(x)$ is $\frac{-D}{4a}$ (2) If $a < 0$, then maximum value of $f(x)$ is $\frac{-D}{4a}$
 (3) If $a > 0$, $D < 0$, then $f(x) > 0$ for all $x \in R$ (4) If $a > 0$, $D > 0$, then $f(x) > 0$ for all $x \in R$

Sol. Answer (4)

$$f(x) = ax^2 + bx + c$$

$$= a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right)$$

$$= a \left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} + \frac{c}{a} \right)$$

$$f(x) = a \left(\left(x + \frac{b}{2a} \right)^2 + \frac{b^2 - 4ac}{4a^2} \right)$$

$$f(x) = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a}$$

$$f(x) = a\left(x + \frac{b}{2a}\right)^2 - \frac{D}{4a}$$

Clearly if $a > 0$ the minimum value of $f(x) = \frac{-D}{4a}$

Similarly if $a < 0$ the maximum value = $\frac{-D}{4a}$

If $ax^2 + bx + c > 0$ then $a > 0$, $D < 0$ for all $x \in R$

Hence option (4) is not true.

86. If the minimum value of $x^2 + 2x + 3$ is m and maximum value of $-x^2 + 4x + 6$ is M then the value of $m + M$ is

(1) 10

(2) 11

(3) 12

(4) 13

Sol. Answer (3)

$$x^2 + 2x + 3 = (x + 1)^2 + 2 \Rightarrow m = 2$$

$$-x^2 + 4x + 6 = -x^2 + 4x + 4 - 4 + 6$$

$$= 6 - (x^2 - 4x + 4) + 4$$

$$= 10 - (x^2 - 4x + 4)$$

$$= 10 - (x - 2)^2$$

$$\Rightarrow M = 10$$

$$m + M = 2 + 10 = 12$$

87. For all $x \in R$ if $mx^2 - 9mx + 5m + 1 > 0$, then m lies in the interval

(1) $\left(-\frac{61}{4}, 0\right)$

(2) $\left(\frac{4}{61}, \frac{61}{4}\right)$

(3) $\left[0, \frac{4}{61}\right)$

(4) $\left(\frac{-4}{61}, 0\right)$

Sol. Answer (3)

$$\text{Let } y = mx^2 - 9mx + 5m + 1$$

We need $y > 0$

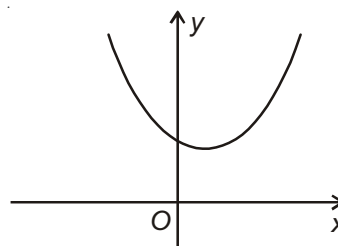
$\Rightarrow$ Upward parabola above x -axis.

$$mx^2 - 9mx + 5m + 1 > 0, \forall x \in R.$$

$$\Rightarrow D < 0, a > 0$$

$$\text{i.e., } 81m^2 - 4(m)(5m + 1) < 0 \text{ and } m > 0$$

$$\Rightarrow m(61m - 4) < 0 \text{ and } m > 0 \Rightarrow 0 < m < \frac{4}{61}$$



Also for $m = 0$,

$$0x^2 - 9(0)x + 0 + 1 = 1 > 0, \forall x \in \mathbb{R}$$

$$\therefore m \in \left[0, \frac{4}{61}\right)$$

88. If one root of the equation $(l - m)x^2 + lx + 1 = 0$ is double the other and if l is real, then the greatest value of m is

(1) $\frac{9}{8}$

(2) $\frac{7}{8}$

(3) $\frac{8}{9}$

(4) $\frac{5}{9}$

Sol. Answer (1)

$$(l - m)x^2 + lx + 1 = 0$$

$$\alpha + 2\alpha = \frac{l}{m - l} \Rightarrow \alpha = \frac{l}{3(m - l)} \quad \dots(1)$$

$$\alpha \cdot (2\alpha) = \frac{1}{l - m} \Rightarrow \alpha^2 = \frac{1}{2(l - m)} \quad \dots(2)$$

From (1) and (2)

$$\frac{1}{2(l - m)} = \frac{l^2}{9(l - m)^2}$$

$$\Rightarrow 2l^2 - 9l + 9m = 0$$

$$\text{For real } l, 81 - 8 \times 9m \geq 0 \Rightarrow m \leq \frac{81}{72}$$

$$\Rightarrow m \leq \frac{9}{8}$$

$$\therefore \text{Greatest value of } m \text{ is } \frac{9}{8}$$

89. If p, q, r are real numbers satisfying the condition $p + q + r = 0$, then the roots of the quadratic equation $3px^2 + 5qx + 7r = 0$ are

(1) Positive

(2) Negative

(3) Real and distinct

(4) Imaginary

Sol. Answer (3)

$$3px^2 + 5qx + 7r = 0$$

$$\begin{aligned}
 \Delta &= (5q)^2 - 4(3p)(7r) \\
 &= 25q^2 - 84pr \\
 &= 25(p+r)^2 - 84pr \\
 &= 25p^2 - 34pr + 25r^2 \\
 &= \left(5p - \frac{17}{5}r\right)^2 + \frac{336}{25}r^2 > 0
 \end{aligned}$$

$\therefore$ roots are real and distinct.

90. The roots of the equation $x^3 - 2x^2 - x + 2 = 0$ are

- (1) 1, 2, 3 (2) -1, 1, 2 (3) -1, 0, 1 (4) -1, -2, 3

Sol. Answer (2)

$$x^3 - 2x^2 - x + 2 = 0$$

As $x = 1$ is the root of the equation

Hence we may write

$$\begin{aligned}
 x^3 - 2x^2 - x + 2 &= x^2(x-1) - x(x-1) - 2(x-1) \\
 &= (x-1)(x^2 - x - 2) \\
 &= (x-1)(x-2)(x+1)
 \end{aligned}$$

Roots = 1, -1, 2.

91. If a and b are rational and α, β be the roots of $x^2 + 2ax + b = 0$, then the equation with rational coefficients one of whose roots is $\alpha + \beta + \sqrt{\alpha^2 + \beta^2}$ is

- (1) $x^2 + 4ax - 2b = 0$ (2) $x^2 + 4ax + 2b = 0$ (3) $x^2 - 4ax + 2b = 0$ (4) $x^2 - 4ax - 2b = 0$

Sol. Answer (2)

$$\alpha + \beta = -2a, \quad \alpha\beta = b$$

$$\alpha + \beta + \sqrt{\alpha^2 + \beta^2} = -2a + \sqrt{4a^2 - 2b}$$

The other root of equation will be $\alpha + \beta - \sqrt{\alpha^2 + \beta^2}$

$$\text{i.e., } -2a - \sqrt{4a^2 - 2b}$$

$\therefore$ Sum of roots, $S = -4a$

$$\text{Product of roots, } P = 4a^2 - (4a^2 - 2b) = 2b$$

$\therefore$ required equation is $x^2 - Sx + P = 0$

$$\text{i.e., } x^2 + 4ax + 2b = 0$$

92. The values of a , for which the quadratic equation $3x^2 + 2(a^2 + 1)x + (a^2 - 3a + 2) = 0$ possesses roots of opposite sign, are

- (1) $1 < a < 2$ (2) $a \in (2, \infty)$ (3) $1 < a < 3$ (4) $-1 < a < 0$

Sol. Answer (1)

For roots of opposite sign, product < 0

$$\Rightarrow \frac{a^2 - 3a + 2}{3} < 0 \Rightarrow (a - 2)(a - 1) < 0$$

$$\Rightarrow 1 < a < 2$$

93. Let $a, b, c \in R$ and $a \neq 0$ be such that $(a + c)^2 < b^2$, then the quadratic equation $ax^2 + bx + c = 0$ has

- (1) Imaginary roots (2) Real roots
(3) Exactly one real root lying in the interval $(-1, 1)$ (4) Exactly two roots in $(-1, 1)$

Sol. Answer (3)

Here we observe that $(a + c)^2 < b^2$

$$\Rightarrow (a - b + c)(a + b + c) < 0$$

$\Rightarrow$ Exactly one real root of the given equation lies in $(-1, 1)$.

$$ax^2 + bx + c = 0$$

$$D = b^2 - 4ac > (a + c)^2 - 4ac = (a - c)^2$$

$$\Rightarrow \Delta \geq 0$$

$\therefore$ Roots are real.

94. If $p + iq$ be one of the roots of the equation

$x^3 + ax + b = 0$, then $2p$ is one of the roots of the equation

- (1) $x^3 + ax + b = 0$ (2) $x^3 - ax - b = 0$ (3) $x^3 + ax - b = 0$ (4) $x^3 + bx + a = 0$

Sol. Answer (3)

$p + iq$ is one root $\Rightarrow p - iq$ is other root.

Let α be third root.

$$\text{Now sum} = \alpha + p + iq + p - iq = 0$$

$$\Rightarrow \alpha = -2p$$

$$\alpha = -2p \text{ is root of } x^3 + ax + b = 0$$

$$\therefore 2p \text{ is root of } (-x)^3 - ax + b = 0$$

$$\Rightarrow x^3 + ax - b = 0$$

95. If $a_1, a_2, a_3, a_4, \dots, a_{n-1}, a_n$ are distinct non-zero real numbers such that $(a_1^2 + a_2^2 + a_3^2 + \dots + a_{n-1}^2)x^2 + 2(a_1a_2 + a_2a_3 + a_3a_4 + \dots + a_{n-1}a_n)x + (a_2^2 + a_3^2 + a_4^2 + \dots + a_n^2) \leq 0$ then $a_1, a_2, a_3, \dots, a_n$ are in

- (1) A.P. (2) G.P. (3) H.P. (4) A.G.P.

Sol. Answer (2)

We have, given expression

$$(a_1^2 + a_2^2 + a_3^2 + \dots + a_{n-1}^2)x^2 + 2(a_1a_2 + a_2a_3 + a_3a_4 + \dots + a_{n-1}a_n)x + (a_2^2 + a_3^2 + a_4^2 + \dots + a_n^2) \leq 0$$

$$\Rightarrow (a_1x + a_2)^2 + (a_2x + a_3)^2 + (a_3x + a_4)^2 + \dots + (a_{n-1}x + a_n)^2 \leq 0$$

$$\Rightarrow (a_1x + a_2)^2 + (a_2x + a_3)^2 + (a_3x + a_4)^2 + \dots + (a_{n-1}x + a_n)^2 = 0,$$

as sum of square can't be negative.

$$\Rightarrow a_1x + a_2 = 0 = a_2x + a_3 = a_3x + a_4 = \dots = a_{n-1}x + a_n$$

$$\Rightarrow -x = \frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_4}{a_3} = \dots = \frac{a_n}{a_{n-1}}$$

$$\Rightarrow a_1, a_2, a_3, \dots, a_{n-1}, a_n \text{ are in G.P.}$$

96. The roots of $ax^2 + bx + c = 0$, whose $a \neq 0, b, c \in R$, are non-real complex and $a + c < b$. Then

(1) $4a + c > 2b$

(2) $4a + c < 2b$

(3) $4a + c = 2b$

(4) None of these

Sol. Answer (2)

$$f(x) = ax^2 + bx + c, \text{ given } f(-1) = a - b + c < 0$$

$$\therefore f(x) < 0, \forall x \in R \text{ as roots are non-real complex}$$

$$\therefore f(-2) < 0 \Rightarrow 4a - 2b + c < 0 \Rightarrow 4a + c < 2b$$

97. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ ($a, b, c \neq 0$) is equal to sum of squaresof their reciprocals, then $\frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in

(1) A.P.

(2) G.P.

(3) H.P.

(4) None of these

Sol. Answer (3)

$$\text{Given, } \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$$

$$\text{Also, } \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$\Rightarrow (\alpha + \beta)(\alpha^2\beta^2) = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \Rightarrow \left(-\frac{b}{a}\right)\left(\frac{c^2}{a^2}\right) = \frac{b^2}{a^2} - \frac{2c}{a}$$

$$\Rightarrow -bc^2 = b^2a - 2ca^2$$

$$\Rightarrow 2ca^2 = bc^2 + b^2a$$

$$\Rightarrow 2\frac{a}{b} = \frac{c}{a} + \frac{b}{c}$$

$$\Rightarrow \frac{a}{c}, \frac{b}{a}, \frac{c}{b} \text{ are in H.P.}$$

98. The number of irrational roots of the equation

$$(x-1)(x-2)(3x-2)(3x+1) = 21 \text{ is}$$

(1) 0

(2) 2

(3) 3

(4) 4

Sol. Answer (2)

$$(x-1)(3x-2)(3x+1)(x-2) = 21$$

$$(3x^2 - 5x + 2)(3x^2 - 5x - 2) = 21$$

$$\text{Put, } 3x^2 - 5x = t$$

$$(t+2)(t-2) = 21 \Rightarrow t^2 = 25 \Rightarrow t = 5, t = -5$$

$$\text{Now, } 3x^2 - 5x = 5 \text{ and } 3x^2 - 5x = -5$$

$$\Rightarrow 3x^2 - 5x - 5 = 0 \text{ and } 3x^2 - 5x + 5 = 0$$

$$3x^2 - 5x - 5 = 0 \text{ has two irrational roots.}$$

$$\text{whereas roots of } 3x^2 - 5x + 5 = 0 \text{ are imaginary.}$$

99. Consider the following statements

$$S_1: \text{ If } x \text{ is in radian and } 2\sin^2 x + 3\sin x - 2 > 0 \text{ and } x^2 - x - 2 < 0 \text{ then } x \in \left(\frac{\pi}{6}, 2\right).$$

$$S_2: \text{ If } x \in \left(0, \frac{\pi}{2}\right) \text{ and } x \notin \{-1, 0\}, \text{ then the expression } \sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}} \text{ is always greater than or equal to } 2 \tan \alpha.$$

$$S_3: \text{ If } a, b, c \text{ are the sides of a triangle } ABC \text{ such that } x^2 - 2(a+b+c)x + 3\lambda(ab+bc+ca) = 0 \text{ has real roots then } \lambda < \frac{4}{3}.$$

$$S_4: \text{ If the equations } x^3 + 3px^2 + 3qx + r = 0 \text{ and } x^2 + 2px + q = 0 \text{ have a common root then } 4(p^2 - q)(q^2 - pr) = (pq - r)^2.$$

The correct combination of true statements is

(1) S_1 only

(2) S_2 only

(3) S_3 only

(4) All S_1, S_2, S_3, S_4

Sol. Answer (4)

$$S_1: x^2 - x - 2 < 0$$

$$\Rightarrow (x-2)(x+1) < 0$$

$$\Rightarrow -1 < x < 2 \quad \dots (i)$$

$$2\sin^2 x + 3\sin x - 2 > 0$$

$$\Rightarrow 2\sin^2 x + 4\sin x - \sin x - 2 > 0$$

$$2\sin x(\sin x + 2) - 1(\sin x + 2) > 0$$

$$\Rightarrow (\sin x + 2)(2\sin x - 1) > 0$$

$$\Rightarrow \sin x > \frac{1}{2}$$

$$\Rightarrow x \in \left(\frac{\pi}{6}, \frac{5\pi}{6}\right) \quad \dots(ii)$$

By (i), (ii) $x \in \left(\frac{\pi}{6}, 2\right)$

S_2 : Using A.M. $\geq$ G.M.

$$\frac{\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}}{2} \geq \sqrt{(x^2 + x) \cdot \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}}$$

$$\Rightarrow \sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}} \geq 2 \tan \alpha$$

S_3 : Using $D \geq 0$

$$\Rightarrow 4(a + b + c)^2 - 4(1)(3\lambda)(ab + bc + ca) \geq 0$$

$$\Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) - 3\lambda(ab + bc + ca) \geq 0$$

$$\Rightarrow a^2 + b^2 + c^2 + (ab + bc + ca)(2 - 3\lambda) \geq 0$$

$$\Rightarrow 3\lambda - 2 \leq \frac{a^2 + b^2 + c^2}{ab + bc + ca} \quad \dots (i)$$

But $(a - b)^2 \leq c^2$

$$(b - c)^2 \leq a^2$$

$$(c - a)^2 \leq b^2$$

$$\Rightarrow \frac{a^2 + b^2 + c^2}{ab + bc + ca} \leq 2$$

$$\Rightarrow 3\lambda - 2 \leq 2$$

$$\Rightarrow \lambda \leq \frac{4}{3}$$

$$S_4 : x^3 + 3px^2 + 3qx + r = 0 \quad \dots(i)$$

Multiply the second equation by x

$$x^3 + 2px^2 + qx = 0 \quad \dots(ii)$$

By (i) - (ii)

$$px^2 + 2qx + r = 0 \quad \dots(iii)$$

But $x^2 + 2px + q = 0$

$$\Rightarrow px^2 + 2p^2x + pq = 0 \quad \dots(iv)$$

By (iii) - (iv)

$$2x(q - p^2) + (r - pq) = 0$$

$$x = \frac{r - pq}{2(q - p^2)}$$

Putting x in

$$x^2 + 2px + q = 0$$

We get

$$4(p^2 - q)(q^2 - pr) = (pq - r^2)^2$$

100. If a, b are real, then the roots of the quadratic equation $(a - b)x^2 - 5(a + b)x - 2(a - b) = 0$ are

- (1) Real and equal (2) Non-real complex (3) Real and unequal (4) None of these

Sol. Answer (3)

The given quadratic equation is $(a - b)x^2 - 5(a + b)x - 2(a - b) = 0$

The discriminant

$$D = (-5(a + b))^2 + 8(a - b)(a - b)$$

$$= 25(a + b)^2 + 8(a - b)^2$$

Hence $D > 0 \forall a \& b$.

So, roots are real and unequal.

101. If α, β are the roots of the equation $ax^2 - bx + c = 0$, then equation $(a + cy)^2 = b^2y$ in y has the roots

- (1) $\frac{1}{\alpha}, \frac{1}{\beta}$ (2) α^2, β^2 (3) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ (4) $\frac{1}{\alpha^2}, \frac{1}{\beta^2}$

Sol. Answer (4)

Since, α, β are the roots of the equation $ax^2 - bx + c = 0$

$$\text{So, } \alpha + \beta = \frac{b}{a}, \alpha\beta = \frac{c}{a}$$

Now, we have to observe root of the equation

$$(a + cy)^2 = b^2y$$

$$\Rightarrow a^2 + 2acy + c^2y^2 = b^2y$$

$$\Rightarrow c^2y^2 + (2ac - b^2)y + a^2 = 0$$

$$\Rightarrow y^2 + \left(\frac{2ac - b^2}{c^2}\right)y + \frac{a^2}{c^2} = 0$$

$$\Rightarrow y^2 - \left(\frac{b^2}{c^2} - \frac{2a}{c}\right)y + \frac{a^2}{c^2} = 0$$

$$\Rightarrow y^2 - \left(\frac{1}{\alpha^2} + \frac{1}{\beta^2}\right)y + \frac{1}{\alpha^2\beta^2} = 0$$

Hence the equation $(a + cy)^2 = b^2y$ has roots $\frac{1}{\alpha^2}, \frac{1}{\beta^2}$

102. If a, b, c are in G.P., then the equation $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root if

$\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in

(1) A.P.

(2) G.P.

(3) H.P.

(4) $ab = cd$

Sol. Answer (1)

Since a, b, c are in G.P.,

So, $b^2 = ac$

$$\Rightarrow 4b^2 - 4ac = 0$$

$D = 0$ for the equation $ax^2 + 2bx + c = 0$

Hence, it will have equal roots, and root will be

$$x = -\frac{b}{a}$$

Now, $ax^2 + 2bx + c$ and $dx^2 + 2ex + f = 0$ have a common root,

So, $x = -\frac{b}{a}$ will satisfy the equation

$$dx^2 + 2ex + f = 0$$

$$\Rightarrow d \cdot \frac{b^2}{a^2} - 2e \cdot \frac{b}{a} + f = 0$$

$$\Rightarrow \frac{db^2 - 2aeb + a^2f}{a^2} = 0$$

$$\Rightarrow db^2 - 2aeb + a^2f = 0$$

$$\Rightarrow dac - 2aeb + a^2f = 0$$

$$\Rightarrow dc + af = 2eb$$

$$\frac{d}{a} + \frac{f}{c} = \frac{2e}{b}$$

So, $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

103. The least integral value of k for which the equation $x^2 - 2(k+2)x + 12 + k^2 = 0$ has two distinct real roots is

(1) 0

(2) 2

(3) 3

(4) 4

Sol. Answer (3)

The given equation is $x^2 - 2(k+2)x + 12 + k^2 = 0$ has distinct real roots when $D > 0$

$$\Rightarrow 4(k+2)^2 - 4(12 + k^2) > 0$$

$$\Rightarrow k^2 + 4 + 4k - 12 - k^2 > 0$$

$$\Rightarrow 4k - 8 > 0$$

$$\Rightarrow k > 2$$

So least integral value of k is 3.

104. The roots x_1 and x_2 of the equation $x^2 + px + 12 = 0$ are such that their difference is 1. Then the positive value of p is

- (1) 1 (2) 2 (3) 3 (4) 7

Sol. Answer (4)

$$\alpha - \beta = \frac{\sqrt{D}}{a} = 1$$

$$\Rightarrow \sqrt{p^2 - 48} = 1$$

$$\Rightarrow p = \pm 7; \text{ but } p \text{ is positive, hence } p = 7.$$

105. If $a < b < c < d$ and $k > 0$, then the quadratic equation $(x - a)(x - c) + k(x - b)(x - d) = 0$ has

- (1) All roots real and distinct (2) All roots real but not necessarily distinct
(3) All root real and negative (4) May be imaginary

Sol. Answer (1)

$$f(x) = (x - a)(x - c) + k(x - b)(x - d)$$

$$f(a) = k(a - b)(a - d) \text{ which is positive}$$

$$f(b) = (b - a)(b - c) \text{ which is negative}$$

$$f(c) = k(c - b)(c - d) \text{ which is negative}$$

$$f(d) = (d - a)(d - c) \text{ which is positive}$$

So, $f(x) = 0$ has a root in the interval (a, b) and another in (c, d) . So the roots are real and distinct.

106. If α, β and γ be the roots of the equation $x^3 - 3x^2 + 3x + 7 = 0$ and ω be non-real cube root of unity, then

the modulus of the expression $\left(\frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\alpha-1} \right)$ is equal to

- (1) 1 (2) 2 (3) 3 (4) 4

Sol. Answer (3)

$$x^3 - 3x^2 + 3x + 7 = 0$$

$$(x - 1)^3 + 8 = 0$$

$$\Rightarrow (x - 1) = -2, -2\omega, -2\omega^2$$

$$x = -1, 1 - 2\omega, 1 - 2\omega^2$$

$$\text{Now, } \left| \frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\alpha-1} \right|$$

$$= \left| \frac{-2}{-2\omega} + \frac{-2\omega}{-2\omega^2} + \frac{-2\omega^2}{-2} \right|$$

$$= \left| \frac{1}{\omega} + \frac{1}{\omega} + \omega^2 \right| = |3\omega^2| = 3$$

107. If the roots of the quadratic equation $x^2 - ax + 2b = 0$ are prime numbers, then the value of $(a - b)$ is

- (1) 0 (2) 2 (3) -2 (4) 4

Sol. Answer (2)

$$x^2 - ax + 2b = 0 \text{ (Here } a, b \text{ are integers)}$$

Let α, β be roots

Now, sum of roots = a

product of roots = $2b$ (an even number)

$\therefore$ '2' is one root

$$\text{Now, } 4 - 2a + 2b = 0 \Rightarrow a - b = 2$$

108. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$.

Then the value of r is

[IIT-JEE 2007]

- (1) $\frac{2}{9}(p-q)(2q-p)$ (2) $\frac{2}{9}(q-p)(2p-q)$ (3) $\frac{2}{9}(q-2p)(2q-p)$ (4) $\frac{2}{9}(2p-q)(2q-p)$

Sol. Answer (4)

Use relation between roots and coefficients

$$\alpha + \beta = p \quad \dots(i)$$

$$\alpha \beta = r \quad \dots(ii)$$

$$\frac{\alpha}{2} + 2\beta = q \quad \dots(iii)$$

$$\left(\frac{\alpha}{2}\right)(2\beta) = r \quad \dots(iv)$$

(ii) and (iv) are same. (i) and (iii) can be solved to obtain α and β in terms of p and q , thereby giving r .

109. A man walks a distance of 3 units from the origin towards the north-east ($N 45^\circ E$) direction. From there, he walks a distance of 4 units towards the north-west ($N 45^\circ W$) direction to reach a point P . Then the position of P in the Argand plane is

[IIT-JEE 2007]

- (1) $3e^{i\pi/4} + 4i$ (2) $(3 - 4i)e^{i\pi/4}$ (3) $(4 + 3i)e^{i\pi/4}$ (4) $(3 + 4i)e^{i\pi/4}$

Sol. Answer (4)

Use the idea of rotation to obtain the desired P as

$$3e^{i\frac{\pi}{4}} \cdot e^{i\phi}, \text{ where } \tan \phi = \frac{4}{3}$$

$$\text{yielding } \sin \phi = \frac{4}{5}, \cos \phi = \frac{3}{5}$$

110. If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on

[IIT-JEE 2007]

- (1) A line not passing through the origin
 (2) $|z| = \sqrt{2}$
 (3) The x-axis
 (4) The y-axis

Sol. Answer (4)

We have,

$$\begin{aligned}\frac{z}{1-z^2} &= \frac{1}{\frac{1}{z} - z} = \frac{1}{\frac{\bar{z}\bar{z}}{z} - z} \\ &= \frac{1}{\bar{z} - z} = \frac{1}{-2i\operatorname{Im}(z)} \\ &= i\left(\frac{1}{z\operatorname{Im}z}\right) = i\beta, \beta \in R\end{aligned}$$

Thus the locus of $\frac{z}{1-z^2}$ is y-axis.

111. A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point z_1 . From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{i} + \hat{j}$ and then it moves through an angle $\frac{\pi}{2}$ in anticlockwise direction on a circle with centre at origin, to reach a point z_2 . The point z_2 is given by

[IIT-JEE 2008]

- (1) $6 + 7i$ (2) $-7 + 6i$ (3) $7 + 6i$ (4) $-6 + 7i$

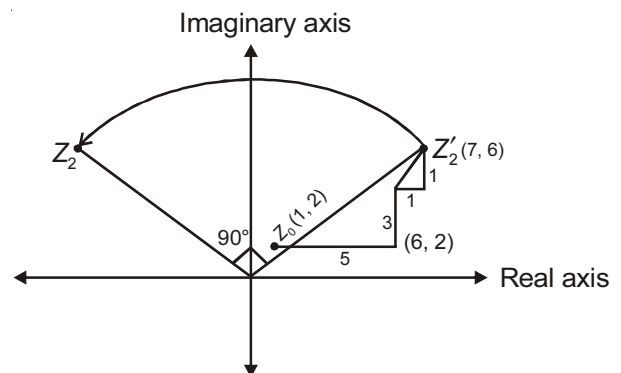
Sol. Answer (4)

$$Z'_2 = (6 + \sqrt{2} \times \cos 45^\circ, 5 + \sqrt{2} \sin 45^\circ) = (7, 6) = 7 + 6i$$

by rotation about $(0, 0)$

$$\frac{Z_2}{Z'_2} = e^{i\frac{\pi}{2}} \Rightarrow Z_2 = Z'_2(e^{i\frac{\pi}{2}})$$

$$Z_2 = (7 + 6i)\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right) = (7 + 6i)(i) = -6 + 7i$$



112. Let $z = x + iy$ be a complex number where x and y are integers. Then the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$ is

[IIT-JEE 2009]

- (1) 48 (2) 32 (3) 40 (4) 80

Sol. Answer (1)

$$z\bar{z}^3 + \bar{z}z^3 = 350$$

$$|z|^2 (\bar{z}^2) + |z|^2 (z^2) = 350$$

$$|z|^2 (z^2 + \bar{z}^2) = 350$$

$$\Rightarrow (x^2 + y^2)(x^2 - y^2 + 2ixy + x^2 - y^2 - 2ixy) = 350$$

$$\Rightarrow 2(x^2 + y^2)(x^2 - y^2) = 350$$

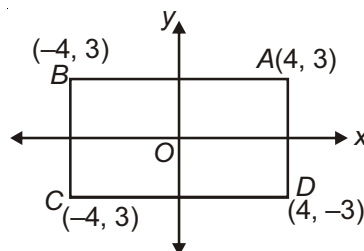
$$\Rightarrow (x^2 + y^2)(x^2 - y^2) = 175 = (3^2 + 4^2)(4^2 - 3^2)$$

Which suggests that points (x, y) satisfying the given equation are $(4, 3)$, $(-4, -3)$, $(-4, 3)$, $(4, -3)$

Required area = $AB \times BC$

$$= 8 \times 6$$

$$= 48 \text{ sq. units}$$



113. Let $z = \cos\theta + i\sin\theta$. Then the value of $\sum_{m=1}^{15} \operatorname{Im}(z^{2m-1})$ at $\theta = 2^\circ$ is

[IIT-JEE 2009]

(1) $\frac{1}{\sin 2^\circ}$

(2) $\frac{1}{3\sin 2^\circ}$

(3) $\frac{1}{2\sin 2^\circ}$

(4) $\frac{1}{4\sin 2^\circ}$

Sol. Answer (4)

$$z = \cos\theta + i\sin\theta = e^{i\theta}$$

$$\text{Now, } \sum_{m=1}^{15} \operatorname{Im} e^{i(2m-1)\theta}$$

$$= \sin\theta + \sin 3\theta + \dots + \sin 29\theta$$

$$= \frac{\sin \frac{15 \cdot 2\theta}{2}}{\sin \left(\frac{2\theta}{2} \right)} \cdot \sin \left(\frac{2\theta + (15-1) \times 2\theta}{2} \right)$$

$$= \frac{\sin 15\theta \cdot \sin 15\theta}{\sin \theta}$$

For $\theta = 2^\circ$, the given expression reduces to

$$= \frac{\sin 30^\circ \cdot \sin 30^\circ}{\sin 2^\circ} = \frac{1}{4\sin 2^\circ}$$

114. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers

satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is [IIT-JEE 2010]

(1) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$

(2) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$

(3) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$

(4) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

Sol. Answer (2)We have $\alpha + \beta = -p$

$$\alpha^3 + \beta^3 = q = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = -p^3 + 3p(\alpha\beta)$$

$$\Rightarrow \alpha\beta = \frac{p^3 + q}{3p}$$

The quadratic equation with $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as roots is

$$\Rightarrow x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right) x + \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 0$$

$$\Rightarrow x^2 - \left(\frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} \right) x + 1 = 0$$

$$\Rightarrow x^2 - \frac{p^2 - 2\frac{p^3 + q}{3p}}{\frac{p^3 + q}{3p}} x + 1 = 0$$

$$\Rightarrow (p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$$

115. Let β and α be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$

[IIT-JEE 2011]

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (3)

We observe that

$$(\alpha + \beta)(\alpha^{n-1} - \beta^{n-1}) = \alpha^n - \beta^n + \alpha\beta(\alpha^{n-2} - \beta^{n-2})$$

$$\Rightarrow 6a_{n-1} = a_n - 2a_{n-2}$$

$$\Rightarrow 6 = \frac{a_n - 2a_{n-2}}{a_{n-1}}$$

$$\Rightarrow 3 = \frac{a_n - 2a_{n-2}}{2a_{n-1}}, \forall n > 2$$

Putting $n = 10$, we get

$$\frac{a_{10} - 2a_8}{2a_9} = 3$$

116. A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0,$$

have one root in common is

[IIT-JEE 2011]

(1) $-\sqrt{2}$

(2) $-i\sqrt{3}$

(3) $i\sqrt{5}$

(4) $\sqrt{2}$

Sol. Answer (2)

Let α be a common root between given equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$

$$\Rightarrow \frac{\alpha^2}{b^2 + 1} = \frac{\alpha}{-1 - b} = \frac{1}{1 - b}$$

$$\Rightarrow \alpha^2 = \frac{b^2 + 1}{1 - b} \text{ and } \alpha = -\left(\frac{1 + b}{1 - b}\right)$$

$$\Rightarrow \frac{b^2 + 1}{1 - b} = \left(\frac{1 + b}{1 - b}\right)^2$$

$$\Rightarrow b^2 + 1 = \frac{(1 + b)^2}{1 - b}$$

$$\Rightarrow b^2 - b^3 + 1 - b = 1 + 2b + b^2$$

$$\Rightarrow b^3 + 3b = 0$$

$$\Rightarrow b = 0, b = \pm\sqrt{3}i$$

$$\Rightarrow b = -\sqrt{3}i$$

117. Let z be a complex number such that the imaginary part of z is nonzero and $a = z^2 + z + 1$ is real. Then a **cannot** take the value

[IIT-JEE 2012]

(1) -1

(2) $\frac{1}{3}$

(3) $\frac{1}{2}$

(4) $\frac{3}{4}$

Sol. Answer (4)

As, a is real,

So $a = \bar{a}$ gives

$$\Rightarrow z^2 + z + 1 = \bar{z}^2 + \bar{z} + 1$$

$$\Rightarrow (z - \bar{z})(z + \bar{z} + 1) = 0$$

As, $z \neq \bar{z}$

$$\text{So, } z + \bar{z} = -1$$

$$\Rightarrow x = -\frac{1}{2} \text{ \{where } z = x + iy\}}$$

Now,

$$a = z^2 + z + 1$$

$$= \left(-\frac{1}{2} + iy\right)^2 + \left(-\frac{1}{2} + iy\right) + 1$$

$$= \frac{3}{4} - y^2$$

As, $y \neq 0$ so, $a \neq \frac{3}{4}$

118. The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has [JEE(Advanced)-2014]

- (1) Only purely imaginary roots (2) All real roots
(3) Two real and two purely imaginary roots (4) Neither real nor purely imaginary roots

Sol. Answer (4)

Let $p(x) = x^2 + a$ ($a > 0$) ($\because$ roots are purely imaginary)

$$p(p(x)) = (x^2 + a)^2 + a \quad (a \in \mathbb{R})$$

$$x^4 + 2a(x^2) + a^2 + a = 0$$

$$\Rightarrow x^2 = \frac{-2a \pm \sqrt{4a^2 - 4a^2 - 4a}}{2}$$

$$= -a \pm \sqrt{a}i$$

$$x = \pm \sqrt{-a \pm \sqrt{a}i} = x_1 + iy_1$$

SECTION - B

Objective Type Questions (More than one options are correct)

1. If $|3z - 1| = 3|z - 2|$, then z lies on
- (1) $6\operatorname{Re}(z) = 7$
- (2) On the perpendicular bisector of line joining $\left(\frac{1}{3}, 0\right)$ and $(2, 0)$
- (3) A line parallel to x -axis
- (4) A line parallel to y -axis

Sol. Answer (1, 2, 4)

$$\text{If } |3z - 1| = 3|z - 2|$$

$$\text{Let } z = x + iy$$

$$|3x + 3iy - 1| = 3|x + iy - 2|$$

$$\Rightarrow (3x - 1)^2 + (3y)^2 = 9[(x - 2)^2 + y^2]$$

$$\Rightarrow 9x^2 + 1 - 6x + 9y^2 = 9x^2 + 9y^2 - 36x + 36$$

$$\Rightarrow 30x = 35$$

$$\Rightarrow x = \frac{7}{6}$$

$$\Rightarrow 6 \operatorname{Re}(z) = 7 \text{ (A line parallel to } y\text{-axis)}$$

$$\text{Also mid-point of } \left(\frac{1}{3}, 0\right) \text{ and } (2, 0) \text{ is } \left(\frac{7}{6}, 0\right)$$

2. If $S = \sum_{k=1}^{10} \left(\sin \frac{2\pi k}{11} - i \cos \frac{2\pi k}{11} \right)$, then

$$(1) S + \bar{S} = 0$$

$$(2) S\bar{S} = 1$$

$$(3) \sqrt{S} = \pm \frac{1}{\sqrt{2}}(1+i) \quad (4) S - \bar{S} = 0$$

Sol. Answer (1, 2, 3)

We can write $\sin \frac{2\pi k}{11} - i \cos \frac{2\pi k}{11}$

$$= -i \left[\cos \frac{2\pi k}{11} + i \sin \frac{2\pi k}{11} \right]$$

Now

$$S = \sum_{k=1}^{10} \left(\sin \frac{2\pi k}{11} - i \cos \frac{2\pi k}{11} \right)$$

$$S = -i \sum_{k=1}^{10} e^{i \frac{2\pi k}{11}}$$

$$= -i \left[\left(e^{\frac{i2\pi}{11}} + e^{\frac{i4\pi}{11}} + \dots + e^{\frac{i20\pi}{11}} + 1 \right) - 1 \right] = i \text{ (as } 1 + \alpha + \alpha^2 + \dots + \alpha^{10} = 0)$$

$$\Rightarrow S = i$$

$$\Rightarrow \bar{S} = -i$$

$$\Rightarrow S + \bar{S} = 0$$

$$\text{and } S\bar{S} = 1$$

$$\Rightarrow S = \frac{1}{2}(1 + i^2 + 2i)$$

$$\Rightarrow \sqrt{S} = \pm \frac{1}{\sqrt{2}}(1+i)$$

3. Let $\cos A + \cos B + \cos C = 0$ and $\sin A + \sin B + \sin C = 0$ then which of the following statement(s) is/are correct ?

- (1) $\Sigma \cos(2A - B - C) = 3$ (2) $\Sigma \cos(2A - B - C) = 0$
 (3) $\Sigma \sin(2A - B - C) = 0$ (4) $\Sigma \sin(2A - B - C) = 3$

Sol. Answer (1, 3)

$$\cos A + \cos B + \cos C = 0$$

$$\sin A + \sin B + \sin C = 0$$

$$\text{Let } a = \cos A + i \sin A = e^{iA}$$

$$b = \cos B + i \sin B = e^{iB}$$

$$c = \cos C + i \sin C = e^{iC}$$

$$a + b + c = 0$$

$$\therefore a^3 + b^3 + c^3 = 3abc$$

$$\Rightarrow \frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} = 3$$

$$\Rightarrow \frac{e^{2iA}}{e^{iB} \cdot e^{iC}} + \frac{e^{2iB}}{e^{iA} e^{iC}} + \frac{e^{2iC}}{e^{iA} e^{iB}} = 3$$

$$e^{2iA-iB-iC} + e^{2iB-iA-iC} + e^{2iC-iA-iB} = 3$$

$$\cos(2A - B - C) + i \sin(2A - B - C) + \cos(2B - A - C) + i \sin(2B - A - C) + \cos(2C - A - B) + i \sin(2C - A - B) = 3$$

On comparing real and imaginary part

$$\cos(2A - B - C) + \cos(2B - A - C) + \cos(2C - A - B) = 3$$

$$\sin(2A - B - C) + \sin(2B - A - C) + \sin(2C - A - B) = 0$$

4. The equation whose roots are n^{th} power of the roots of the equation, $x^2 - 2x \cos \phi + 1 = 0$ is given by

- (1) $(x + \cos n\phi)^2 + \sin^2 n\phi = 0$ (2) $(x - \cos n\phi)^2 + \sin^2 n\phi = 0$
 (3) $x^2 + 2x \cos n\phi + 1 = 0$ (4) $x^2 - 2x \cos n\phi + 1 = 0$

Sol. Answer (2, 4)

$$x^2 - 2x \cos \phi + 1 = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

$$x = \frac{2 \cos \phi \pm \sqrt{4 \cos^2 \phi - 4}}{2}$$

$$= \cos \phi \pm i \sin \phi$$

$$\alpha = \cos\phi + i\sin\phi, \beta = \cos\phi - i\sin\phi$$

$$\Rightarrow \alpha^n = \cos n\phi + i\sin n\phi; \beta^n = \cos n\phi - i\sin n\phi$$

$$S = \alpha^n + \beta^n = 2\cos n\phi; P = 1$$

$$\text{Equation is } x^2 - Sx + P = 0$$

$$\text{i.e., } x^2 - 2x\cos n\phi + 1 = 0$$

$$\Rightarrow \text{Option (4)}$$

$$x^2 - 2x\cos n\phi + \cos^2 n\phi + 1 - \cos^2 n\phi = 0$$

$$\Rightarrow (x - \cos n\phi)^2 + \sin^2 n\phi = 0$$

$$\Rightarrow \text{Option (2)}$$

5. If a, b, c are real numbers and z is a complex number such that, $a^2 + b^2 + c^2 = 1$ and $b + ic = (1 + a)z$, then

$$\frac{1+iz}{1-iz} \text{ equals.}$$

$$(1) \frac{b-ic}{1-ia}$$

$$(2) \frac{a+ib}{1+c}$$

$$(3) \frac{1-c}{a-ib}$$

$$(4) \frac{1+a}{b+ic}$$

Sol. Answer (2, 3)

$$a^2 + b^2 + c^2 = 1$$

$$b + ic = (1 + a)z$$

$$\Rightarrow z = \frac{b+ic}{1+a} \Rightarrow \frac{1+iz}{1-iz} = \frac{1+i\left(\frac{b+ic}{1+a}\right)}{1-i\left(\frac{b+ic}{1+a}\right)}$$

$$\Rightarrow \frac{1+iz}{1-iz} = \frac{1+a-c+ib}{1+a+c-ib}$$

$$= \frac{(1+a-c+ib)}{(1+a+c-ib)} \times \frac{(1+a+c+ib)}{(1+a+c+ib)}$$

$$= \frac{(1+a+ib)^2 - c^2}{(1+a+c)^2 + b^2} = \frac{a+ib}{1+c}$$

Similarly,

$$\frac{1+a-c+ib}{1+a+c-ib} \times \frac{1+a-c-ib}{1+a+c+ib} = \frac{1-c}{a+ib}$$

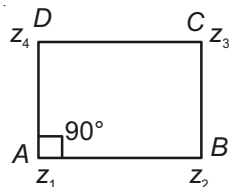
6. If z_1, z_2, z_3, z_4 are the four complex numbers represented by the vertices of a quadrilateral taken in order such that $z_1 - z_4 = z_2 - z_3$ and $\arg\left(\frac{z_4 - z_1}{z_2 - z_1}\right) = \frac{\pi}{2}$, then the quadrilateral is a

- (1) Rhombus (2) Square (3) Rectangle (4) A cyclic quadrilateral

Sol. Answer (3, 4)

$$\arg\left(\frac{z_4 - z_1}{z_2 - z_1}\right) = \frac{\pi}{2}$$

$$\Rightarrow \angle A = 90^\circ$$



$$\text{also } z_1 - z_4 = z_2 - z_3 \Rightarrow |z_1 - z_4| = |z_2 - z_3|$$

$$\Rightarrow AD = BC \text{ and } AD \parallel BC$$

$$\text{So } AB \parallel CD \text{ and } AB = CD$$

$$\Rightarrow ABCD \text{ is a rectangle or cyclic quadrilateral}$$

7. If α, β, γ are cube roots of $p < 0$, then for any x, y, z , $\frac{\alpha^2 x^2 + \beta^2 y^2 + \gamma^2 z^2}{\beta^2 x^2 + \gamma^2 y^2 + \alpha^2 z^2}$ is

(1) 1

(2) $\frac{\alpha}{\gamma}$

(3) $\frac{\beta}{\alpha}$

(4) $\frac{\gamma}{\beta}$

Sol. Answer (2, 3, 4)

$$\text{Cube roots of } p \text{ are } \frac{1}{p^{\frac{1}{3}}}, \frac{1}{p^{\frac{1}{3}}}\omega, \frac{1}{p^{\frac{1}{3}}}\omega^2$$

$$\text{i.e., } \alpha = \frac{1}{p^{\frac{1}{3}}}, \beta = \frac{1}{p^{\frac{1}{3}}}\omega, \gamma = \frac{1}{p^{\frac{1}{3}}}\omega^2$$

$$\text{Now, } \frac{\alpha^2 x^2 + \beta^2 y^2 + \gamma^2 z^2}{\beta^2 x^2 + \gamma^2 y^2 + \alpha^2 z^2} = \frac{x^2 + \omega^2 y^2 + \omega^4 z^2}{\omega^2 x^2 + \omega^4 y^2 + z^2} = \frac{x^2 + \omega^2 y^2 + \omega z^2}{\omega^2 (x^2 + \omega^2 y^2 + \omega z^2)}$$

$$= \omega = \frac{\gamma}{\beta} \Rightarrow \text{Option (4)}$$

We can assign the α, β, γ different value we get other options also.

8. If z is a complex number satisfying $z + z^{-1} = 1$, then $z^n + z^{-n}$, $n \in \mathbb{N}$, has the value
- (1) $2(-1)^n$, when n is a multiple of 3 (2) $(-1)^{n-1}$, when n is not a multiple of 3
- (3) $(-1)^{n+1}$, when n is a multiple of 3 (4) 0 when n is not a multiple of 3

Sol. Answer (1, 2)

$$z + z^{-1} = 1$$

$$z + \frac{1}{z} = 1$$

$$z^2 - z + 1 = 0$$

$$z = \frac{1 \pm i\sqrt{3}}{2} = -\omega, -\omega^2$$

$$z^n + z^{-n} = (-\omega)^n + (-\omega)^{-n}$$

Case (1), $n = 3m$

$$(-\omega)^{3m} + (-\omega)^{-3m} = (-1)^n + (-1)^n = 2(-1)^n$$

when $n = 3m + 1$

$$(-\omega)^{3m+1} + (-\omega)^{-3m-1}$$

$$= (-1)^n \omega^{3m+1} + \frac{(-1)^n}{(\omega)^{3m+1}}$$

$$= (-1)^n \left(\omega + \frac{1}{\omega} \right)$$

$$= (-1)^n (-1) = (-1)^{n+1}$$

9. If z satisfies $|z - 1| < |z + 3|$, then $\omega = 2z + 3 - i$ satisfies

$$(1) |\omega - 5 - i| < |\omega + 3 + i|$$

$$(2) |\omega - 5| < |\omega + 3|$$

$$(3) \operatorname{Im}(i\omega) > 1$$

$$(4) |\arg(\omega - 1)| < \frac{\pi}{2}$$

Sol. Answer (2, 3, 4)

$$|z - 1| < |z + 3|$$

$$\text{Let } z = x + iy$$

$$(x - 1)^2 + y^2 < (x + 3)^2 + y^2$$

$$x^2 + y^2 - 2x + 1 < (x^2 + y^2 + 6x + 9)$$

$$\Rightarrow 8x > -8$$

$$x > -1$$

$$i\omega = i((2x + i2y) + 3 - i)$$

$$= i2x - 2y + 3i + 1$$

$$= i(3 + 2x) + (1 - 2y)$$

As $3 + 2x > 1 \Rightarrow$ option

$$\text{Also, } \omega - 1 = 2z + 3 - i - 1$$

$$= 2z + 2 - i = 2x + 2iy + 2 - i$$

$$= 2(x + 1) + i(2y - 1)$$

As $x > -1$

$$\therefore 2(x+1) > 0 \therefore \arg(\omega-1) < \frac{\pi}{2}$$

$\Rightarrow$ option (4)

$$\omega - 5 = 2(x-1) + i(2y-1)$$

$$\omega + 3 = 2(x+3) + i(2y-1) \text{ as } x > -1$$

$$\Rightarrow |\omega + 3| > |\omega - 5|$$

$\Rightarrow$ Option (2)

10. If $|z + \omega|^2 = |z|^2 + |\omega|^2$, where z and ω are complex numbers, then

$$(1) \frac{z}{\omega} \text{ is purely real} \quad (2) \frac{z}{\omega} \text{ is purely imaginary} \quad (3) z\bar{\omega} + \bar{z}\omega = 0 \quad (4) \operatorname{amp}\left(\frac{z}{\omega}\right) = \frac{\pi}{2}$$

Sol. Answer (2, 3, 4)

$$|z + \omega|^2 = |z|^2 + |\omega|^2$$

$$\text{Since } |z + \omega|^2 = |z|^2 + |\omega|^2 + z\bar{\omega} + \omega\bar{z}$$

$$\Rightarrow |z|^2 + |\omega|^2 = |z|^2 + |\omega|^2 + z\bar{\omega} + \omega\bar{z} \Rightarrow z\bar{\omega} + \omega\bar{z} = 0$$

$$\Rightarrow z\bar{\omega} = -\omega\bar{z} \Rightarrow \frac{z}{\omega} = -\frac{\bar{z}}{\bar{\omega}}$$

$$\Rightarrow \frac{z}{\omega} \text{ is purely imaginary}$$

$$\text{Therefore, } \operatorname{amp}\left(\frac{z}{\omega}\right) = \frac{\pi}{2}$$

11. Let z_1, z_2 be two complex numbers represented by points on the circle $|z_1| = 1$ and $|z_2| = 2$ respectively then

$$(1) \min |z_1 - z_2| = 1 \quad (2) \max |2z_1 + z_2| = 4 \quad (3) \left| z_2 + \frac{1}{z_1} \right| \leq 3 \quad (4) \min |z_1 - z_2| = 2$$

Sol. Answer (1, 2, 3)

$$\text{Clearly } |z_1 - z_2|_{\min} = 2 - 1 = 1$$

$$|z_1 - z_2|_{\max} = 3$$

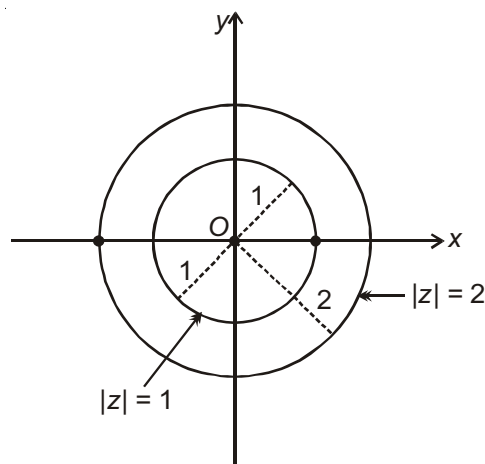
$$\max |2z_1 + z_2| = |2 + 2| = 4$$

$$\text{Now, } |z| = 1$$

$$|z_1| = 1$$

$$z_1 \bar{z}_1 = 1$$

$$\bar{z}_1 = \frac{1}{z_1}$$



Now, $\left| z_2 + \frac{1}{z_1} \right| = |z_2 + \bar{z}_1|$

and $|z_2 + \bar{z}_1| \leq |z_2| + |\bar{z}_1|$
 ≤ 3 .

12. Let complex number z satisfy $\left| z - \frac{2}{z} \right| = 1$, then $|z|$ can take all values except

(1) 1

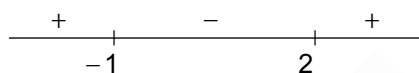
(2) 2

(3) 3

(4) 4

Sol. Answer (3, 4)

$$\left| z - \frac{2}{z} \right| = 1$$



$$\left| z \right| - \frac{2}{|z|} \leq 1$$

$$|z|^2 - |z| - 2 \leq 0$$

$$(|z| - 2)(|z| + 1) \leq 0$$

$$\Rightarrow -1 \leq |z| \leq 2$$

$$\text{But } |z| \geq 0$$

$$\therefore 0 \leq |z| \leq 2$$

$$\Rightarrow |z| \neq 3 \text{ and } |z| \neq 4$$

$$\Rightarrow \text{Option (3) and (4)}$$

13. If $z = x + iy$, then the equation $\left| \frac{2z-i}{z+1} \right| = m$ represent a circle when

(1) $m = \frac{1}{2}$

(2) $m = 1$

(3) $m = 2$

(4) $m = 3$

Sol. Answer (1, 2, 4)

The given equation is $|2z - i| = m|z + i|$

$$\Rightarrow (2z - i)(2\bar{z} + i) = m^2(z + i)(\bar{z} - i)$$

$$\Rightarrow 4z\bar{z} + 2iz - 2i\bar{z} + 1 = m^2(z\bar{z} - iz + i\bar{z} + 1)$$

$$\Rightarrow (4 - m^2)z\bar{z} + (2 + m^2)iz - (2 + m^2)i\bar{z} + (1 - m^2) = 0$$

The above equation will does not represent a circle, when,

$$4m^2 = 0 \Rightarrow m = 2 ; \text{ since } m \text{ cannot be negative}$$

Hence answer is (1, 2, 4)

14. If $\sqrt[3]{-1} = -1, -\omega, -\omega^2$, then roots of the equation $(x + 1)^3 + 64 = 0$ are

- (1) $-1 - 4\omega$ (2) $-1 - 4\omega^2$ (3) -5 (4) -4

Sol. Answer (1, 2, 3)

$$(x + 1)^3 = (-4)^3$$

$$\Rightarrow x + 1 = -4, -4\omega, -4\omega^2$$

$$\Rightarrow x = -5, -1 - 4\omega, -1 - 4\omega^2.$$

Hence, roots are $-5, -1 - 4\omega, -1 - 4\omega^2$.

15. If $z_1 = a + ib$ and $z_2 = c + id$ are complex numbers such that $|z_1| = |z_2| = 1$ and $\operatorname{Re}(z_1 \bar{z}_2) = 0$, then the pair of complex numbers $\omega_1 = a + ic$ and $\omega_2 = b + id$ satisfy

- (1) $|\omega_1| = 1$ (2) $|\omega_2| = 1$ (3) $\operatorname{Re}(\omega_1 \bar{\omega}_2) = 0$ (4) $|\omega_1| = 2|\omega_2|$

Sol. Answer (1, 2, 3)

$$|z_1| = \sqrt{a^2 + b^2} = 1 \Rightarrow a^2 + b^2 = 1 \quad \dots(i)$$

$$|z_2| = \sqrt{c^2 + d^2} = 1 \Rightarrow c^2 + d^2 = 1 \quad \dots(ii)$$

$$\operatorname{Re}(z_1 \bar{z}_2) = \operatorname{Re}[(a + ib)(c - id)]$$

$$= \operatorname{Re}[ac + bd + i(bc - ad)] = 0$$

$$\Rightarrow ac + bd = 0 \quad \dots(iii)$$

Now, using (i) & (iii) we can prove that $b = c, a = d$.

$$\text{Hence, } |\omega_1| = \sqrt{a^2 + c^2} = \sqrt{a^2 + b^2} = 1$$

Similarly we can observe, $|\omega_2| = 1$

$$\operatorname{Re}(\omega_1 \bar{\omega}_2) = 0$$

16. If a complex number z satisfies $\log_{\frac{1}{\sqrt{2}}} \left(\frac{|z|^2 + 2|z| + 6}{2|z|^2 - 2|z| + 1} \right) < 0$, then locus/region of the point represented by z is

- (1) $|z| = 5$ (2) $|z| < 5$ (3) $|z| > 1$ (4) $2 < |z| < 3$

Sol. Answer (2, 4)

Given inequality is,

$$\log_{\frac{1}{\sqrt{2}}} \frac{|z|^2 + 2|z| + 6}{2|z|^2 - 2|z| + 1} < 0$$

$$\Rightarrow \frac{|z|^2 + 2|z| + 6}{2|z|^2 - 2|z| + 1} > 1$$

$$\Rightarrow |z|^2 + 2|z| + 6 > 2|z|^2 - 2|z| + 1$$

$$\Rightarrow |z|^2 - 4|z| - 5 < 0$$

$$\Rightarrow |z| \in (-1, 5), \text{ but } |z| > 0$$

$$\therefore 0 < |z| < 5$$

17. If z_1, z_2 be two complex numbers satisfying the equation $\left| \frac{z_1 + z_2}{z_1 - z_2} \right| = 1$, then

(1) $z_1 \bar{z}_2 + z_2 \bar{z}_1 = 1$ (2) $\left(\frac{z_1}{z_2} \right) = -\frac{z_1}{z_2}$ (3) $z_1 \bar{z}_2 + z_2 \bar{z}_1 = 0$ (4) $\operatorname{Re}(z_1 \bar{z}_2) = 0$

Sol. Answer (2, 3, 4)

z_1, z_2 are the complex numbers satisfying,

$$\left| \frac{z_1 + z_2}{z_1 - z_2} \right| = 1$$

$$\Rightarrow |z_1 + z_2| = |z_1 - z_2|$$

$$\Rightarrow (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$$

$$\Rightarrow z_1 \bar{z}_1 + z_1 \bar{z}_2 + z_2 \bar{z}_1 + z_2 \bar{z}_2 = z_1 \bar{z}_1 - z_1 \bar{z}_2 - z_2 \bar{z}_1 + z_2 \bar{z}_2$$

$$\Rightarrow 2(z_1 \bar{z}_2 + z_2 \bar{z}_1) = 0$$

$$\Rightarrow \left(\frac{\bar{z}_1}{z_2} \right) = -\frac{z_1}{z_2}$$

$$\Rightarrow z_1 \bar{z}_2 + z_2 \bar{z}_1 = 0$$

$$\Rightarrow z_1 \bar{z}_2 + \overline{z_1 \bar{z}_2} = 0$$

$$\Rightarrow \operatorname{Re}(z \bar{z}_2) = 0$$

18. If $\sin \alpha, \cos \alpha$ are the roots of the equation $ax^2 + bx + c = 0$ ($c \neq 0$), then

(1) $a^2 - b^2 + 2ac = 0$ (2) $(a + c)^2 = b^2 + c^2$ (3) $\frac{b}{a} \in [-\sqrt{2}, \sqrt{2}]$ (4) $\frac{c}{a} \in \left[-\frac{1}{2}, \frac{1}{2} \right]$

Sol. Answer (1, 2, 3, 4)

$$\Rightarrow \sin \alpha + \cos \alpha = -\frac{b}{a}$$

$$\Rightarrow \sin \alpha \cdot \cos \alpha = \frac{c}{a}$$

$$\Rightarrow \sin^2 \alpha + \cos^2 \alpha = 1$$

$$\Rightarrow \left(-\frac{b}{a} \right)^2 - \frac{2c}{a} = 1 \Rightarrow b^2 - 2ca = a^2$$

$$\Rightarrow a^2 - b^2 + 2ac = 0$$

$$\text{Also } (a + c)^2 = b^2 + c^2.$$

19. Let a, b, c be real numbers in G.P. such that a and c are positive, then the roots of the equation $ax^2 + bx + c = 0$
- (1) Are real and are in the ratio $b : ac$
 - (2) Are real
 - (3) Are imaginary and are in ratio $1 : \omega$ where ω is a non-real complex cubic root of constant
 - (4) Are imaginary and are in the ratio $\omega^2 : 1$ with usual notation

Sol. Answer (3, 4)

$$ax^2 + bx + c = 0$$

Let a, b, c is a, ar, ar^2

$$\text{Now, } ax^2 + arx + ar^2 = 0$$

$$\Rightarrow x^2 + rx + r^2 = 0 \Rightarrow x = r \left(\frac{-1 \pm \sqrt{3}i}{2} \right) \Rightarrow x = \omega r \text{ or } \omega^2 r$$

$\therefore$ Roots are imaginary and are in the ratio $1 : \omega$ or $\omega^2 : 1$.

20. Let $\cos \alpha$ be a root of the equation $25x^2 + 5x - 12 = 0$, $-1 < x < 0$, then the value of $\sin 2\alpha$ is

(1) $\frac{20}{25}$

(2) $-\frac{12}{25}$

(3) $\frac{24}{25}$

(4) $-\frac{24}{25}$

Sol. Answer (3, 4)

$$25x^2 + 5x - 12 = 0 \Rightarrow (5x - 3)(5x + 4) = 0$$

$$\Rightarrow x = \frac{3}{5}, x = -\frac{4}{5} \quad \left(x \neq \frac{3}{5} \text{ as } -1 < x < 0 \right)$$

$$\therefore \cos \alpha = -\frac{4}{5}$$

$$\Rightarrow \sin \alpha = \frac{3}{5} \text{ or } -\frac{3}{5}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha = 2 \times \frac{3}{5} \left(-\frac{4}{5} \right) = -\frac{24}{25}$$

$$\text{or } 2 \left(-\frac{3}{5} \right) \left(-\frac{4}{5} \right) = \frac{24}{25}$$

21. If the quadratic equations $x^2 + pqx + r = 0$ and $x^2 + prx + q = 0$ have a common root then the equation containing their other roots is/are

(1) $x^2 - p(q + r)x + p^2qr = 0$

(2) $x^2p(q + r) + (q + r)x - pqr = 0$

(3) $p(q + r)x^2 - (q + r)x + pqr = 0$

(4) $x^2 + p(q + r)x - p^2qr = 0$

Sol. Answer (1, 2)

Let α be common root

$$\alpha^2 + pq\alpha + r = 0, \quad \alpha + \beta = -pq, \quad \alpha\beta = r$$

$$\alpha^2 + pr\alpha + q = 0, \quad \alpha + \gamma = -pr, \quad \alpha\gamma = q$$

$$\overline{\alpha p(q-r)+r-q=0} \Rightarrow \alpha = \frac{1}{p}$$

Common root is $\alpha = \frac{1}{p}$

Other roots are, $\beta = rp$ and $\gamma = qp$

$\therefore$ Equation containing other roots is

$$x^2 - p(r+q)x + p^2rq = 0$$

$$\therefore \frac{1}{p} \text{ is common root} \Rightarrow \left(\frac{1}{p}\right)^2 + pq\left(\frac{1}{p}\right) + r = 0$$

$$\Rightarrow \frac{1}{p^2} = -(q+r)$$

Now, $x^2 - p(q+r)x + p^2qr = 0$

$$\Rightarrow -p \left[-\frac{p}{p^2}x^2 + (q+r)x - pqr \right] = 0$$

$$\Rightarrow p(q+r)x^2 + (q+r)x - pqr = 0$$

22. The quadratic equation $x^2 - (m-3)x + m = 0$ has
- (1) Real distinct roots if and only if $m \in (-\infty, 1) \cup (9, \infty)$
 - (2) Both positive roots if and only if $m \in (9, \infty)$
 - (3) Both negative roots if and only if $m \in (0, 1)$
 - (4) No roots

Sol. Answer (1, 2, 3)

$$x^2 - (m-3)x + m = 0$$

For real distinct roots, $(m-3)^2 - 4m > 0$

$$\Rightarrow m^2 - 10m + 9 > 0$$

$$\Rightarrow (m-9)(m-1) > 0 \Rightarrow m \in (-\infty, 1) \cup (9, \infty) \quad \dots(1)$$

For positive roots,

Sum > 0 , product > 0

$$\Rightarrow m-3 > 0, m > 0 \quad \dots(2)$$

From (1) and (2), $m \in (9, \infty)$

For negative roots

sum < 0 , product > 0

$$\Rightarrow m-3 < 0, m > 0 \quad \dots(3)$$

From (1) and (3), $m \in (0, 1)$

23. If both roots of the equation $x^2 - 2ax + a^2 - 1 = 0$ lie between -3 and 4 , then $[a]$ is/are, where $[]$ represents the greatest integer function

- (1) 1 (2) -1 (3) 2 (4) 0

Sol. Answer (1, 2, 3, 4)

$$x^2 - 2ax + a^2 - 1 = 0$$

$$(x - a)^2 = 1 \Rightarrow x = a + 1, a - 1.$$

$$\text{Now, } -3 \leq [a + 1] \leq 4 \text{ and } -3 \leq [a - 1] \leq 4$$

$$\Rightarrow -3 \leq [a] + 1 \leq 4 \text{ and } -3 \leq [a] - 1 \leq 4$$

$$\Rightarrow -4 \leq [a] \leq 3 \text{ and } -2 \leq [a] \leq 5$$

$$\Rightarrow -2 \leq [a] \leq 3 \Rightarrow [a] = -2, -1, 0, 1, 2, 3.$$

24. Let α, β be the roots of $x^2 - 4x + A = 0$ and γ, δ be the roots of $x^2 - 36x + B = 0$. If $\alpha, \beta, \gamma, \delta$ forms an increasing G.P. Then

- (1) $B = 81A$ (2) $A = 3$ (3) $B = 243$ (4) $A + B = 251$

Sol. Answer (1, 2, 3)

$$\alpha + \beta = 4, \alpha\beta = A$$

$$\gamma + \delta = 36, \gamma\delta = B$$

$$\text{Let } \alpha, \beta, \gamma, \delta \text{ be } a, ar, ar^2, ar^3$$

$$a + ar = 4$$

$$ar^2 + ar^3 = 36 \Rightarrow \frac{1+r}{r^2(1+r)} = \frac{1}{9}$$

$$\Rightarrow r^2 = 9 \Rightarrow r = \pm 3, a = 1.$$

$$A = \alpha\beta \Rightarrow a(ar) = A \Rightarrow A = 3$$

$$B = \gamma\delta \Rightarrow B = (ar^2)(ar^3) \Rightarrow B = 243 \Rightarrow B = 81A.$$

25. For the equation $x^{\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}$, which one of the following is true?

- (1) Has at least one real solution (2) Has exactly three real solutions
(3) Has exactly one irrational solutions (4) Has non-real complex roots

Sol. Answer (1, 2, 3)

$$x^{\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4}} = \sqrt{2}$$

Taking log with base 2 on both side.

$$\left[\frac{3}{4}(\log_2 x)^2 + \log_2 x - \frac{5}{4} \right] \log_2 x = \log_2 \sqrt{2} = \frac{1}{2}$$

$$\text{Put } \log_2 x = t, \left(\frac{3}{4}t^2 + t - \frac{5}{4} \right) t = \frac{1}{2}$$

$$\Rightarrow 3t^3 + 4t^2 - 5t - 2 = 0 \Rightarrow (t-1)(3t^2 + 7t + 2) = 0$$

$$\Rightarrow (t-1)(3t+1)(t+2) = 0$$

$$\Rightarrow t = 1, -\frac{1}{3}, -2 \Rightarrow \log_2 x = 1, -\frac{1}{3}, -2 \Rightarrow x = 2, 2^{-1/3}, 2^{-2}$$

26. If $f(x) = ax^2 + bx + c$, $g(x) = -ax^2 + bx + c$, where $ac \neq 0$ then $f(x).g(x) = 0$ has

(1) At least three real roots

(2) No real roots

(3) At least two real roots

(4) At most two imaginary roots

Sol. Answer (3, 4)

Let D_1 and D_2 be the respective discriminates.

$$\text{then } D_1 = b^2 - 4ac \quad \text{and } D_2 = b^2 + 4ac$$

$$\text{Adding we get } D_1 + D_2 = 2b^2$$

$\therefore D_1 + D_2$ is positive

$\Rightarrow$ at least one of D_1 or D_2 is positive.

$\Rightarrow$ at least 2 real roots.

27. The value of a for which the equation $x^2 + ax + a^2 + 6a < 0$ is satisfied for all $x \in (1, 2)$

$$(1) \left(\frac{-7-4\sqrt{5}}{2}, -2 \right)$$

$$(2) \left(\frac{-7-4\sqrt{5}}{2}, -3 \right)$$

$$(3) \left(\frac{-7-4\sqrt{5}}{2}, -4+2\sqrt{3} \right)$$

$$(4) \left(\frac{-7+4\sqrt{5}}{2}, 4+3\sqrt{3} \right)$$

Sol. Answer (1, 2, 3)

Let α, β be the roots of the corresponding equation

$$x^2 + ax + a^2 + 6a = 0$$



As the coefficient of $x^2 = 1 > 0$ $x^2 + ax + a^2 + 6a < 0$ will be satisfied for all values of $x \in (\alpha, \beta)$ if α, β are real and unequal (let $\alpha < \beta$).

Hence the inequality will hold for all real $x \in (1, 2)$ if the interval $(1, 2)$ is a subset of the interval (α, β) . Thus for (1) we should have $D > 0$ and $\alpha < 1, \beta > 1$ as well as $\alpha < 2, \beta > 2$.

$$\text{Now, } D > 0 \Rightarrow a^2 - 4(a^2 + 6a) > 0$$

$$\Rightarrow a^2 + 8a < 0$$

$$\Rightarrow a \in (-8, 0)$$

... (ii)

$$\alpha < 1, \beta > 1 \Rightarrow \alpha - 1 < 0, \beta - 1 > 0$$

$$(\alpha - 1)(\beta - 1) < 0$$

$$\Rightarrow a^2 + 7a + 1 < 0$$

$$a \in \left(\frac{-7 - \sqrt{45}}{2}, \frac{-7 + \sqrt{45}}{2} \right) \quad \dots \text{(iii)}$$

$$\alpha < 2, \beta > 2 \Rightarrow (\alpha - 2)(\beta - 2) < 0$$

$$\Rightarrow a^2 + 8a + 4 < 0$$

$$\Rightarrow a \in (-4 - 2\sqrt{3}, -4 + 2\sqrt{3}) \quad \dots \text{(iv)}$$

Common values of a satisfying (ii), (iii) and (iv) are

$$a \in \left(\frac{-7 - \sqrt{45}}{2}, -4 + 2\sqrt{3} \right) \quad \dots \text{(v)}$$

Hence answer is (1), (2), (3) those are subject of (v)

28. If the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but opposite in sign and its product is α

$$(1) \quad p + q = r$$

$$(2) \quad p + q = 2r$$

$$(3) \quad \alpha^2 = \frac{p^2 + q^2}{2}$$

$$(4) \quad \alpha = -\left(\frac{p^2 + q^2}{2} \right)$$

Sol. Answer (2, 3)

The given equation is

$$\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$$

$$\Rightarrow \frac{x+q+x+p}{(x+p)(x+q)} = \frac{1}{r}$$

$$\Rightarrow (2x + p + q)r = x^2 + (p + q)x + pq$$

$$\Rightarrow x^2 + (p + q - 2r)x + (pq - qr - rp) = 0$$

According to the question the given equation has roots equal in magnitude but opposite in sign, hence

Coefficient of $x = 0$

$$\Rightarrow p + q - 2r = 0$$

$$\Rightarrow r = \frac{p+q}{2}$$

Product of roots

$$= + [(p + q)r - pq]$$

$$= \frac{(p+q)^2}{2} - pq$$

$$= \frac{1}{2}(p^2 + q^2)$$

29. The integral values of a for which the equation $(a + 2)x^2 + 2(a + 1)x + a = 0$ will have both roots integers

(1) 0

(2) -1

(3) -2

(4) -3

Sol. Answer (1, 2, 4)

$$(a + 2)x^2 + 2(a + 1)x + a = 0$$

Let α, β be roots

$$\alpha + \beta = \frac{-2(a+1)}{a+2} \text{ (integer)}$$

$$\alpha\beta = \frac{a}{a+2} \text{ (integer)}$$

$$\frac{a}{a+2} \text{ will integer if}$$

For $a = 0, a = -1, a = -3$

Also for $a = 0, a = -1, a = -3$

$$\alpha + \beta = \frac{-2(a+1)}{a+2} \text{ is integer}$$

30. If $(x - 1)^2$ is a factor of $ax^3 + bx^2 + c$, then roots of the equation $cx^3 + bx + a = 0$ may be

(1) 1

(2) -1

(3) -2

(4) 0

Sol. Answer (1, 3)

Since 1 is the repeated roots of $ax^3 + bx^2 + c = 0$

$$\text{So, } 1 + 1 + \alpha = -\frac{b}{a}$$

$$1.1 + \alpha + \alpha = 0 \Rightarrow \alpha = -\frac{1}{2}$$

$$1.1.\alpha = -\frac{c}{a} = -\frac{1}{2} \Rightarrow \frac{c}{a} = \frac{1}{2}$$

$$\frac{b}{a} = -\frac{3}{2}$$

$$\therefore \frac{b}{c} = -3$$

Now, by the equation,

$$cx^3 + bx + a = 0$$

$$\Rightarrow x^3 + \frac{b}{c}x + \frac{a}{c} = 0$$

$$\Rightarrow x^3 - 3x + 2 = 0$$

$$x^3 - x^2 + x^2 - x - 2x + 2 = 0$$

$$\Rightarrow x^2(x-1) + x(x-1) - 2(x-1) = 0$$

$$\Rightarrow (x-1)(x^2 + x - 2) = 0$$

$$\Rightarrow (x-1)(x^2 + 2x - x - 2) = 0$$

$$\Rightarrow (x-1)(x-1)(x+2) = 0$$

$$\Rightarrow x = +1, -2$$

Hence answer is (1), (3)

31. If $b^2 \geq 4ac$ for the equation $ax^4 + bx^2 + c = 0$ then all roots of the equation will be real if

(1) $b > 0, a < 0, c > 0$

(2) $b > 0, a > 0, c > 0$

(3) $b < 0, a > 0, c > 0$

(4) $b > 0, a < 0, c < 0$

Sol. Answer (3, 4)

Let $x^2 = y$

So the equation $ay^2 + by + c = 0$ should have both roots non-negative in order to all roots of the equation $ax^4 + bx^2 + c = 0$ are real for this

$$\alpha + \beta = -\frac{b}{a} > 0 \Rightarrow \frac{b}{a} < 0 \quad \dots(i)$$

$$\alpha\beta = \frac{c}{a} > 0 \quad \dots(ii)$$

From (i) and (ii)

$$b > 0, a < 0, c < 0$$

or $b < 0, a > 0, c > 0$

32. The difference between the roots of the equation $x^2 + kx + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of k is

(1) $(-3, 0)$

(2) $(0, 3)$

(3) $(-3, 3)$

(4) $(3, \infty)$

Sol. Answer (1, 2, 3)

$$|\alpha - \beta| < \sqrt{5}$$

$$\Rightarrow (\alpha - \beta)^2 < 5$$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta < 5$$

$$\Rightarrow k^2 - 4 < 5$$

$$\Rightarrow k^2 < 9 \Rightarrow k \in (-3, 3)$$

Hence answers is (1, 2, 3)

33. The set of real values of a for which $a^2 + 2a$, $2a + 3$ and $a^2 + 3a + 8$ are the sides of a triangle may be

(1) $\left(6, \frac{13}{2}\right)$

(2) $(5, 7)$

(3) $(5, \infty)$

(4) $(0, 5)$

Sol. Answer (1, 2, 3)

We know that in a triangle sum of two sides of a triangle is greater than third side.

$$\text{So, } a^2 + 2a + 2a + 3 > a^2 + 3a + 8 \Rightarrow 4a > 3a + 5 \Rightarrow a > 5$$

$$a^2 + 2a + a^2 + 3a + 8 > 2a + 3 \Rightarrow 2a^2 + 3a + 5 > 0 \Rightarrow a \in R$$

$$2a + 3 + a^2 + 3a + 8 > a^2 + 2a \Rightarrow 3a > -11 \Rightarrow a > -\frac{11}{3}$$

Combining these three,

$$a \in (5, \infty)$$

Hence answer is (1, 2, 3)

34. Let z_1 and z_2 be two distinct complex numbers and let $z = (1 - t)z_1 + tz_2$ for some real number t with $0 < t < 1$. If $\arg(w)$ denotes the principal argument of a non-zero complex number w , then [IIT-JEE 2010]

$$(1) |z - z_1| + |z - z_2| = |z_1 - z_2|$$

$$(2) \arg(z - z_1) = \arg(z - z_2)$$

$$(3) \begin{vmatrix} z - z_1 & \bar{z} - \bar{z}_1 \\ z_2 - z_1 & \bar{z}_2 - \bar{z}_1 \end{vmatrix} = 0$$

$$(4) \arg(z - z_1) = \arg(z_2 - z_1)$$

Sol. Answer (1, 3, 4)

$$z_1 \xrightarrow[t=1-t]{z} z_2$$

we have $z = (1 - t)z_1 + tz_2$, $0 < t < 1$,

$$\Rightarrow z = \frac{tz_2 + (1-t)z_1}{t+1-t}$$

$$\Rightarrow z = tz_2 + (1-t)z_1$$

z_1, z, z_2 are collinear

$$\text{so } \arg(z - z_1) = \arg(z - z_2) = \arg(z_2 - z_1)$$

35. Let $w = \frac{\sqrt{3} + i}{2}$ and $P = \{w^n : n = 1, 2, 3, \dots\}$. Further $H_1 = \left\{z \in \mathbb{C} : \operatorname{Re} z > \frac{1}{2}\right\}$ and

$$H_2 = \left\{z \in \mathbb{C} : \operatorname{Re} z < \frac{-1}{2}\right\},$$

where $\mathbb{C}$ is the set of all complex numbers. If $z_1 \in P \cap H_1$, $z_2 \in P \cap H_2$ and O represents the origin, then $\angle z_1 O z_2 =$ [JEE(Advanced) 2013]

$$(1) \frac{\pi}{2}$$

$$(2) \frac{\pi}{6}$$

$$(3) \frac{2\pi}{3}$$

$$(4) \frac{5\pi}{6}$$

Sol. Answer (3, 4)

Note that $|w| = 1$

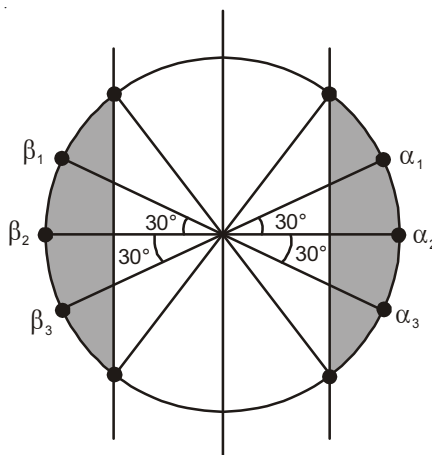
α_i are possible value of z_1

β_i are possible value of z_2

($i = 1, 2, 3$)

$$w = \frac{\sqrt{3}}{2} + \frac{i}{2}$$

$$w = e^{i\frac{\pi}{6}}$$



$$\omega^2 = e^{i\frac{\pi}{3}}$$

$$\omega^3 = e^{i\frac{\pi}{2}}$$

$$\omega^4 = e^{2i\frac{\pi}{3}}$$

$$\omega^5 = e^{i\frac{5\pi}{6}}$$

$$\text{So, } \angle z_1 o z_2 \text{ can be } \Rightarrow \frac{2\pi}{3}, \frac{5\pi}{6}$$

36. Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ? [JEE(Advanced) 2015]

(1) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$

(2) $\left(-\frac{1}{\sqrt{5}}, 0\right)$

(3) $\left(0, \frac{1}{\sqrt{5}}\right)$

(4) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

Sol. Answer (1, 4)

$$\alpha x^2 - x + \alpha = 0$$

$$|x_1 - x_2| < 1$$

$$\left| \frac{\sqrt{1-4\alpha^2}}{\alpha} \right| < 1$$

$$\sqrt{1-4\alpha^2} < |\alpha|$$

$$1 - 4\alpha^2 < \alpha^2$$

$$\alpha^2 - \frac{1}{5} > 0$$

$$\alpha \in \left(-\infty, -\frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \infty\right)$$

$$\text{Also, } 1 - 4\alpha^2 > 0$$

$$\alpha^2 - \frac{1}{4} < 0$$

$$\left(\alpha - \frac{1}{2}\right)\left(\alpha + \frac{1}{2}\right) < 0$$

$$\alpha \in \left(-\frac{1}{2}, \frac{1}{2}\right)$$

Taking intersection

$$\alpha \in \left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$$

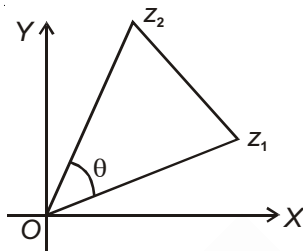
SECTION - C

Linked Comprehension Type Questions

Comprehension I

Let z_1, α, β be complex numbers of which α and β are constants and z_1 varies. If z_2 is given in terms of z_1 by one of the following equations, it is required to find z_2 corresponding to z_1 , then

1. The construction indicated in the diagram is given by



(1) $z_2 = e^{i\theta} \cdot z_1$

(2) $z_2 = (\cos\theta - i \sin\theta)z_1$

(3) $z_2 = \frac{\cos\theta + i \sin\theta}{z_1}$

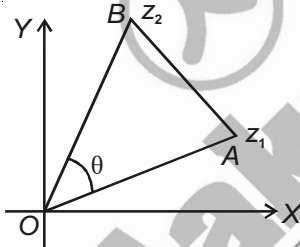
(4) $z_2 = \frac{|z_2|}{|z_1|} (\cos\theta + i \sin\theta)z_1$

Sol. Answer (4)

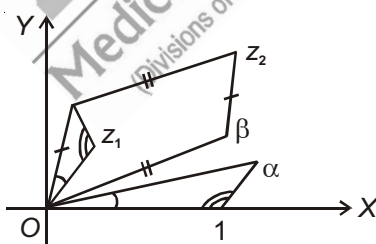
From rotation formula

$$z_2 - 0 = (z_1 - 0) \cdot \frac{OB}{OA} e^{i\theta}$$

$$z_2 = z_1 \cdot \left| \frac{z_2}{z_1} \right| (\cos\theta + i \sin\theta)$$



2. In the given figure



(1) $z_2 = \alpha z_1 - \beta$

(2) $z_2 = \frac{\alpha z_1}{\beta}$

(3) $z_2 = \alpha \beta z_1$

(4) $z_2 = \alpha z_1 + \beta$

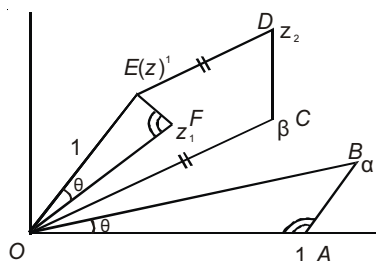
Sol. Answer (4)

In $\triangle OAB$,

$$\alpha = |\alpha| e^{i\theta}$$

In $\triangle OFE$

Using rotation formula



$$z^1 = z_1 e^{i\theta} \cdot \frac{OE}{OF}$$

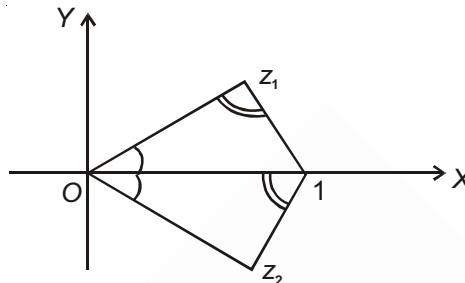
$$\Delta OFE \sim \Delta OAB$$

$$\therefore \frac{OE}{OF} = \frac{OB}{OA}$$

$$z^1 = z_1 e^{i\theta} \cdot |\alpha| = z_1 \alpha$$

$$\therefore z_2 = z_1 \alpha + \beta$$

3. The given figure illustrates



(1) $z_2 = 1 + z_1$

(2) $z_2 = 2z_1$

(3) $z_2 = \frac{1}{z_1}$

(4) $z_2 = \frac{1}{z_1^2}$

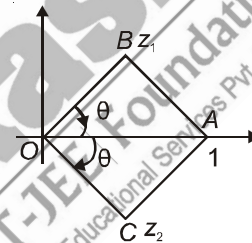
Sol. Answer (3)

From rotation formula

$$z_1 = \frac{OB}{OA} \cdot (1-0) \cdot e^{i\theta} \quad \dots(i)$$

$$z_2 = \frac{OC}{OA} \cdot (1-0) \cdot e^{-i\theta} \quad \dots(ii)$$

$$z_1 z_2 = 1 \quad \Rightarrow \quad z_2 = \frac{1}{z_1}$$



Comprehension II

If x is the root of the equation $x^2 - ix - 1 = 0$, then

1. The value of x^{51} is

(1) 1

(2) -1

(3) i

(4) $-i$

2. The value of $x^{20} + \frac{1}{x^{20}}$ may be

(1) -1

(2) 1

(3) i

(4) $-i$

3. $x^{2013} - \frac{1}{x^{2013}}$ may be equal to

(1) -1

(2) 1

(3) $-2i$

(4) $-i$

Solution of Comprehension II

Given equation is $x^2 - ix - 1 = 0$

$$\Rightarrow x^2 - 1 = ix$$

$$\Rightarrow x - \frac{1}{x} = i \quad \dots(i)$$

Now let $x = \cos\theta + i\sin\theta$, then equation (i) becomes,

$$\cos\theta + i\sin\theta - \cos\theta + i\sin\theta = i$$

$$\Rightarrow 2\sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

1. Answer (3)

We have to find out the value of x^{51}

$$= \left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6} \right)^{51}$$

$$= \cos\frac{17\pi}{2} + i\sin\frac{17\pi}{2}$$

$$= 0 + i = i$$

2. Answer (1)

We have to evaluate $x^{20} + \frac{1}{x^{20}}$

$$= \cos\frac{20\pi}{6} + i\sin\frac{20\pi}{6} + \cos\frac{20\pi}{6} - i\sin\frac{20\pi}{6}$$

$$= 2\cos\frac{10\pi}{3} = 2\cos\frac{4\pi}{3}$$

$$= 2\cos\left(\pi + \frac{\pi}{3}\right)$$

$$= -2\cos\frac{\pi}{3} = -1$$

3. Answer (3)

For finding $x^{2013} - \frac{1}{x^{2013}}$

$$= \cos\frac{2013\pi}{6} + i\sin\frac{2013\pi}{6} - \cos\frac{2013\pi}{6} + i\sin\frac{2013\pi}{6}$$

$$= 2i\sin\frac{2013\pi}{6}$$

$$= 2i\sin\frac{671\pi}{2}$$

$$= 2i\sin\frac{\pi}{2} = i$$

Comprehension III

If $(1 + x)^n = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, then

1. Find the sum of the series $a_0 + a_2 + a_4 + \dots$

- (1) 2^n (2) 2^{n-1} (3) 2 (4) 2^{n-2}

2. Value of the series $a_0 - a_2 + a_4 - a_6 + \dots$ is

- (1) $2^n \cos \frac{n\pi}{4}$ (2) $2^{n-1} \sin \frac{n\pi}{4}$ (3) $2^{n-1} \cos \frac{n\pi}{4}$ (4) $2^{\frac{n}{2}} \cos \frac{n\pi}{4}$

3. The sum of the series $a_0 + a_4 + a_8 + a_{12} + \dots$ is

- (1) $2^{n-1} \cos \frac{n\pi}{4}$ (2) $2^{n-2} + 2^{\frac{n}{2}-1} \cos \frac{n\pi}{4}$ (3) $2^{n-1} + 2^{\frac{n}{2}} \sin \frac{n\pi}{4}$ (4) $2^{n-1} \sin \frac{n\pi}{4}$

Solution of Comprehension III

1. Answer (2)

2. Answer (4)

3. Answer (2)

Given expression is

$$(1 + x)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n \quad \dots(i)$$

Putting $x = \pm 1$, we get

$$(1 + 1)^n = a_0 + a_1 + a_2 + \dots + a_n$$

$$(1 - 1)^n = a_0 - a_1 + a_2 - \dots \pm a_n$$

Adding these,

$$2^n = 2(a_0 + a_2 + a_4 + \dots)$$

$$\Rightarrow a_0 + a_2 + a_4 + \dots = 2^{n-1} \quad \dots(ii)$$

Hence, answer of question 1 is (2)

Again, putting $x = \pm i$ in (i), we get

$$(1 + i)^n = a_0 + a_1i - a_2 - a_3i + a_4 + a_5i - a_6 - a_7i + a_8 + \dots$$

$$(1 - i)^n = a_0 - a_1i - a_2 + a_3i + a_4 - a_5i - a_6 + a_7i + a_8 + \dots$$

Adding these,

$$2(a_0 - a_2 + a_4 - a_6 + \dots) = \frac{(1+i)^n + (1-i)^n}{2} = \frac{2 \cdot 2^{n/2} \cdot \cos \frac{n\pi}{4}}{2} = 2^{n/2} \cos \frac{n\pi}{4} \quad \dots(iii)$$

Hence, answer of question 2 is (4)

Now, adding (ii) & (iii), we can get

$$2(a_0 + a_4 + a_8 + \dots) = 2^{n-1} + 2^{n/2} \cos \frac{n\pi}{4}$$

$$\Rightarrow a_0 + a_4 + a_8 + a_{12} + \dots = 2^{n-2} + 2^{\frac{n}{2}-1} \cos \frac{n\pi}{4}$$

Hence, answer is (2)

Comprehension-IV

Let us consider an equation $f(x) = x^3 - 3x + k = 0$. Then the values of k for which the equation has

1. Exactly one root which is positive, then k belongs to

- (1) $(-\infty, -2)$ (2) $(2, \infty)$ (3) $(0, 2)$ (4) $(-2, 0)$

Sol. Answer (1)

$$f(x) = x^3 - 3x + k$$

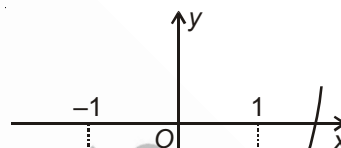
$$f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$$

For exactly one positive root

$$f(-1) < 0 \text{ and } f(1) < 0$$

$$\Rightarrow -1 + 3 + k < 0 \text{ and } 1 - 3 + k < 0$$

$$\Rightarrow k < -2 \text{ and } k < 2 \Rightarrow k \in (-\infty, -2)$$



2. Exactly one root which is negative, then k belongs to

- (1) $(2, \infty)$ (2) $(0, 2)$ (3) $(-2, 0)$ (4) $(-\infty, -2)$

Sol. Answer (1)

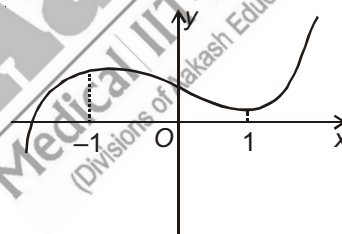
For exactly one negative root,

$$f(-1) > 0, f(1) > 0$$

$$\Rightarrow -1 + 3 + k > 0, 1 - 3 + k > 0$$

$$\Rightarrow k > -2, k > 2$$

$$\Rightarrow k \in (2, \infty)$$



3. One negative and two positive root if k belongs to

- (1) $(2, \infty)$ (2) $(0, 2)$ (3) $(-2, 0)$ (4) $(2, 3)$

Sol. Answer (2)

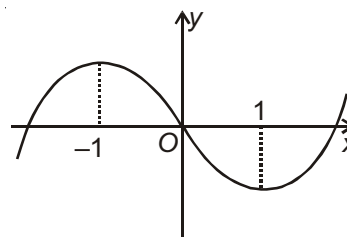
For one negative and two positive root

$$f(-1) > 0, f(0) > 0, f(1) < 0$$

$$\Rightarrow -1 + 3 + k > 0, k > 0, 1 - 3 + k < 0$$

$$\Rightarrow k > -2, k > 0, k < 2$$

$$\Rightarrow 0 < k < 2 \text{ i.e., } k \in (0, 2)$$



Comprehension-V

The values of 'k' for which the equation $|x|^2 (|x|^2 - 2k + 1) = 1 - k^2$, has

1. No real root, when k belongs to

- (1) $(-1, 1)$ (2) $\left(-1, \frac{5}{4}\right)$ (3) $(-\infty, -1) \cup \left(\frac{5}{4}, \infty\right)$ (4) R

Sol. Answer (3)

$$|x|^2 (|x|^2 - 2k + 1) = 1 - k^2$$

$$\Rightarrow x^2(x^2 - 2k + 1) = 1 - k^2 \Rightarrow x^4 - (2k - 1)x^2 + k^2 - 1 = 0$$

All roots are imaginary, if $D = b^2 - 4ac < 0$

$$\Rightarrow (2k - 1)^2 - 4(k^2 - 1) < 0$$

$$\Rightarrow k > \frac{5}{4} \quad \dots(1)$$

Also roots are imaginary if $D > 0$, but x^2 is negative, i.e. roots of $(x^2)^2 - (2k - 1)(x^2) + k^2 - 1 = 0$ are both negative.

$\therefore$ Sum < 0 , and product > 0

$$\Rightarrow 2k - 1 < 0 \text{ and } k^2 - 1 > 0 \Rightarrow k \in (-\infty, -1)$$

$$\therefore \text{All roots are imaginary if } k \in (-\infty, -1) \cup \left(\frac{5}{4}, \infty\right)$$

2. Exactly two real roots, when k belongs to

- (1) $(-\infty, -1)$ (2) $(-1, 1)$ (3) $\left(1, \frac{5}{4}\right)$ (4) R

Sol. Answer (2)

For exactly two real roots of

$$t^2 - (2k - 1)t + k^2 - 1 = 0$$

$D > 0$ and one value of $t = x^2$ is positive and one is negative.

$$\Rightarrow (2k - 1)^2 - 4(k^2 - 1) > 0$$

$$\Rightarrow k < \frac{5}{4} \quad \dots(i)$$

$$\text{Product} = k^2 - 1 < 0 \Rightarrow -1 < k < 1 \quad \dots(ii)$$

From (1) and (2) $k \in (-1, 1)$.

3. Repeated roots, when k belongs to

(1) $\{1, -1\}$

(2) $\{0, 1\}$

(3) $\{0, -1\}$

(4) $\{2, 3\}$

Sol. Answer (1)

For repeated roots $\Delta = 0$, $k = \frac{5}{4}$

or $P = 0$, $\Rightarrow k^2 - 1 = 0$, $k = \pm 1$

When product = 0, $x = 0$ is repeated root.

Comprehension-VI

Let A, B, C be three sets of complex numbers as defined below

$$A = \{z : \operatorname{Im} z \geq 1\}$$

$$B = \{z : |z - 2 - i| = 3\}$$

$$C = \{z : \operatorname{Re}((1-i)z) = \sqrt{2}\}.$$

[IIT-JEE 2008]

1. The number of elements in the set $A \cap B \cap C$ is

(1) 0

(2) 1

(3) 2

(4) ∞

Sol. Answer (2)

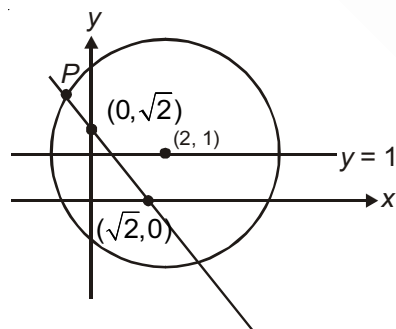
Let $z = x + iy$

Set A corresponds to the region $y \geq 1$... (i)

Set B consists of points lying on the circle, centred at $(2, 1)$ and radius 3,

i.e., $x^2 + y^2 - 4x - 2y = 4$... (ii)

Set C consists of points lying on the $x + y = \sqrt{2}$... (iii)



Clearly, there is only one point of intersection of the line $x + y = \sqrt{2}$ and circle $x^2 + y^2 - 4x - 2y = 4$

2. Let z be any point in $A \cap B \cap C$. The $|z + 1 - i|^2 + |z - 5 - i|^2$ lies between

(1) 25 and 29

(2) 30 and 34

(3) 35 and 39

(4) 40 and 44

Sol. Answer (3)

$$|z + 1 - i|^2 + |z - 5 - i|^2$$

$$= (x + 1)^2 + (y - 1)^2 + (x - 5)^2 + (y - 1)^2$$

$$= 2(x^2 + y^2 - 4x - 2y) + 28$$

$$= 2(4) + 28 \quad (\because x^2 + y^2 - 4x - 2y = 4)$$

$$= 36$$

3. Let z be any point in $A \cap B \cap C$ and let w be any point satisfying $|w - 2 - i| < 3$. Then, $|z| - |w| + 3$ lies between

(1) -6 and 3

(2) -3 and 6

(3) -6 and 6

(4) -3 and 9

Sol. Answer (4)

$$|w - (2 + i)| < 3$$

$$\Rightarrow ||w| - |2 + i|| < 3$$

$$\Rightarrow -3 + \sqrt{5} < |w| < 3 + \sqrt{5}$$

$$\Rightarrow -3 - \sqrt{5} < -|w| < 3 - \sqrt{5} \quad \dots(i)$$

$$\text{Also, } |z - (2 + i)| = 3$$

$$\Rightarrow -3 + \sqrt{5} \leq |z| \leq 3 + \sqrt{5} \quad \dots(ii)$$

$$\therefore -3 < |z| - |w| + 3 < 9$$

Comprehension-VII

Let $S = S_1 \cap S_2 \cap S_3$, where

$$S_1 = \{z \in \mathbb{C} : |z| < 4\},$$

$$S_2 = \left\{ z \in \mathbb{C} : \operatorname{Im} \left[\frac{z - 1 + \sqrt{3}i}{1 - \sqrt{3}i} \right] > 0 \right\} \text{ and}$$

$$S_3 = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}.$$

[JEE(Advanced) 2013]

1. Area of $S =$

(1) $\frac{10\pi}{3}$

(2) $\frac{20\pi}{3}$

(3) $\frac{16\pi}{3}$

(4) $\frac{32\pi}{3}$

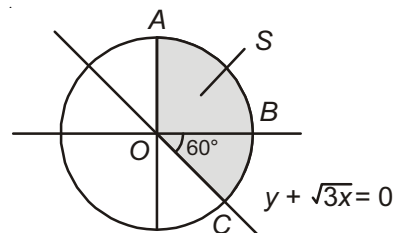
Sol. Answer (2)

S_1 represent circle with centre $(0, 0)$ and radius 4

$$S_1 : |z| < 4 \Rightarrow x^2 + y^2 < 16$$

$$S_2 : \operatorname{Im} \left[\frac{z - 1 + \sqrt{3}i}{1 - \sqrt{3}i} \right] > 0$$

$$\operatorname{Im} \left(\frac{[(x - 1) + (y + \sqrt{3}i)][1 + \sqrt{3}i]}{2} \right) > 0$$



$$S_2 \equiv y + \sqrt{3}x > 0$$

$$S_3 \text{ Re}(z) > 0, \text{ i.e., } x > 0$$

$$S = S_1 \cap S_2 \cap S_3$$

$$\begin{aligned} \text{Area of shaded region is } OAB + OBC &= \frac{\pi(4)^2}{4} + \frac{60}{360} \times \pi(4)^2 \\ &= 4\pi + \frac{16\pi}{6} \\ &= 4\pi + \frac{8\pi}{3} \\ &= \frac{20\pi}{3} \end{aligned}$$

2. $\min_{z \in S} |1 - 3i - z| =$

(1) $\frac{2 - \sqrt{3}}{2}$

(2) $\frac{2 + \sqrt{3}}{2}$

(3) $\frac{3 - \sqrt{3}}{2}$

(4) $\frac{3 + \sqrt{3}}{2}$

Sol. Answer (3)

$$\min |z - (1 - 3i)|$$

Minimum distance of z from $(1, -3)$

From question, minimum distance of $(1, -3)$ from $y + \sqrt{3}x = 0$ is $\left| \frac{-3 + \sqrt{3}}{2} \right| = \frac{3 - \sqrt{3}}{2}$.

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : Let z_1 and z_2 be two complex numbers such that $\arg(z_1) = \frac{\pi}{3}$ and $\arg(z_2) = \frac{\pi}{6}$ then $\arg(z_1 z_2) = \frac{\pi}{2}$.

and

STATEMENT-2 : $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) + 2k\pi, k \in \{0, 1, -1\}$

Sol. Answer (1)

Clearly Statement-1 is true and Statement-2 is its explanation. (standard results)

2. STATEMENT-1 : The locus of z , if $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$ is a circle.

and

STATEMENT-2 : $\left| \frac{z-2}{z+2} \right| = \frac{\pi}{2}$, then the locus of z is a circle.

Sol. Answer (4)

$$\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$$

Locus of z is a semicircle $\Rightarrow$ statement-1 is false and statement-2 is true. $\Rightarrow$ Option (4)**OR**Let $z = x + iy$

$$\frac{z-1}{z+1} = \frac{(x-1)+iy}{(x+1)+iy}$$

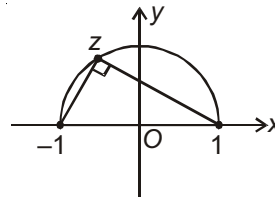
$$\frac{z-1}{z+1} = \frac{(x-1)+iy}{(x+1)+iy} \times \frac{(x+1)-iy}{(x+1)-iy}$$

$$\frac{z-1}{z+1} = \frac{(x^2-1+y^2)+i(2y)}{(x+1)^2+y^2}$$

$$\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$$

$$\Rightarrow x^2 + y^2 - 1 = 0$$

$$x^2 + y^2 = 1 \text{ and } y > 0$$

 $\Rightarrow$ Locus of z is semicircle.

3. STATEMENT-1 : If $e^{i\theta} = \cos\theta + i\sin\theta$ and the value of $e^{iA} \cdot e^{iB} \cdot e^{iC}$ is equal to -1 , where A, B, C are the angles of a triangle.

andSTATEMENT-2 : In any $\triangle ABC$, $A + B + C = 180^\circ$.**Sol.** Answer (1)

$$e^{iA} \cdot e^{iB} \cdot e^{iC} = e^{i(A+B+C)}$$

$$= e^{i\pi} = \cos\pi + i\sin\pi$$

$$= -1$$

4. STATEMENT-1 : $z_1^2 + z_2^2 + z_3^2 + z_4^2 = 0$ where z_1, z_2, z_3 and z_4 are the fourth roots of unity.

and

$$\text{STATEMENT-2 : } (1)^{\frac{1}{4}} = (\cos 0^\circ + i\sin 0^\circ)^{\frac{1}{4}}.$$

Sol. Answer (1)

$$(1)^{1/4} = (\cos 2\pi r + i \sin 2\pi r)^{1/4}$$

$$= \cos \frac{\pi r}{2} + i \sin \frac{\pi r}{2}$$

where $r = 0, 1, 2, 3$

$$\therefore 1^{1/4} = 1, i, -1, -i$$

$$\begin{aligned} \therefore z_1^2 + z_2^2 + z_3^2 + z_4^2 &= 1 + i^2 + 1 + i^2 \\ &= 2 - 1 - 1 = 0 \end{aligned}$$

5. STATEMENT-1 : For any four complex numbers z_1, z_2, z_3 and z_4 , it is given that the four points are concyclic, then $|z_1| = |z_2| = |z_3| = |z_4|$.

and

STATEMENT-2 : Modulus of a complex number represents the distance from origin.

Sol. Answer (4)

$$|z_1| = |z_2| = |z_3| = |z_4|$$

This may not be the case if centre of the circle is not origin.

6. STATEMENT-1 : The expression $\left(\frac{2i}{1+i}\right)^n$ is a positive integer for all the values of n .

and

STATEMENT-2 : Here $n = 8$ is the least positive for which the above expression is a positive integer.

Sol. Answer (4)

$$\begin{aligned} \left(\frac{2i}{1+i}\right)^n &= \left(\frac{2i(1-i)}{2}\right)^n = (1+i)^n \\ &= (\sqrt{2})^n e^{i\frac{n\pi}{4}} \end{aligned}$$

Now clearly the least integral value for which the given number is a positive integer is 8.

7. STATEMENT-1 : If $1 - i, 1 + i, z_1$ and z_2 are the vertices of a square taken in order in the anti-clockwise sense then z_1 is $i - 1$.

and

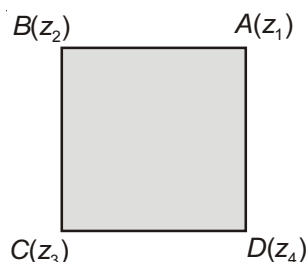
STATEMENT-2 : If the vertices of a square are z_1, z_2, z_3, z_4 taken in order in the anti-clockwise sense, then $z_3 = -iz_1 + (1+i)z_2$.

Sol. Answer (1)

$$\angle ABC = \frac{\pi}{2} = \arg \frac{z_1 - z_2}{z_3 - z_2}$$

$$\text{And } AB = |z_1 - z_2| = BC = |z_3 - z_2|$$

$$\therefore \left| \frac{z_1 - z_2}{z_3 - z_2} \right| = 1$$



$$\text{Hence, } \frac{z_1 - z_2}{z_3 - z_2} = 1 \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right)$$

$$\Rightarrow z_3 - z_2 = (z_1 - z_2) \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right)$$

$$= -i(z_1 - z_2)$$

$$= -iz_1 + iz_2$$

$$\Rightarrow z_3 = -iz_1 + (1 + i)z_2$$

$$\begin{aligned} \Rightarrow z_3 &= -i(1 - i) + (1 + i)(1 + i) = -i + i^2 + 1 + i^2 + 2i \\ &= i - 1 \end{aligned}$$

8. STATEMENT-1 : If $\left| z + \frac{1}{z} \right| = a$ where z is a complex number and a is a real number, the least and greatest

values of $|z|$ are $\frac{\sqrt{a^2 + 4} - a}{2}$ and $\frac{\sqrt{a^2 + 4} + a}{2}$.

and

STATEMENT-2 : For a equal to zero the greatest and the least values of $|z|$ are equal.

Sol. Answer (2)

Let $z = r(\cos \theta + i \sin \theta)$

$$\left| z + \frac{1}{z} \right|^2 = \left(r + \frac{1}{r} \right)^2 \cos^2 \theta + \left(r - \frac{1}{r} \right)^2 \sin^2 \theta$$

$$= r^2 + \frac{1}{r^2} + 2 \cos 2\theta$$

$$\Rightarrow a^2 = r^2 + \frac{1}{r^2} + 2 - 4 \sin^2 \theta$$

$$\Rightarrow \left(r + \frac{1}{r} \right)^2 = a^2 + 4 \sin^2 \theta$$

$$\therefore \left(r + \frac{1}{r} \right)^2 \leq a^2 + 4$$

$$\Rightarrow r + \frac{1}{r} \leq \sqrt{a^2 + 4}$$

$$r^2 - \sqrt{a^2 + 4}r + 1 \leq 0$$

$$\Rightarrow r \text{ lies between } \frac{\sqrt{a^2 + 4} - a}{2} \text{ and } \frac{\sqrt{a^2 + 4} + a}{2}$$

This is true for all real $a \geq 0$.

$$|z|_{\max} = \frac{\sqrt{a^2 + 4} + a}{2}, |z|_{\min} = \frac{\sqrt{a^2 + 4} - a}{2}$$

Hence, for $a = 0$

$$|z|_{\min} = |z|_{\max}$$

Hence both statement-1 and statement-2 are true. But statement-2 is not the correct explanation of statement-1.

9. STATEMENT-1 : The locus of complex number z , satisfying $(z - 2)^n = z^n$ is a straight line.

and

STATEMENT-2 : The equation of the form $ax + by + c = 0$ in $x - y$ plane is the general equation of straight line.

Sol. Answer (1)

Given equation is $(z - 2)^n = z^n$

$$\Rightarrow \left| \frac{z - 2}{z} \right|^n = 1$$

$$\Rightarrow \left| \frac{z - 2}{z} \right| = 1^{1/n} = 1$$

$$\Rightarrow |z - 2| = |z|$$

Hence z is the locus of a straight line perpendicular bisector of the segment joining the points $(2, 0)$ and $(0, 0)$, i.e., $x = 1$.

10. STATEMENT-1 : A root of the equation $(2^{10} - 3)x^2 - 2^{11}x + (2^{10} + 3) = 0$ is 1.

and

STATEMENT-2 : The sum of the coefficients of a quadratic equation is zero, then 1 is a root of the equation.

Sol. Answer (1)

$$ax^2 + bx + c = 0$$

$$x = 1, a + b + c = 0$$

If sum of coefficient is 0 then 1 is the root of the equation.

$$(2^{10} - 3) - 2^{11} + 2^{10} + 3 = 0$$

$\therefore$ Both are true and Statement-2 is correct explanation of Statement-1

11. STATEMENT-1 : The equation whose roots are reciprocal of the roots of the equation $10x^2 - x - 5 = 0$ is $5x^2 + x - 10 = 0$.

and

STATEMENT-2 : To obtain a quadratic equation whose roots are reciprocal of the roots of the given equation $ax^2 + bx + c = 0$ change the coefficients a, b, c to c, b, a . ($c \neq 0$)

Sol. Answer (1)

For reciprocal roots, replacing x by $\frac{1}{x}$ in $ax^2 + bx + c = 0$ $\frac{a}{x^2} + \frac{b}{x} + c = 0 \Rightarrow cx^2 + bx + a = 0$

Statement-2 is correct and is correct explanation of Statement-1

$$10x^2 - x - 5 = 0 \Rightarrow \frac{10}{x^2} - \frac{1}{x} - 5 = 0 \Rightarrow 5x^2 + x - 10 = 0$$

12. STATEMENT-1 : The equation $x^2 - 2009x + 2008 = 0$ has rational roots.

and

STATEMENT-2 : The quadratic equation $ax^2 + bx + c = 0$ has rational roots iff $b^2 - 4ac$ is a perfect square.

Sol. Answer (3)

The equation in first statement is $x^2 - 2009x + 2008 = 0$ can be written as $(x - 2008)(x - 1) = 0$

$\Rightarrow x = 1, 2008$ are roots of the equation where are rationals also.

$\Rightarrow$ Statement 1 is True.

Statement 2 is not always true. When $D = b^2 - 4ac = a$ perfect square than roots of the equation $ax^2 + bx + c = 0$ are rational only when a, b, c are rationals, otherwise roots are irrationals.

To this end, let us consider an equation $4x^2 - 4\sqrt{3}x - 1 = 0$

whose discriminant $= 48 + 16 = 64 = 8^2 = a$ perfect square but roots are

$$x = \frac{4\sqrt{3} \pm \sqrt{48+16}}{2 \times 4}$$

$$= \frac{\sqrt{3} \pm 2}{2} \text{ which are not rationals.}$$

Thus statement 2 is false.

Hence option (3) is correct.

13. STATEMENT-1 : One root of the equation $x^2 + 5x - 7 = 0$ lie in the interval $(1, 2)$.

and

STATEMENT-2 : For a polynomial $f(x)$, if $f(p)f(q) < 0$, then there exists at least one real root of $f(x) = 0$ in (p, q) .

Sol. Answer (1)

We observe that $f(1)f(2) = (1 + 5 - 7)(4 + 10 - 7) < 0$

Hence there exists a root of $x^2 + 5x - 7 = 0$ in $(1, 2)$.

Clearly option (1) is correct.

14. STATEMENT-1 : The quadratic equation $ax^2 + bx + c = 0$ has real roots if $(a + c)^2 > b^2, \forall a, b, c \in R$.

and

STATEMENT-2 : The quadratic equation $ax^2 + bx + c = 0$ has real roots if $b^2 - 4ac \geq 0$.

Sol. Answer (4)

We observe that

$$(a + c)^2 > b^2$$

$$\Rightarrow (a + c)^2 - b^2 > 0$$

$$\Rightarrow (a - b + c)(a + b + c) > 0$$

$$\Rightarrow f(-1)f(1) > 0, \text{ where } f(x) = ax^2 + bx + c$$

$$\Rightarrow f(x) = ax^2 + bx + c = 0 \text{ has either no root in } (-1, 1) \text{ or if real roots exist, then both roots lie in } (-1, 1)$$

$$\Rightarrow \text{Statement 1 is not necessarily true}$$

Hence statement 1 is false.

Answer is 4

15. STATEMENT-1 : There is just one quadratic equation with real coefficients, one of whose roots is $\frac{1}{3+\sqrt{7}}$.

and

STATEMENT-2 : In a quadratic equation with rational coefficients the irrational roots occur in pair.

Sol. Answer (4)

Statement-1 is wrong.

$$\frac{1}{3+\sqrt{7}} \text{ can be root of infinite equations with real coefficients, e.g. } (x-1)\left(x-\frac{1}{3+\sqrt{7}}\right) = 0$$

$$(x-2)\left(x-\frac{1}{3+\sqrt{7}}\right) = 0 \dots$$

16. STATEMENT-1 : The roots of $x^2 + 2\sqrt{2008}x + 501 = 0$ are irrational.

and

STATEMENT-2 : If the discriminant of the equation $ax^2 + bx + c = 0$, $a \neq 0$ ($a, b, c, \in R$) is a perfect square, then the roots are rational.

Sol. Answer (3)

$$x^2 + 2\sqrt{2008}x + 501 = 0$$

$$x = \frac{-2\sqrt{2008} \pm \sqrt{4 \times 2008 - 4(501)}}{2}$$

$$= -\sqrt{2008} \pm \sqrt{1507}$$

Roots are rational.

STATEMENT-2 is wrong as roots are rational only when coefficients are rational and $b^2 = 4ac$ is perfect square.

17. STATEMENT-1 : If a, b, c not all equal and $a \neq 0$, $a^3 + b^3 + c^3 = 3abc$, then the equation $ax^2 + bx + c = 0$ has two real roots of opposite sign.

and

STATEMENT-2 : If roots of a quadratic equation $ax^2 + bx + c = 0$ are real and of opposite sign then $ac < 0$.

Sol. Answer (4)

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 0,$$

$a + b + c = 0 \therefore a, b, c$ are not equal. The sign of other root depends on sign of ac .

Hence (4) is answer.

18. STATEMENT-1 : There is just one quadratic equation with real coefficient one of whose roots is $\frac{1}{\sqrt{2}+1}$.

and

STATEMENT-2 : In a quadratic equation with rational coefficients the irrational roots are in conjugate pairs.

Sol. Answer (4)

$(\sqrt{2}-1)$ is a root if coefficients are real then other root can be rational.

19. STATEMENT-1 : Let a quadratic equation has a root $3 - 9i$ then the sum of roots is 6.

and

STATEMENT-2 : If one root of $ax^2 + bx + c = 0$, $a \neq 0$, $a, b, c \in R$ is $\alpha + i\beta$, $\alpha, \beta \in R$ then the other root must be $\alpha - i\beta$.

Sol. Answer (4)

For, $ax^2 + bx + c = 0$, roots are conjugate of each other if a, b, c are real, which is not mentioned in statement-1.

20. Let a, b, c, p, q be real numbers. Suppose α, β are the roots of the equation $x^2 + 2px + q = 0$ and $\alpha, \frac{1}{\beta}$ are the roots of the equation $ax^2 + 2bx + c = 0$, where $\beta^2 \in \{-1, 0, 1\}$ [IIT-JEE 2008]

STATEMENT-1 : $(p^2 - q)(b^2 - ac) \geq 0$

and

STATEMENT-2 : $b \neq pa$ or $c \neq qa$

Sol. Answer (2)

$$x^2 + 2px + q = 0$$

$$\alpha + \beta = -2p \quad \dots(i)$$

$$\alpha\beta = q \quad \dots(ii)$$

$$ax^2 + 2bx + c = 0$$

$$\alpha + \frac{1}{\beta} = \frac{-2b}{a} \quad \dots(iii)$$

$$\frac{\alpha}{\beta} = \frac{c}{a} \quad \dots(iv)$$

$$(p^2 - q)(b^2 - ac) = \left(\left(\frac{\alpha + \beta}{-2} \right)^2 - \alpha\beta \right) \left(\left(\frac{\alpha + \frac{1}{\beta}}{2} \right)^2 - \frac{\alpha}{\beta} \right) a^2$$

$$= \frac{(\alpha - \beta)^2}{16} \left(\alpha - \frac{1}{\beta} \right)^2 \cdot a^2 \geq 0 \text{ statement-1 is true}$$

$$\text{Now } pa = - \left(\frac{\alpha + \beta}{2} \right) a = \frac{a}{2} (\alpha + \beta)$$

$$b = - \frac{a}{2} \left(\alpha + \frac{1}{\beta} \right)$$

$$pa \neq b \Rightarrow \alpha + \frac{1}{\beta} \neq \alpha + \beta \Rightarrow \beta^2 \neq 1, \beta \neq \{-1, 0, 1\}, \text{ correct}$$

Similarly

$$\text{If } c \neq qa \Rightarrow a \frac{\alpha}{\beta} \neq a\alpha\beta$$

$$\Rightarrow \alpha \left(\beta - \frac{1}{\beta} \right) \neq 0$$

$$\Rightarrow \alpha \neq 0, \text{ and } \beta - \frac{1}{\beta} \neq 0 \Rightarrow \beta \neq \{-1, 0, 1\}$$

STATEMENT-2 is true.

Both statement 1 and statement 2 are true, But statement 2 do not explains statement 1.

SECTION - E

Matrix-Match Type Questions

1. Match the following

Column-I

Column-II

(A) The value of $\sum_{k=1}^{2007} \left(\sin \frac{2k\pi}{9} - i \cos \frac{2k\pi}{9} \right)$ is

(p) -1

(B) If z_1, z_2 and z_3 are unimodular complex numbers such

(q) 2

that $|z_1 + z_2 + z_3| = 1$, then $\left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right|$ is equal to

(C) If the complex numbers z_1, z_2 and z_3 represent the vertices of an equilateral triangle such that

(r) 1

$|z_1| = |z_2| = |z_3|$, then $(z_1 + z_2 + z_3) - 1$ is equal to

(D) If α is an imaginary fifth root of unity, then

(s) 0

$4 \log_4 \left| 1 + \alpha + \alpha^2 + \alpha^3 - \frac{1}{\alpha} \right|$ is

Sol. Answer A(s), B(r), C(p), D(q)

$$(A) \sum_{k=1}^{2007} \left(\sin \frac{2k\pi}{9} - i \cos \frac{2k\pi}{9} \right) = (-i) \sum_{k=1}^{2007} \left(\cos \frac{2k\pi}{9} + i \sin \frac{2k\pi}{9} \right) = (-i) \sum_{k=1}^{2007} e^{i \frac{2k\pi}{9}}$$

$$= -i \left[e^{\frac{i2\pi}{9}} + e^{\frac{i4\pi}{9}} + \dots + e^{\frac{i2(2007)\pi}{9}} \right]$$

Which is G.P.

$$= -i \left[e^{\frac{i2\pi}{9}} \frac{\left(e^{\frac{i2\pi(2007)}{9}} - 1 \right)}{e^{\frac{i2\pi}{9}} - 1} \right] = -i \left[\frac{e^{\frac{i2\pi}{9}} (1-1)}{e^{\frac{i2\pi}{9}} - 1} \right] = 0$$

$$(B) |z_1| = 1, |z_2| = 1, |z_3| = 1$$

$$z_1 \bar{z}_1 = 1, z_2 \bar{z}_2 = 1, z_3 \bar{z}_3 = 1$$

$$\frac{1}{z_1} = \bar{z}_1, \frac{1}{z_2} = \bar{z}_2, \frac{1}{z_3} = \bar{z}_3$$

$$\text{Now, } \left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right| = |\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = |\overline{z_1 + z_2 + z_3}| = 1$$

$$(C) |z_1| = |z_2| = |z_3| \text{ and } z_1, z_2, z_3 \text{ are vertices of equilateral triangle}$$

$\Rightarrow$ Origin is its centroid

$$\therefore z_1 + z_2 + z_3 = 0$$

$$\text{Now, } |z_1 + z_2 + z_3| - 1 = -1$$

$$(D) \text{ Let } (1)^{\frac{1}{5}} = 1, \alpha, \alpha^2, \alpha^3, \alpha^4$$

$$1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0$$

$$1 + \alpha + \alpha^2 + \alpha^3 = -\alpha^4$$

$$\text{and } \alpha^5 = 1$$

$$\Rightarrow \frac{1}{\alpha} = \alpha^4 \text{ and } |\alpha| = 1$$

$$\text{Now } 4 \log_4 |-\alpha^4 - \alpha^4| = 4 \log_4 |-2\alpha^4| = 4 \log_4 |-2| |\alpha^4| = 4 \times \log_4 2 = 4 \times \frac{1}{2} \log_2 2 = 2$$

2. Match the following

Column-I

Column-II

(A) The smallest positive integer for which

(p) 1

$$(1 + i)^n = (1 - i)^n \text{ is}$$

(B) If $\sqrt[3]{a + ib} = x + iy$ and $\frac{b}{y} - \frac{a}{x} = k(x^2 + y^2)$, then

(q) -3

k is equal to

(C) If $x = \frac{1+i}{\sqrt{2}}$, then the value of

(r) 2

$$1 + x^2 + x^4 + x^6 + x^8 + x^{10} + \dots + x^{2004} + x^{2006} + x^{2008}$$

(D) If the minimum value of

(s) 4

$$|z+1+i| + |z-1-i| + |2-z| + |3-z| \text{ is } k \text{ then } (k-8) \text{ equals}$$

Sol. Answer A(s), B(r), C(p), D(q)

(A) $(1 + i)^n = (1 - i)^n$

$$\Rightarrow \left(\frac{1+i}{1-i} \right)^n = 1$$

$$\Rightarrow i^n = 1$$

$$\Rightarrow n = 4$$

(B) $(a + ib) = (x + iy)^3 = x^3 - iy^3 + 3ix^2y - 3xy^2$

$$a = x(x^2 - 3y^2) \text{ and } b = -y^3 + 3x^2y$$

$$\frac{a}{x} = x^2 - 3y^2 \quad \frac{b}{y} = -y^2 + 3x^2$$

$$\frac{b}{y} - \frac{a}{x} = 2(x^2 + y^2)$$

$$k = 2$$

(C) $x = \frac{1+i}{\sqrt{2}}$

$$\Rightarrow x^2 = \frac{(1+i)^2}{2} = \frac{1-1+2i}{2} = i$$

$$1 + x^2 + x^4 + x^6 + x^8 + x^{10} + \dots + x^{2004} + x^{2006} + x^{2008}$$

$$\frac{1[1-x^{2010}]}{1-x^2} = \frac{1-i^2}{1-i^2} = \frac{1+1}{1+1} = 1$$

3. Let z_1 and z_2 be two given complex numbers. The locus of z such that

Column-I

(A) $|z - z_1| + |z - z_2| = \text{constant} = k$, where $k \neq |z_1 - z_2|$

(B) $|z - z_1| - |z - z_2| = k$, where $k \neq |z_1 - z_2|$

(C) $\arg\left(\frac{z - z_1}{z - z_2}\right) = \pm \frac{\pi}{2}$

(D) If ω lies on $|\omega| = 1$ then $\frac{2007}{\omega}$ lies on

Column-II

(p) Circle with z_1 and z_2 as the vertices of diameter.

(q) Circle

(r) Hyperbola

(s) Ellipse

Sol. Answer A(s), B(r), C(p, q), D(q)

A) $|z - z_1| + |z - z_2| = \text{constant} = k$,

where $k \neq |z_1 - z_2|$, represents an ellipse.

(B) $|z - z_1| - |z - z_2| = k$, where $k \neq |z_1 - z_2|$

is a hyperbola having foci at z_1 and z_2 .

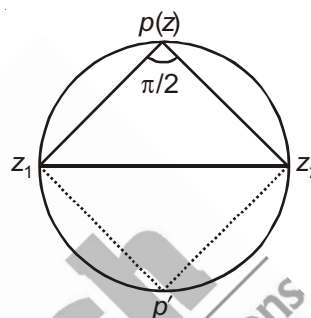
(C) $\arg\left(\frac{z - z_1}{z - z_2}\right) = \pm \frac{\pi}{2}$

This represents a circle with z_1 and z_2 as the vertices of diameter.

(D) If ω lies on $|\omega| = 1$, then

$$\left|\frac{2007}{\omega}\right| = \frac{2007}{|\omega|} = 2007$$

$$\Rightarrow \frac{2007}{\omega} \text{ lies on the circle.}$$



4. Match the following

Column-I

(A) $|z - 6i| + |z - 8| = k$ will represent an ellipse for k equals to

(B) $||z - 12i + 3| - |z - 2|| = k$ will represent hyperbola if k equals to

(C) $|z - ki| + |z - 4| = \sqrt{10k}$ will represent line segment if k equals to

(D) $\left|\frac{z - k + 2ki}{z - 2 + 4i}\right| = k$ will represent circle if k equals to

Column-II

(p) 2

(q) 8

(r) 12

(s) 11

(t) 10

Sol. Answer A(r, s), B(p, q, r, s, t), C(p, q), D(q, r, s, t)

(A) $|z - 6i| + |z - 8| = k$ will represent ellipse if $\sqrt{6^2 + 8^2} < k \Rightarrow k > 10$

(B) $||z - 12i + 3| - |z - 2||$ will represent ellipse if $\sqrt{(5)^2 + 12^2} > k \Rightarrow 13 > k$

(C) $|z - ki| + |z - 4| = \sqrt{10k}$ will represent line segment if $k^2 + 16 = 10k \Rightarrow k = 2, 8$

(D) To represent circle $k \neq 1$ and also $k = 2$

5. Match the entries of column-I with those of column-II

Column-I

Column-II

(A) $t \in R$ such that there is at least one z satisfying
 $|z| = 3$, $|z - \{t(1+i) - i\}| \leq 3$ and $|z + 2t - (t+1)i| > 3$. (p) 6

(B) Solve for x : $\frac{(1+i)x - 2i}{3+i} + \frac{(2-3i)y + i}{3-i} = i$ (q) 0

(C) The integer n for which $\left(\frac{1+i}{1-i}\right)^n$ is real (r) 3

(D) The greatest and least absolute value of $z + 1$,
 where $|z + 4| \leq 3$ are (s) 4

(t) 8

Sol. Answer A(r, s), B(r), C(p, q, s, t), D(p, q)

(A) To satisfy all at a time z should lie on the circle $|z| = 3$.

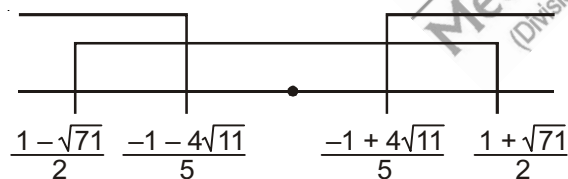
Inside the circle $|z - \{(1+i) - i\}| = 3$ and outside the circle $|z + 2t - (t+1)i| = 3$

For this,

$$\sqrt{(t-0)^2 + (t-1-0)^2} \leq 3+3 \text{ and } \sqrt{4t^2 + (4+1)^2} > 3+3$$

$$\Rightarrow 2t^2 - 2t - 35 \leq 0 \text{ and } 5t^2 + 2t - 35 > 0$$

Using sign scheme we have,



$$\text{Hence, } 1 - \left(1 - \frac{\sqrt{71}}{2}, \frac{-1 - 4\sqrt{11}}{5}\right) \cup \left(\frac{-1 + 4\sqrt{11}}{5}, \frac{1 + \sqrt{71}}{2}\right)$$

Hence, 3, 4 lies in above interval.

(B) We have to solve for x, y

$$\frac{(1+i)x - 2i}{3+i} + \frac{(2-3i)y + i}{3-i} = i$$

$$\Rightarrow \frac{(1+i)(3-i)x - 2i(3-i) + (2-3i)(3+i)y + i(3+1)}{9-i^2} = i$$

$$\Rightarrow (4 + 2i)x - 6i - 2 + (9 - 7i)y + 3i - 1 = 10i$$

$$\Rightarrow (4x + 9y - 3) + i(2x - 7y - 3) = 0 + 10i$$

Comparing the real and imaginary parts

$$4x + 9y - 3 = 0 \quad \dots(i)$$

$$2x - 7y - 13 = 0 \quad \dots(ii)$$

(i) - 2 × (ii) gives,

$$9y + 14y - 3 + 26 = 0$$

$$\Rightarrow 23y = -23 \Rightarrow y = -1$$

Putting $y = -1$ in (i), we get

$$4x - 9 - 3 = 0 \Rightarrow x = 3$$

$$(C) \left(\frac{1+i}{1-i} \right)^n = \left(\frac{1+i^2+2i}{1+1} \right) = i^n$$

Hence $n = 0, 4, 8, 6$

(D) Greatest and least absolute values of $z + 1$ are 1 and 6.

$$\therefore |z + 4| \leq 3$$

6. For the quadratic equation $x^2 - (k-3)x + k = 0$, match the condition in column I with the corresponding values of k in column II.

Column-I

- (A) Both the roots are positive
 (B) Both the roots are negative
 (C) Both the roots are real
 (D) One root is less than -1 and other is greater than 1

Column-II

- (p) ϕ
 (q) $(-\infty, 1) \cup (9, \infty)$
 (r) $[10, \infty)$
 (s) $(0, 1]$
 (t) $[125, 1250]$

Sol. Answer A(r, t), B(s), C(q, r, s, t), D(p)

$$x^2 - (k-3)x + k = 0$$

For roots to be real

$$(k-3)^2 - 4k \geq 0$$

$$\Rightarrow k^2 - 6k + 9 - 4k \geq 0 \Rightarrow k^2 - 10k + 9 \geq 0 \Rightarrow (k-1)(k-9) \geq 0$$

$$\Rightarrow k \in (-\infty, 1] \cup [9, \infty) \quad \dots(i)$$

(A) For both roots to be positive,

$$f(0) > 0 \text{ and } \frac{k-3}{2} > 0$$

$$k > 0 \quad \dots(ii)$$

$$\text{and } k > 3 \quad \dots(iii)$$

From (i), (ii) and (iii)

$$k \in [9, \infty)$$

(B) For both roots to be negative

$$D \geq 0$$

$$k > 0, (k - 3) / 2 < 0, \Rightarrow k < 3, \Rightarrow k \in (0, 1]$$

(C) For both roots to be real $k \in (-\infty, 1] \cup [9, \infty)$

(D) $f(-1) < 0, f(1) < 0$

$$1 + (k - 3) + k < 0 \text{ also } 1 - (k - 3) + k < 0$$

$$\Rightarrow 2k - 2 < 0 \Rightarrow k < 1, 4 < 0$$

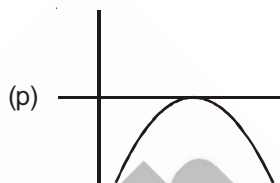
No such value is possible

7. Let $y = f(x) = ax^2 + bx + c, a \neq 0$ be a given quadratic expression. Then for real values of x , match the appropriate graph of $y = f(x)$ indicated in the Column-II from the conditions given in Column-I.

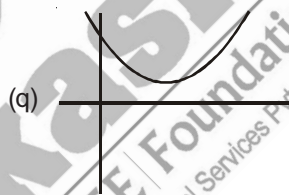
Column-I

Column-II

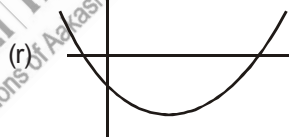
(A) $f(x) > 0, a > 0, c > 0$



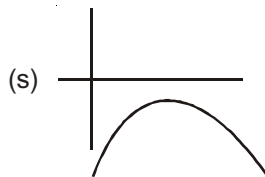
(B) $D > 0, c < 0$ and $ab < 0$



(C) $D < 0, a < 0$



(D) $D = 0, a < 0, b > 0$



Sol. Answer A(q), B(r), C(s), D(p)

(A) $f(x) > 0, \forall x \in \mathbb{R}, f(0) = c > 0$

Parabola is upward $\Rightarrow a > 0$.

(B) Roots are real and distinct $\Rightarrow D > 0$

$$f(0) < 0 \Rightarrow c < 0$$

One root is positive and one negative $\Rightarrow ab < 0$.

(C) Roots are imaginary $\Rightarrow D < 0$

Parabola is downward $\Rightarrow a < 0$.

(D) Parabola touches x-axis $\Rightarrow D = 0$

Parabola is downward $\Rightarrow a < 0$

Both roots are positive, sum $= -\frac{b}{a} > 0 \Rightarrow b > 0$

8. Match the following

Column-I

Column-II

(A) If the roots of the equation

$$x^2 - 2ax + a^2 + a - 3 = 0 \text{ are real}$$

and less than 3, then a belongs to

(p) a does not exist

(B) If the roots of the equation

$$x^2 - 2ax + a^2 + a - 3 = 0 \text{ are real}$$

and greater than 3, then a belongs to

(q) $\left(\frac{3}{2}, 3\right)$

(C) If exactly one root of the above equation

lies in the interval $(1, 3)$, then a belongs to

(r) $(-1, 2) \cup (2, 3)$

(D) If the roots of the above equation are such

that one root is greater than 3 and other smaller than 1, then a belongs to

(s) $(-\infty, 2)$

Sol. Answer A(s), B(p), C(r), D(p)

(A) For real roots, $D \geq 0$

$$\Rightarrow (-2a)^2 - 4(a^2 + a - 3) \geq 0$$

$$\Rightarrow 4[a^2 - a^2 - a + 3] \geq 0 \Rightarrow a \leq 3 \quad \dots(i)$$

$$f(3) > 0 \Rightarrow 9 - 6a + a^2 + a - 3 > 0$$

$$\Rightarrow a^2 - 5a + 6 > 0$$

$$\Rightarrow a \in (-\infty, 2) \cup (3, \infty) \quad \dots(ii)$$

$$-\frac{b}{2a} < 3 \Rightarrow \frac{2a}{2} < 3 \Rightarrow a < 3 \quad \dots(iii)$$

from (i), (ii) and (iii) $a \in (-\infty, 2)$

(B) $D \geq 0 \Rightarrow a \leq 3 \quad \dots(i)$

$$f(3) > 0 \Rightarrow a \in (-\infty, 2) \cup (3, \infty) \quad \dots(ii)$$

For greater than 3,

$$-\frac{b}{2a} > 3 \Rightarrow a > 3 \dots \text{(iii)}$$

From (i), (ii) and (iii), $a \in \phi$.

$$\text{(C) } D > 0 \Rightarrow a < 3 \dots \text{(i)}$$

$$f(1) f(3) < 0$$

$$(1 - 2a + a^2 + a - 3)(9 - 6a + a^2 + a - 3) < 0$$

$$\Rightarrow (a^2 - a - 2)(a^2 - 5a + 6) < 0 \Rightarrow (a - 2)^2(a + 1)(a - 3) < 0$$

$$\Rightarrow a \in (-1, 3) - \{2\} \dots \text{(ii)}$$

From (i) and (ii)

$$a \in (-1, 2) \cup (2, 3)$$

$$\text{(D) } D \geq 0 \Rightarrow a \leq 3 \dots \text{(i)}$$

$$f(1) < 0 \Rightarrow (1 - 2a + a^2 + a - 3) < 0$$

$$\Rightarrow (a^2 - a - 2) < 0$$

$$\Rightarrow -1 < a < 2 \dots \text{(ii)}$$

$$f(3) < 0 \Rightarrow (9 - 6a + a^2 + a - 3) < 0$$

$$\Rightarrow a^2 - 5a + 6 < 0$$

$$\Rightarrow 2 < a < 3 \dots \text{(iii)}$$

From (i), (ii) and (iii)

$$a \in \phi.$$

9. Let $f(x) = |x - 1| + |x - 2| + |x - 3|$, match the column I for the value of k in column II.

Column I

- (A) $f(x) = k$ has no solution
 (B) $f(x) = k$ has only one solution
 (C) $f(x) = k$ has two solution of same sign
 (D) $f(x) = k$ has two solution of opposite sign

Column II

- (p) 1
 (q) 2
 (r) 4
 (s) 5
 (t) 8

Sol. Answer A(p), B(q), C(r, s), D(t)

We have,

$$f(x) = |x - 1| + |x - 2| + |x - 3| = k$$

when $x < 1$, then

$$f(x) = -3x + 6 = k$$

$$\Rightarrow x = 2 - \frac{k}{3}$$

$$\text{Now, } 2 - \frac{k}{3} < 1$$

$$\Rightarrow k > 3 \quad \dots(i)$$

For $1 \leq x < 2$

$$-x = k - 4$$

$$x = 4 - k$$

$$\Rightarrow 2 < k \leq 3 \quad \dots(ii)$$

For $2 \leq x < 3$

$$x = k, \quad 2 \leq k < 3$$

For $x \geq 3$

$$3x - 6 = k$$

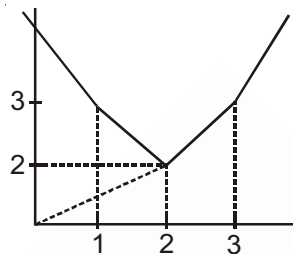
$$x = \frac{k+6}{3}$$

$$\Rightarrow \frac{k+6}{3} \geq 3$$

$$k \geq 3$$

Clearly

- (A) for $k < 2$, there is no solution
 (B) for $k = 2$, there is only one solution
 (C) for $2 < k < 6$, there are two solution of same sign
 (D) for $k > 6$, there are two solution of opposite sign



10. For given equation $x^2 - ax + b = 0$, match conditions in column I with possible values in column II

Column-I

- (A) If roots differ by unity then a^2 is equal to
 (B) If roots differ by unity then $1 + a^2$ is equal to
 (C) If one of the root be twice the other then $2a^2$ is equal to
 (D) If the sum of roots of the equation equal to the sum of squares of their reciprocal then a^2 is equal to

Column-II

- (p) $b(ab + 2)$
 (q) $1 + 4b$
 (r) $2(1 + 2b)$
 (s) $9b$

Sol. Answer A(q), B(r), C(s), D(p)

$$(A) \rightarrow (q), (B) \rightarrow (r)$$

$$x^2 - ax + b = 0$$

$$\alpha + \beta = a, \quad \alpha\beta = b$$

$$|\alpha - \beta| = 1 = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{a^2 - 4b}$$

$$\Rightarrow a^2 - 4b = 1$$

$$\Rightarrow a^2 = 1 + 4b$$

$$\Rightarrow a^2 + 1 = 2 + 4b = 2(1 + 2b)$$

$$(C) \quad \alpha + 2\alpha = a \Rightarrow \alpha = \frac{a}{3}$$

$$2\alpha \cdot \alpha = b \Rightarrow \alpha^2 = \frac{b}{2}$$

$$\left(\frac{a}{3}\right)^2 = \frac{b}{2} \Rightarrow a^2 = \frac{9b}{2} \Rightarrow 2a^2 = 9b$$

$$(D) \quad \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

$$(\alpha + \beta)(\alpha^2\beta^2) = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$\Rightarrow a(b)^2 = a^2 - 2b$$

$$\Rightarrow a^2 = ab^2 + 2b = b(ab + 2)$$

11. If α, β, γ be the roots of the equation $x(1 + x^2) + x^2(6 + x) + 2 = 0$, then match the entries of column-I with those of column-II.

Column I

- (A) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ is less than or equal to
- (B) $\alpha^2 + \beta^2 + \gamma^2$ equals
- (C) $(\alpha^{-1} + \beta^{-1} + \gamma^{-1}) - (\alpha + \beta + \gamma)$ is less than or equal to
- (D) $[\alpha^{-1} + \beta^{-1} + \gamma^{-1}]$ equals where $[\cdot]$ denotes the greatest integer less than or equal to

Column II

- (p) 8
- (q) $-\frac{1}{2}$
- (r) -1
- (s) 3
- (t) $\frac{5}{2}$

Sol. Answer A(p, q, s, t), B(p), C(p, s, t), D(r)

α, β, γ be the roots of the equation

$$x(1 + x^2) + x^2(6 + x) + 2 = 0$$

$$\Rightarrow x^3 + x + 6x^2 + x^3 + 2 = 0$$

$$\Rightarrow 2x^3 + 6x^2 + x + 2 = 0$$

So, $\alpha + \beta + \gamma = -3$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{1}{2}$$

$$\alpha\beta\gamma = -1$$

Now,

$$(A) \alpha^{-1} + \beta^{-1} + \gamma^{-1} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = -\frac{1}{2}$$

$$(B) \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ = 9 - 2 \cdot \frac{1}{2} = 8$$

$$(C) (\alpha^{-1} + \beta^{-1} + \gamma^{-1}) - (\alpha + \beta + \gamma)$$

$$= \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} - (\alpha + \beta + \gamma)$$

$$= -\frac{1}{2} + 3 = \frac{5}{2}$$

$$(D) [\alpha^{-1} + \beta^{-1} + \gamma^{-1}] = \left[-\frac{1}{2}\right] = -1$$

12. Match the statements in **Column-I** with those in **Column-II**.

[IIT-JEE 2010]

[Note: Here z takes values in the complex plane and $\text{Im } z$ and $\text{Re } z$ denote, respectively, the imaginary part and the real part of z .]

Column I

- (A) The set of points z satisfying

$$|z - i| |z| = |z + i| |z| \text{ is contained in or equal to}$$

- (B) The set of points z satisfying $|z + 4| + |z - 4| = 10$ is contained in or equal to

- (C) If $|w| = 2$, then the set of points $z = w - \frac{1}{w}$ is contained in or equal to

- (D) If $|w| = 1$, then the set of points $z = w + \frac{1}{w}$ is contained in or equal to

Column II

- (p) An ellipse with eccentricity $\frac{4}{5}$

- (q) The set of points z satisfying $\text{Im } z = 0$

- (r) The set of points z satisfying $|\text{Im } z| \leq 1$

- (s) The set of points z satisfying $|\text{Re } z| \leq 2$

- (t) The set of points z satisfying $|z| \leq 3$

Sol. Answer A(q, r), B(p), C(p, s, t), D(q, r, s, t)

- (A) z is equidistant from the points $i|z|$ and $-i|z|$, whose perpendicular bisector is $\text{Im}(z) = 0$.
- (B) Sum of distance of z from $(4, 0)$ and $(-4, 0)$ is a constant 10, hence locus of z is ellipse with semi-major axis 5 and focus at $(\pm 4, 0)$.

$$ae = 4$$

$$\therefore e = \frac{4}{5}$$

$$(C) |z| \leq |w| + \left| \frac{1}{w} \right| = \frac{5}{2} < 3$$

$$(D) |z| \leq |w| + \left| \frac{1}{w} \right| = 2$$

$$\therefore \text{Re } z \leq |z| \leq 2$$

13. Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i \sin\left(\frac{2k\pi}{10}\right)$; $k = 1, 2, \dots, 9$.

[JEE(Advanced)-2014]

Column I

Column II

(A) For each z_k there exists a z_j such that $z_k \cdot z_j = 1$ (p) True

(B) There exists a $k \in \{1, 2, \dots, 9\}$ such that $z_1 \cdot z = z_k$ (q) False

has no solution z in the set of complex numbers

(C) $\frac{|1 - z_1| |1 - z_2| \dots |1 - z_9|}{10}$ equals (r) 1

(D) $1 - \sum_{k=1}^9 \cos\left(\frac{2k\pi}{10}\right)$ equals (s) 2

Sol. Answer A(p), B(q), C(r), D(s)

$$z_k = \left(\cos \frac{2\pi}{10} + i \sin \frac{2\pi}{10} \right)^k = e^{\frac{i2\pi k}{10}}$$

(A) $z_k \cdot z_j = 1$

$$\Rightarrow e^{\frac{2\pi}{10}(k+j)} = 1$$

$$\Rightarrow \cos\left(\frac{2\pi(k+j)}{10}\right) = 1 \text{ and } \sin\left(\frac{2\pi(k+j)}{10}\right) = 0$$

$$\Rightarrow k + j = 10n \text{ and } k + j = 5m; \text{ so true}$$

$$(B) \quad z = \frac{z_k}{z_1} = e^{i\frac{2\pi}{10}(k-1)}$$

So, if $k \in \{1, 2, \dots, 9\}$, z has solution; i.e., false

$$(B) \rightarrow 2$$

$$(C) \quad z^{10} - 1 = 0$$

$$\Rightarrow (z - z_1)(z - z_2) \dots (z - z_9) = 1 + z + z^2 + \dots + z^9$$

$$\text{So, } |1 - z_1||1 - z_2| \dots |1 - z_9| = 10$$

$$\text{i.e., } (C) \rightarrow 3$$

$$(D) \quad 1 - \left\{ \cos \frac{2\pi}{10} + \cos \frac{4\pi}{10} + \dots + \cos \frac{18\pi}{10} \right\}$$

$$= 1 - \cos \left(\frac{\frac{4\pi}{10} + 8 \times \frac{2\pi}{10}}{2} \right) = 2$$

SECTION - F

Integer Answer Type Questions

1. P is a point satisfying $\arg z = \pi/4$, such that sum of its distance from two given points $(0, 1)$ and $(0, 2)$ is minimum, then P must be $\frac{k}{3}(1 + i)$. Then numerical value of k is _____.

Sol. Answer (2)

$PA + PB$ will be minimum

where A and A' are mirror image

A', P, B are collinear

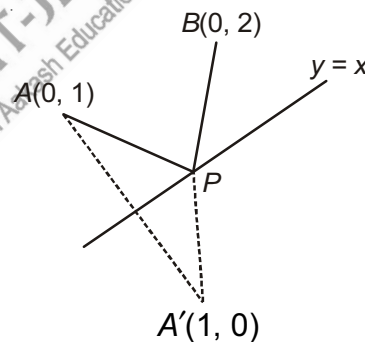
Equation of line $A'B$: $2x + y = 2$

Solve $A'B$ with $y = x$

$$\therefore x = \frac{2}{3}, y = \frac{2}{3}$$

$$\therefore P = \frac{2}{3}(1 + i)$$

$$\therefore k = 2$$



2. If $\sum_{r=1}^{2002+(2k-1)} \cos\left(\frac{2r\pi}{7}\right) + i \sin\left(\frac{2r\pi}{7}\right) = 0$, then the non-negative integral values of k less than 10 may be _____.

Sol. Answer (4)

$$\sum_{r=1}^{2002+(2k-1)} \cos\left(\frac{2r\pi}{7}\right) + i \sin\left(\frac{2r\pi}{7}\right) = 0$$

It is possible only when $2002 + (2k - 1)$ should be multiple of 7.

3. If $z = x + iy$ and roots of $z\bar{z}^3 + \bar{z}z^3 = 30$ are the vertices of a rectangle and z_0 is centre of rectangle. Let d be distance of z_0 from the point on circle $|z - 3| \leq 2$ then maximum value of d is _____.

Sol. Answer (5)

We have

$$\Rightarrow z\bar{z}^3 + \bar{z}z^3 = 30$$

$$\Rightarrow z\bar{z}(\bar{z}^2 + z^2) = 30$$

$$\Rightarrow (x^2 + y^2)((x^2 - y^2) - 2i xy + x^2 - y^2 + 2i xy) = 0$$

$$\Rightarrow (x^2 + y^2)(x^2 - y^2) = 15 = (2^2 + 1)(2^2 - 1)$$

Which suggests the possible values of x and y are

$$x = 2, y = 1$$

or

$$x = -2, y = -1$$

or

$$x = -2, y = 1$$

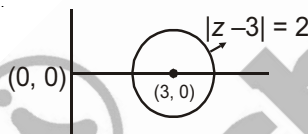
or

$$x = 2, y = -1$$

$\therefore$ Centre of rectangle is $(0, 0)$

Now,

Maximum distance of $(0, 0)$ from the circle is 5 and minimum is 1.



4. If the complex number $A(z_1)$, $B(z_2)$ and origin forms an isosceles triangle such that $\angle AOB = \frac{2\pi}{3}$, then

$$\frac{z_1^2 + z_2^2 + 4z_1z_2}{z_1z_2} \text{ equals } \underline{\hspace{2cm}}.$$

Sol. Answer (3)

If origin z_1 , z_2 forms an isosceles triangle then $z_1^2 + z_2^2 + z_1z_2 = 0$

$$\text{Hence } \frac{z_1^2 + z_2^2 + 4z_1z_2}{z_1z_2} = \frac{3z_1z_2}{z_1z_2} = 3$$

5. The area of the triangle formed by three points $\sqrt{3} + i$, $-1 + \sqrt{3}i$ and $(\sqrt{3} - 1) + (\sqrt{3} + 1)i$ is _____.

Sol. Answer (2)

Let $\sqrt{3} + i = z$, hence other two vertices are iz and $z + iz$

$$\text{So, area of such triangle is } = \frac{1}{2}|z|^2 = \frac{1}{2} \cdot (2)^2 = \frac{1}{2} \times 4 = 2$$

6. The number of value(s) of k , for which both the roots of the equation $x^2 - 6kx + 9(k^2 - k + 1) = 0$ are real, distinct and have values atmost 3 is _____.

Sol. Answer (0)

$$x^2 - 6kx + 9(k^2 - k + 1) = 0$$

For real and distinct roots

$$D > 0$$

$$\Rightarrow k^2 - (k^2 - k + 1) > 0$$

$$\Rightarrow k - 1 > 0 \quad \dots(i)$$

$$f(3) \geq 0$$

$$9 - 18k + 9(k^2 - k + 1) \geq 0$$

$$\Rightarrow 1 - 2k + k^2 - k + 1 \geq 0$$

$$\Rightarrow k^2 - 3k + 2 \geq 0$$

$$k \in (-\infty, 1] \cup [2, \infty) \quad \dots(ii)$$

$$\text{Also } \frac{\text{sum of roots}}{2} < 3$$

$$\Rightarrow 3k < 3$$

$$k < 1 \quad \dots(iii)$$

From (i), (ii), (iii)

We observe that there does not exist any real value of k .

7. The possible greatest integral value of a for which the expression $\frac{ax^2 + 3x + 4}{x^2 + 2x + 2}$ is less than 5 for all real x is _____.

Sol. Answer (2)

We have

$$y = \frac{ax^2 + 3x + 4}{x^2 + 2x + 2}$$

$$\text{As } \frac{ax^2 + 3x + 4}{x^2 + 2x + 2} < 5$$

$$\Rightarrow \frac{ax^2 + 3x + 4 - 5x^2 - 10x - 10}{x^2 + 2x + 2} < 0$$

$$\Rightarrow (a - 5)x^2 - 7x - 6 < 0 \quad (\text{as } x^2 + 2x + 2 > 0, \forall x \in \mathbb{R})$$

It is satisfied for all x if

$$a - 5 < 0, 49 + 24(a - 5) < 0$$

$$a < \frac{71}{24}$$

$$\Rightarrow a < 3$$

The possible greatest integral value of a is 2.

8. Let $f(x) = ax^2 + bx + c$, where a, b, c are real numbers. If the numbers $2a, a + b$ and c are all integers, then the number of integral values between 1 and 5 that $f(x)$ can take is _____.

Sol. Answer (5)

Firstly, let $f(x) = ax^2 + bx + c$; $a, b, c \in \mathbb{R}$ be an integer whenever x is an integer

$$\Rightarrow f(0), f(1), f(-1) \text{ are integers}$$

$$\Rightarrow c, a + b + c, a - b + c \text{ are integers}$$

$$\Rightarrow c, a + b + c - c, a - b + c - c \text{ are integers}$$

$$\Rightarrow c, a + b, a - b \text{ are integers}$$

$\Rightarrow c, a + b, a + b + a - b$ are integers

$\Rightarrow c, a + b, 2a$ are integers

Secondly let $2a, a + b$ and c be integers. Let x be an integer.

$$\text{Then } f(x) = ax^2 + bx + c = 2a\left(\frac{x(x-1)}{2}\right) + (a+b)x + c$$

Since x is an integer $\Rightarrow x(x-1)$ is an even integer.

$$\Rightarrow 2a\left(\frac{x(x-1)}{2}\right) + (a+b)x + c \text{ is an integer as } 2a, a + b, c \text{ are integers.}$$

$$\Rightarrow f(x) \text{ is an integer for all integer } x.$$

9. The smallest value of k , for which both the roots of the equation $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is _____.

[IIT-JEE 2009]

Sol. Answer (2)

$$x^2 - 8kx + 16(k^2 - k + 1) = 0$$

Roots are real and distinct

$$\text{Let } f(x) = x^2 - 8kx + 16(k^2 - k + 1)$$

$$\Rightarrow D > 0 \Rightarrow x > 1$$

$$\text{Let } f(4) > 0 \Rightarrow k \in (-\infty, 1] \cup [2, \infty)$$

$\therefore$ Least value of k can be 2.

10. If z is any complex number satisfying $|z - 3 - 2i| \leq 2$, then the minimum value of $|2z - 6 + 5i|$ is _____.

[IIT-JEE 2011]

Sol. Answer (5)

From the given condition,

$$|z - 3 - 2i| \leq 2$$

$$\Rightarrow |2z - 6 - 4i| \leq 4$$

$$\Rightarrow 4 \geq |(2z - 6 + 5i) - 9i|$$

$$\geq ||9i| - |2z - 6 + 5i||$$

$$\Rightarrow 4 \geq 9 - |2z - 6 + 5i|$$

$$\Rightarrow |2z - 6 + 5i| \geq 5$$

Minimum value of $|2z - 6 + 5i|$ is = 5

11. The value of $6 + \log_3 \left(\frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \sqrt{4 - \frac{1}{3\sqrt{2}} \dots}}} \right)$ is _____.

[IIT-JEE 2012]

Sol. Answer (4)

$$\text{As, } y = \frac{1}{3\sqrt{2}} \sqrt{4 - y}$$

$$\Rightarrow y = \frac{4}{9} \text{ (as } y > 0)$$

$$\text{So, } 6 + \log_3 \left(\frac{4}{9} \right) = 4$$

12. For any integer k , let $\alpha_k = \cos\left(\frac{k\pi}{7}\right) + i \sin\left(\frac{k\pi}{7}\right)$, where $i = \sqrt{-1}$. The value of the expression

$$\frac{\sum_{k=1}^{12} |\alpha_{k+1} - \alpha_k|}{\sum_{k=1}^3 |\alpha_{4k-1} - \alpha_{4k-2}|} \text{ is}$$

[JEE(Advanced)-2015]

Sol. Answer (4)

$$S = \frac{\sum_{k=1}^{12} |\alpha_{k+1} - \alpha_k|}{\sum_{k=1}^3 |\alpha_{4k-1} - \alpha_{4k-2}|}$$

$|\alpha_{r+1} - \alpha_r|$ = side length of the polygon having 14 sides = a

$$\text{So, } S = \frac{12a}{3a} = 4$$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1: If $\alpha = e^{\frac{i2\pi}{7}}$ and $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k$, then $f(x) + f(\alpha x) + f(\alpha^2 x) + \dots + f(\alpha^6 x) = 5(A_0 + A_7 x^7 + A_{14} x^{14})$.

STATEMENT-2: The least argument of a complex number z , satisfying $|z - 16i| \leq 8$ is $\frac{\pi}{3}$.

STATEMENT-3: Maximum argument of the complex number satisfying $|z - 16i| \leq 8$ is $\frac{2\pi}{3}$.

(1) F T T

(2) F T F

(3) F F F

(4) T F T

Sol. Answer (1)

Statement-1:

$$f(x) + f(\alpha x) + f(\alpha^2 x) + \dots + f(\alpha^6 x)$$

$$= 7A_0 + \sum_{k=1}^{20} A_k x^k (1 + \alpha^k + \dots + \alpha^{6k}) \text{ but when } k \neq 7$$

and $k \neq 14$, then $1 + \alpha^k + \alpha^{2k} + \dots + \alpha^{6k} = 0$

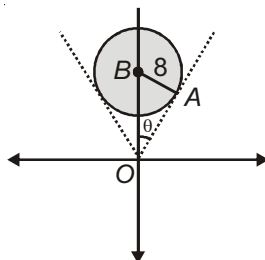
$$f(x) + f(\alpha x) + \dots + f(\alpha^6 x) = 7(A_0 + A_7 x^7 + A_{14} x^{14})$$

Statement-2:

Clearly $AB = 8$ $OB = 16$

$$\sin \theta = \frac{AB}{OB} = \frac{8}{16} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

So, the argument is $\frac{\pi}{3}$



Statement-3:

It can be seen from the figure that the maximum value of argument is $\frac{2\pi}{3}$.

2. STATEMENT-1 : Two regular polygons are inscribed in the same circle. The first polygon has 1982 sides and second has 2973 sides. If the polygons have a common vertex, then the number of vertex common to both of them is 991.

STATEMENT-2 : The total number of complex numbers z satisfying $|z - 1| = |z + 1| = |z|$ is one.

STATEMENT-3 : The locus represented by $|2011z + 1| = 2011|z + 1|$ is a straight line

(1) T T T

(2) T F T

(3) F T T

(4) F F F

Sol. Answer (2)

Statement-1 : The number of common vertices is equal to the number of common roots of $z^{1982} - 1 = 0$ and $z^{2973} - 1 = 0$, which is H.C.F. of 1982, 2973, i.e., 991. Here consider both vertex has (1, 0) as one vertex.

Statement-2 : Since two parallel lines never meet so no solution

Statement-3 : Clearly the locus is a straight line.

3. STATEMENT-1 : If x is real and $y = \frac{x^2 - x + 3}{x + 2}$, then $y \in (-\infty, \infty) - (-11, 1)$.

STATEMENT-2 : If $[]$ represents the greatest integer function and $f(x) = x - [x]$ then number of real roots of the equation $f(x) + f\left(\frac{1}{x}\right) = 1$ are infinite.

STATEMENT-3 : If the difference of the roots of the equation $x^2 + hx + 7 = 0$ is 6, then possible value(s) of h are -8 and 8.

(1) T T T

(2) T T F

(3) T F F

(4) F F F

Sol. Answer (1)

Let $y = \frac{x^2 - x + 3}{x + 2}$ so that $x^2 - (y + 1)x + 3 - 2y = 0$

For real values of x

$$(y + 1)^2 - 4(3 - 2y) \geq 0$$

$$\Rightarrow (y + 11)(y - 1) \geq 0$$

$$\Rightarrow y \geq 1 \text{ or } y \leq -11$$

$$\Rightarrow y \text{ lies in } R - (-11, 1)$$

$$\Rightarrow \text{Statement-1 is true}$$

The roots of $x^2 - 4|x| + 3 = 0$ are

1, 3, -1, -3 and hence their sum is zero

Statement-3 is also true

4. STATEMENT-1 : The solution of the equations $(3|x| - 3)^2 = |x| + 7$ which belongs to the domain of the function $y = \sqrt{x(x-3)}$ are given by $-\frac{1}{9}, -2$.

STATEMENT-2 : For all x , $x^2 + 2ax + 10 - 3a > 0$, then $a \in [-5, 2]$.

STATEMENT-3 : If $x^2 - 2x + \sin^2\alpha = 0$ then $x \in [0, 2]$.

(1) F F T

(2) T T F

(3) T F F

(4) T T T

Sol. Answer (4)

Clearly statement 1 and 2 are false

We have $(x - 1)^2 = \cos^2 \alpha$

$$\Rightarrow x = 1 \pm \cos \alpha$$

$$\Rightarrow x \in [0, 2]$$

$\Rightarrow$ statement-3 is true

SECTION - H

Aakash Challengers Questions

1. Let z_1 and z_2 be two given complex numbers such that $\frac{z_1}{z_2} + \frac{z_2}{z_1} = 1$, and $|z_1| = 3$, then $|z_1 - z_2|^2$ is equal to

Sol. $\frac{z_1}{z_2} + \frac{z_2}{z_1} = 1$

$$z_1^2 + z_2^2 = z_1 z_2$$

$$\Rightarrow z_1^2 + z_2^2 + 0^2 = z_1 z_2 + 0z_1 + 0z_2$$

$\Rightarrow 0, z_1, z_2$ form an equilateral triangle

$$\text{Hence, } |z_1| = |z_2| = |z_1 - z_2| = 3$$

$$\Rightarrow |z_1 - z_2|^2 = 9$$

2. The locus of the centre of a circle which touches the given circles $|z - z_1| = |3 + 4i|$ and $|z - z_2| = |1 + i\sqrt{3}|$ is a hyperbola, then the length of its transverse axis is

Sol. $|z - z_1| = 5$

$$|z - z_2| = 2$$

$$|z - z_1| = r + 5$$

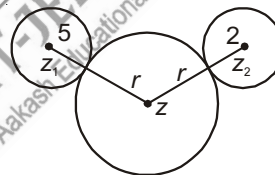
$$|z - z_2| = r + 2$$

$$|z - z_1| - |z - z_2| = 3 \quad \text{which is a constant}$$

$\rightarrow$ locus is hyperbola and we know

$$PF_1 - PF_2 = 2a = \text{length of transverse axis.}$$

$$\text{length of transverse axis} = 3$$



3. If $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$, the $\frac{6}{\pi} \text{amp} \left(\frac{z_1}{z_2} \right)$ is equal to

Sol. $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$ (Given)

$$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2 \operatorname{Re}(z_1 \bar{z}_2) = |z_1|^2 + |z_2|^2 \dots \dots \dots (\text{Relation})$$

$$\Rightarrow \operatorname{Re}(z_1 \bar{z}_2) = 0$$

$$\operatorname{amp} \left(\frac{z_1}{z_2} \right) = \frac{\pi}{2}$$

$$\therefore \frac{6}{\pi} \times \operatorname{amp} \left(\frac{z_1}{z_2} \right) = \frac{6}{\pi} \times \left(\pm \frac{\pi}{2} \right) = \pm 3$$

4. For every real $a \geq 0$, find all the complex numbers z that satisfy the equation $2|z| - 4az + 1 + ia = 0$.

Sol. Let $z = x + iy$

So, then equation becomes,

$$2\sqrt{x^2 + y^2} - 4a(x + iy) + 1 + ia = 0 \quad \dots(i)$$

Comparing real and imaginary part,

$$2\sqrt{x^2 + y^2} - 4ax + 1 = 0 \quad \dots(ii)$$

$$-4ay + a = 0 \quad \dots(iii)$$

$$(3) \Rightarrow a(-4y + 1) = 0 \Rightarrow a = 0 \text{ or } y = \frac{1}{4}$$

But a cannot be zero. Hence $y = \frac{1}{4}$ gives

$$2\sqrt{x^2 + \frac{1}{16}} = 4ax - 1$$

$$4\left(x^2 + \frac{1}{16}\right) = (4ax - 1)^2$$

$$4x^2 + \frac{1}{4} = 16a^2x^2 + 1 - 8ax$$

$$(4 - 16a^2)x^2 + 8ax - \frac{3}{4} = 0$$

$$= \frac{-8a \pm \sqrt{16a^2 + 12}}{8(1 - 4a^2)}$$

$$= \frac{4a \pm 2\sqrt{4a^2 + 3}}{4(4a^2 - 1)}$$

Now, we can observe that

if $0 \leq a \leq \frac{1}{2}$, there is no solution

if $a > \frac{1}{2}$, solution is $z = x + iy$

$$= \frac{4a + \sqrt{4a^2 + 3}}{4(4a^2 - 1)} + i \cdot \frac{1}{4}$$

5. Let z_1, z_2 be complex numbers with $|z_1| = |z_2| = 1$. Prove that $|z_1 + 1| + |z_2 + 1| + |z_1 z_2 + 1| \geq 2$.

Sol. We have,

$$|z_1 + 1| + |z_2 + 1| + |z_1 z_2 + 1| \geq |z_1 + 1| + |z_1 z_2 + 1 - (z_2 + 1)|$$

$$= |z_1 + 1| + |z_1 z_2 - z_2| \geq |z_1 + 1| + |z_2| |z_1 - 1|$$

$$= |z_1 + 1| + |z_1 - 1| \geq |z_1 + 1 + z_1 - 1| = 2|z_1| = 2$$

$$\therefore |z_1 + 1| + |z_2 + 1| + |z_1 z_2 + 1| \geq 2$$

6. If one root of the quadratic equation $ax^2 + bx + c = 0$ is equal to the n^{th} power of the other, then show that

$$(ac^n)^{\frac{1}{n+1}} + (a^n c)^{\frac{1}{n+1}} + b = 0.$$

Sol. Let α and α^n be the roots of the given quadratic equation $ax^2 + bx + c = 0$

$$\Rightarrow \alpha + \alpha^n = -\frac{b}{a} \text{ and } \alpha \cdot \alpha^n = \frac{c}{a} = \alpha^{n+1} \Rightarrow \alpha = \left(\frac{c}{a}\right)^{\frac{1}{n+1}}$$

$$\Rightarrow \left(\frac{c}{a}\right)^{\frac{1}{n+1}} + \left(\frac{c}{a}\right)^{\frac{n}{n+1}} = -\frac{b}{a}$$

$$\Rightarrow \frac{1}{c^{\frac{1}{n+1}}} a^{1-\frac{1}{n+1}} + \frac{n}{c^{\frac{n}{n+1}}} a^{1-\frac{n}{n+1}} + b = 0$$

$$\Rightarrow \frac{1}{c^{\frac{1}{n+1}}} a^{\frac{n}{n+1}} + \frac{n}{c^{\frac{n}{n+1}}} a^{\frac{1}{n+1}} + b = 0$$

$$\Rightarrow (a^n c)^{\frac{1}{n+1}} + (ac^n)^{\frac{1}{n+1}} + b = 0$$

7. Let α and β be the roots of the equation $x^2 - px + q = 0$ and $V_n = \alpha^n + \beta^n$. Show that $V_{n+1} = pV_n - qV_{n-1}$. Find V_5 .

Sol. We have $\alpha + \beta = p$ and $\alpha\beta = q$.

$$\text{Now, } pV_n - qV_{n-1} = (\alpha + \beta)(\alpha^n + \beta^n) - \alpha\beta(\alpha^{n-1} + \beta^{n-1})$$

$$= \alpha^{n+1} + \alpha\beta^n + \alpha^n\beta + \beta^{n+1} - \alpha^n\beta - \alpha\beta^n$$

$$= \alpha^{n+1} + \beta^{n+1} = V_{n+1}$$

$$\text{Also } V_5 = \alpha^5 + \beta^5 = pV_4 - qV_3 = p[pV_3 - qV_2] - qV_3$$

$$= (p^2 - q)(\alpha^3 + \beta^3) - pqV_2$$

$$= (p^2 - q)[(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)] - pq[(\alpha + \beta)^2 - 2\alpha\beta]$$

$$= (p^2 - q)[p^3 - 3pq] - pq[p^2 - 2q]$$

8. If p, q are roots of the quadratic equation $x^2 - 10rx - 11s = 0$ and r, s are roots of $x^2 - 10px - 11q = 0$, then find the value of $p + q + r + s$.

Sol. According to the question

$$p + q = 10r; pq = -11s$$

$$r + s = 10p; rs = -11q.$$

$$\text{On subtraction, } (p - r) + (q - s) = 10(r - p)$$

$$\Rightarrow (q - s) = 11(r - p) \quad \dots(i)$$

$$\text{Also, } p \text{ is a root of } x^2 - 10rx - 11s = 0$$

$$\Rightarrow p^2 - 10pr - 11s = 0$$

$$\text{Similarly } r^2 - 10pr - 11q = 0$$

$$\text{On subtraction,}$$

$$p^2 - r^2 = -11(q - s)$$

$$\Rightarrow (p - r)(p + r) = -11 \times 11(r - p)$$

$$\Rightarrow p + r = 121$$

$$\text{Now } p + q + r + s = 10(p + r) = 10 \times 121 = 1210.$$

9. Solve the equation $4^{x+1.5} + 9^{x+0.5} = 10 \times 6^x$.

Sol. The given equation can be written as

$$2^{2x+3} + 3^{2x+1} = 10 \times 2^x \times 3^x.$$

$$\Rightarrow 8 + 3 \cdot \left(\frac{3}{2}\right)^{2x} = 10 \cdot \left(\frac{3}{2}\right)^x$$

$$\Rightarrow 3 \left(\frac{3}{2}\right)^{2x} - 10 \left(\frac{3}{2}\right)^x + 8 = 0$$

$$\Rightarrow 3 \cdot \left(\frac{3}{2}\right)^{2x} - 6 \left(\frac{3}{2}\right)^x - 4 \left(\frac{3}{2}\right)^x + 8 = 0$$

$$\Rightarrow 3 \cdot \left(\frac{3}{2}\right)^x \left[\left(\frac{3}{2}\right)^x - 2 \right] - 4 \left[\left(\frac{3}{2}\right)^x - 2 \right] = 0$$

$$\Rightarrow \left[\left(\frac{3}{2}\right)^x - 2 \right] \left[3 \cdot \left(\frac{3}{2}\right)^x - 4 \right] = 0$$

$$\Rightarrow \text{Either } \left(\frac{3}{2}\right)^x = 2 \text{ or } \left(\frac{3}{2}\right)^x = \frac{4}{3}$$

$$\Rightarrow \text{When } \left(\frac{3}{2}\right)^x = 2, \text{ Taking logarithm of both sides we get } x(\log 3 - \log 2) = \log 2$$

$$\Rightarrow x = \frac{\log 2}{\log 3 - \log 2}$$

$$\text{Also when } \left(\frac{3}{2}\right)^x = \frac{4}{3}$$

$$\Rightarrow x(\log 3 - \log 2) = \log 4 - \log 3$$

$$\Rightarrow x = \frac{\log 4 - \log 3}{\log 3 - \log 2} \quad \therefore x = \frac{\log 2}{\log 3 - \log 2} \text{ or } \frac{\log 4 - \log 3}{\log 3 - \log 2}$$

10. The twice of the product of real roots of the equation $(2x + 3)^2 - 3|2x + 3| + 2 = 0$ is _____.

Sol. $|2x + 3|^2 - 3|2x + 3| + 2 = 0$

$$\Rightarrow (|2x + 3| - 2)(|2x + 3| - 1) = 0 \Rightarrow |2x + 3| = 2, |2x + 3| = 1$$

$$\Rightarrow 2x + 3 = \pm 2, 2x + 3 = \pm 1 \Rightarrow x = -\frac{5}{2}, -\frac{1}{2}, -2, -1$$

$$\text{Product} = \left(-\frac{5}{2}\right)\left(-\frac{1}{2}\right)(-2)(-1) = \frac{5}{2}$$

$$2(\text{Product}) = 5$$

11. The equation $ax^2 + bx + c = 0$ and $x^3 - 4x^2 + 8x - 8 = 0$ have two roots in common. Then $2b + c$ is equal to _____.

Sol. $x^3 - 4x^2 + 8x - 8 = 0 \Rightarrow (x-2)(x^2 - 2x + 4) = 0$

$\Rightarrow x = 2, x^2 - 2x + 4 = 0$ will give imaginary roots.

$\therefore x^2 - 2x + 4 = 0$ and $ax^2 + bx + c = 0$ will have both roots common.

$\Rightarrow \frac{a}{1} = \frac{b}{-2} = \frac{c}{4} = k \Rightarrow b = -2k, c = 4k \Rightarrow 2b + c = 0$

12. If $x = 3 + 3^{2/3} + 3^{1/3}$, then the value of the expression $x^3 - 9x^2 + 18x - 10$ is equal to _____.

Sol. $x = 3 + 3^{2/3} + 3^{1/3}$

$x - 3 = 3^{1/3}(1 + 3^{1/3})$

$\Rightarrow (x-3)^3 = 3(1 + 3^{1/3})^3$

$\Rightarrow x^3 + 27x - 9x^2 - 27 = 3(1 + 3 + 3(3^{1/3} + 3^{2/3})) \Rightarrow x^3 + 27x - 9x^2 - 27 = 3(4 + 3(x-3))$

$\Rightarrow x^3 - 9x^2 + 18x - 12 = 0 \Rightarrow x^3 - 9x^2 + 18x - 10 = 2$

13. Let n be an even positive integer such that $\frac{n}{2}$ is odd and let $\alpha_0, \alpha_1, \dots, \alpha_{n-1}$ be the complex roots of unity

of order n . Prove that $\prod_{k=0}^{n-1} (a + b\alpha_k^2) = (a^2 + b^2)^{n/2}$ for any complex numbers a and b .

Sol. If $b = 0$ result can be shown easily.

But if $b \neq 0$.

Let $n = 2(2m + 1)$

Let the complex number α such that $\alpha^2 = \frac{a}{b}$ and the polynomial.

$f(x) = x^n - 1 = (x - \alpha_0)(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1})$

We have,

$f\left(\frac{\alpha}{i}\right) = \left(\frac{1}{i}\right)^n \cdot (\alpha - i\alpha_0)(\alpha - i\alpha_1) \dots (\alpha - i\alpha_{n-1})$

and $f\left(\frac{\alpha}{-i}\right) = \left(-\frac{1}{i}\right)^n \cdot (\alpha + i\alpha_0)(\alpha + i\alpha_1) \dots (\alpha + i\alpha_{n-1})$

Hence,

$f\left(\frac{\alpha}{i}\right)f\left(\frac{\alpha}{-i}\right) = (\alpha^2 + \alpha_0^2)(\alpha^2 - \alpha_1^2) \dots (\alpha^2 - \alpha_{n-1}^2)$

Therefore $\prod_{k=0}^{n-1} (a + b\alpha_k^2) = b^n \prod_{k=0}^{n-1} \left(\frac{a}{b} + \alpha_k^2\right)$

$= b^n \prod_{k=0}^{n-1} (\alpha + \alpha_k^2)$

$$= b^n f\left(\frac{\alpha}{i}\right) \cdot f\left(-\frac{\alpha}{i}\right) = b^n \cdot [(\alpha^2)^{2m+1} + 1]^2$$

$$= b^n \left[\left(\frac{a}{b}\right)^{2m+1} + 1 \right]^2$$

$$= b^{2(2m+1)} \left(\frac{a^{2m+1} + b^{2m+1}}{b^{2m+1}} \right)^2$$

$$= (a^{n/2} + b^{n/2}) \text{ hence proved}$$

14. Let z_1, z_2, z_3 be complex numbers, not all real, such that $|z_1| = |z_2| = |z_3| = 1$ and $2(z_1 + z_2 + z_3) - 3z_1z_2z_3 \in \mathbb{R}$.

Prove that $\max(\arg z_1, \arg z_2, \arg z_3) \geq \frac{\pi}{6}$. Where $0 < \arg(z_1), \arg(z_2), \arg(z_3) < 2\pi$.

Sol. Let $z_k = \cos t_k + i \sin t_k : k \in \{1, 2, 3\}$

The condition $2(z_1 + z_2 + z_3) - 3z_1z_2z_3 \in \mathbb{R}$

$$\Rightarrow 2(\sin t_1 + \sin t_2 + \sin t_3) = 3\sin(t_1 + t_2 + t_3) \quad \dots(i)$$

Assume by the way contradiction that $\max(t_1, t_2, t_3) < \frac{\pi}{6}$,

$$\text{Hence } t_1, t_2, t_3 < \frac{\pi}{6}$$

$$\text{Let } t = \frac{t_1 + t_2 + t_3}{3} \in \left(0, \frac{\pi}{6}\right)$$

$$\frac{1}{3}(\sin t_1 + \sin t_2 + \sin t_3) \leq \sin \frac{t_1 + t_2 + t_3}{3} \quad \dots(ii)$$

From (i) & (ii) we have,

$$\frac{\sin(t_1 + t_2 + t_3)}{2} \leq \sin\left(\frac{t_1 + t_2 + t_3}{3}\right)$$

Then, $\sin 3t \leq 2\sin t$

$$\Rightarrow 4\sin^3 t - \sin t \geq 0$$

i.e., $\sin^2 t \geq \frac{1}{4}$, hence $\sin t \geq \frac{1}{2}$ and $t \geq \frac{\pi}{6}$ which contradicts that

$$t \in \left(0, \frac{\pi}{6}\right)$$

$$\therefore \max(t_1, t_2, t_3) \geq \frac{\pi}{6}$$

Hence proved.

15. Let z_1, z_2, z_3 be complex numbers such that

$$(i) |z_1| = |z_2| = |z_3| = 1$$

$$(ii) z_1 + z_2 + z_3 \neq 0$$

$$(iii) z_1^2 + z_2^2 + z_3^2 = 0$$

Prove that for all integers $n \geq 2$, $|z_1^n + z_2^n + z_3^n| \in \{0, 1, 2, 3\}$.

Sol. Let $s_1 = z_1 + z_2 + z_3$, $s_2 = z_1z_2 + z_2z_3 + z_3z_1$, $s_3 = z_1z_2z_3$ and take a cubic equation.

$$z^3 - s_1z^2 + s_2z - s_3 = 0 \text{ with roots } z_1, z_2, z_3$$

$$\text{It is given that } z_1^2 + z_2^2 + z_3^2 = 0$$

$$\text{Hence, } s_1^2 = 2s_2 \quad \dots(i)$$

Again we have,

$$s_2 = s_3 \left(\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right)$$

$$= s_3 (\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = s_3 \cdot \bar{s}_1 \quad \dots(ii)$$

Now, from (i) and (ii)

$$s_1^2 = 2s_3 \cdot \bar{s}_1 \text{ and } |s_1|^2 = 2(s_3)(\bar{s}_1)$$

$$= 2|s_1|$$

$$\therefore |s_1| = 2 \Rightarrow s_1 = 2\lambda \text{ with } |\lambda| = 1$$

Now, again from relation (i) and (ii) it follows that

$$s_2 = \frac{1}{2}s_1^2 = 2\lambda^2 \text{ and } s_3 = \frac{s_2}{s_1} = \frac{2\lambda^2}{2\lambda} = \lambda$$

Now, the given equation becomes

$$z^3 - 2\lambda z^2 + 2\lambda^2 z - \lambda^3 = 0$$

$$\Rightarrow (2 - \lambda)(z^2 - \lambda z + \lambda^2) = 0$$

The roots are

$$z = \lambda, \lambda\omega = -\lambda\omega^2$$

Now,

$$R_n = |z_1^n + z_2^n + z_3^n| = |\lambda^n + \lambda^n\omega^n + (-1)^n\lambda^n\omega^{2n}|$$

$$= |\lambda|^n |1 + \omega^n + (-1)^n\omega^{2n}|$$

$$R_0 = 3, R_1 = |1 + \omega - \omega^2|$$

$$= |-2\omega^2| = 2$$

$$R_2 = |1 + \omega^2 + \omega| = 0$$

$$R_3 = |1 + 1 - 1| = 1$$

$$R_4 = 0$$

$$R_5 = 2$$

$$R_6 = 3$$

$$\therefore R_{k+6} = R_k \text{ for all integers } k.$$

$$\therefore R_n \in \{0, 1, 2, 3\}$$

16. Find all complex numbers z such that

$$|z - |z + 1|| = |z + |z - 1||$$

Sol. $|z - |z + 1|| = |z + |z - 1||$

$$\Rightarrow |z - |z + 1||^2 = |z + |z - 1||^2$$

$$\Rightarrow (z - |z + 1|)(\bar{z} - |z + 1|) = (z + |z - 1|)(\bar{z} + |z - 1|)$$

$$\Rightarrow z\bar{z} - |z + 1|(z + \bar{z}) + |z + 1|^2 = z\bar{z} + (z + \bar{z})|z - 1| + |z - 1|^2$$

$$|z + 1|^2 - |z - 1|^2 = (z + \bar{z})(|z + 1| + |z - 1|)$$

$$(z + 1)(\bar{z} + 1) - (z - 1)(\bar{z} - 1) = (z + \bar{z})(|z + 1| + |z - 1|)$$

$$z\bar{z} + z + \bar{z} + 1 - z\bar{z} + z + \bar{z} - 1 = (z + \bar{z})(|z + 1| + |z - 1|) \quad 2(z + \bar{z}) - (z + \bar{z})(|z + 1| + |z - 1|) = 0$$

$$(z + \bar{z})(2 - (|z + 1| + |z - 1|)) = 0$$

$$\text{i.e., } z + \bar{z} = 0 \text{ or } |z + 1| + |z - 1| = 2$$

$$2 = |(z + 1) - (z - 1)| \leq |z + 1| + |z - 1|$$

$$\therefore \text{Solution of the equation } |z + 1| + |z - 1| = 2 \text{ satisfy } z + 1 = t(1 - z) \text{ where } t \in \mathbb{R}, t \geq 0$$

$$z = \frac{t-1}{t+1}, \text{ so, } z \text{ is any real number with } -1 \leq z \leq 1$$

$$\text{The equation } z + \bar{z} = 0 \text{ has the solutions } z = bi, b \in \mathbb{R}.$$

$$\text{Hence the solutions to the equation are } \{bi : b \in \mathbb{R}\} \cup \{a \in \mathbb{R} : a \in [-1, 1]\}$$

17. Suppose p is a polynomial with complex coefficients and an even degree. If all the roots of p are complex non-real numbers with modulus 1, prove that $p(1) \in \mathbb{R}$ iff $p(-1) \in \mathbb{R}$.

Sol. In order to prove the result it will be sufficient to prove that $\frac{p(1)}{p(-1)} \in \mathbb{R}$

Let $x_1, x_2, x_3, \dots, x_{2n}$ be roots of p . Then

$$p(x) = \lambda(x - x_1)(x - x_2)(x - x_3) \dots (x - x_{2n})$$

For some $\lambda \in \mathbb{C}$, and

$$\frac{p(1)}{p(-1)} = \frac{\lambda(1 - x_1)(1 - x_2) \dots (1 - x_{2n})}{\lambda(-1 - x_1) \dots (-1 - x_{2n})} = \prod_{k=1}^{2n} \frac{1 - x_k}{1 + x_k}$$

It is given that $|x_k| = 1$ for all $k = 1, 2, \dots, 2n$. Then

$$\left(\frac{1-x_k}{1+x_k} \right) = \frac{1-\bar{x}_k}{1+\bar{x}_k} = \frac{1-\frac{1}{x_k}}{1+\frac{1}{x_k}} = \frac{x_k-1}{x_k+1} = -\frac{1-x_k}{1+x_k}$$

$$\begin{aligned} \text{Now, } \left(\frac{p(1)}{p(-1)} \right) &= \prod_{k=1}^{2n} \left(\frac{1-x_k}{1+x_k} \right) = \prod_{k=1}^{2n} -\left(\frac{1-x_k}{1+x_k} \right) \\ &= (-1)^{2n} \prod_{k=1}^{2n} \frac{1-x_k}{1+x_k} = \frac{p(1)}{p(-1)} \end{aligned}$$

$\therefore \frac{p(1)}{p(-1)}$ is a real number.

18. The points $A_1, A_2, \dots, A_{10}$ are equally distributed on a circle of radius R (taken in order). Prove that $A_1 A_4 - A_1 A_2 = R$.

Sol. Without loss of generalization, we take $A_1 = R + i_0$

Now, in triangle $A_1 O A_4$

$$\cos 108^\circ = \frac{R^2 + R^2 - (A_1 A_4)^2}{2R^2}$$

$$\begin{aligned} (A_1 A_4)^2 &= 4R^2 \sin^2 54^\circ \\ &= 2R \sin 54^\circ \end{aligned}$$

Similarly in triangle

$A_1 O A_2$

$$\cos 36^\circ = \frac{R^2 + R^2 - (A_1 A_2)^2}{2R^2}$$

$$(A_1 A_2)^2 = 2R^2 \sin^2 18^\circ$$

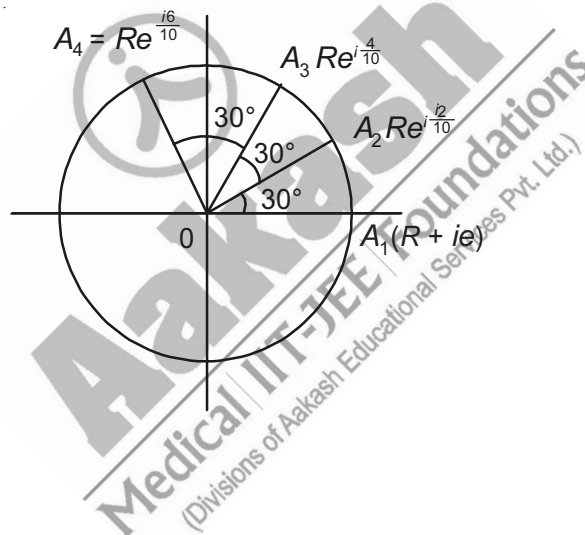
$$A_1 A_2 = 4R \sin 18^\circ$$

$$\text{Now } A_1 A_4 - A_1 A_2 = 2R(\sin 54^\circ - \sin 18^\circ)$$

$$= 2R \cdot 2 \cos 36^\circ \cdot \sin 18^\circ$$

$$= 4R \cdot \left(\frac{\sqrt{5}+1}{4} \right) \left(\frac{\sqrt{5}-1}{4} \right)$$

$$= R \text{ (proved)}$$



19. Let a and b be positive real numbers with $a^3 + b^3 = a - b$, and $k = a^2 + 4b^2$, then

(1) $k < 1$

(2) $k > 1$

(3) $k = 1$

(4) $k > 2$

Sol. Answer (1)

Given, $a^3 + b^3 = a - b$

$$\text{let } \frac{a}{b} = k \ (a > b)$$

$$\text{Now } a^3 + b^3 = a - b \Rightarrow b^3 k^3 + b^3 = bk - b$$

$$\Rightarrow b^2 = \frac{k-1}{k^3+1}$$

$$\text{Now, } a^2 + 4b^2 = b^2 k^2 + 4b^2 = b^2 (k^2 + 4)$$

$$= \frac{(k^2 + 4)(k-1)}{k^3 + 1}$$

$$= 1 - \frac{k^2 - 4k + 5}{k^3 + 1}$$

$$= 1 - \frac{(k-2)^2 + 1}{k^3 + 1} \Rightarrow \text{as } k > 0$$

$$\Rightarrow a^2 + 4b^2 < 1$$

20. Let k be a real number such that the inequality $\sqrt{x-3} + \sqrt{6-x} \geq k$ has a solution then the maximum value of k is

(1) $\sqrt{3}$

(2) $\sqrt{6} - \sqrt{3}$

(3) $\sqrt{6}$

(4) $\sqrt{6} + \sqrt{3}$

Sol. Answer (3)

$$f(x) = \sqrt{x-3} + \sqrt{6-x}$$

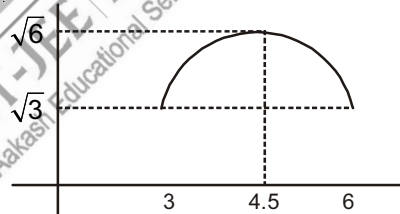
$$Df = [3, 6]$$

$$f(3) = \sqrt{3}, f(6) = \sqrt{3}$$

from symmetry $f(x)$ have maxima

$$\text{at } x = \frac{6+3}{2} = 4.5$$

$$f(4.5) = \sqrt{4.5-3} + \sqrt{6-4.5} = 2\sqrt{1.5} = \sqrt{6}$$



21. Let α and β be the root of $x^2 + px - \frac{1}{2p^2} = 0, p \in R$. The minimum value of $\alpha^4 + \beta^4$ is

(1) 2

(2) $2\sqrt{2}$

(3) $2 + \sqrt{2}$

(4) $2 - \sqrt{2}$

Sol. Answer (3)

$$x^2 + px - \frac{1}{2p^2} = 0, p \in R$$

Let α, β be roots

$$\alpha + \beta = -p, \alpha\beta = -\frac{1}{2p^2}$$

$$\alpha^4 + \beta^4 = (\alpha + \beta)^4 - 4\alpha\beta^3 - 4\alpha^3\beta - 6\alpha^2\beta^2$$

$$(\alpha + \beta)^4 - 4\alpha\beta(\alpha^2 + \beta^2) - 6\alpha^2\beta^2$$

$$(\alpha + \beta)^4 - 4\alpha\beta((\alpha + \beta)^2 - 2\alpha\beta) - 6\alpha^2\beta^2$$

$$p^4 + \frac{2}{p^2}\left(p^2 + \frac{1}{p^2}\right) - \frac{6}{4p^4}$$

$$p^4 + 2 + \frac{2}{p^4} - \frac{6}{4p^4}$$

$$p^4 + \frac{2}{4p^4} + 2$$

$$\text{Now, } \frac{p^4 + \frac{1}{2p^4}}{2} \geq \left(\frac{1}{2}\right)^{\frac{1}{2}} \text{ (A.M. - G.M. inequality)}$$

$$\Rightarrow p^4 + \frac{1}{2p^4} \geq 2 \cdot \left(\frac{1}{2}\right)^{\frac{1}{2}}$$

$$\geq \sqrt{2}$$

$$\therefore \text{Minimum value of } \alpha^4 + \beta^4 = \sqrt{2} + 2$$

22. The number of real solutions of the equation $\sqrt[4]{97-x} + \sqrt[4]{x} = 5$

(1) 0

(2) 1

(3) 2

(4) 4

Sol. Answer (3)

$$\sqrt[4]{97-x} + \sqrt[4]{x} = 5$$

$$\frac{a^m + b^m}{2} < \left(\frac{a+b}{2}\right)^m \text{ if } m \in (0, 1)$$

Put $a = 97 - x$, $b = x$, $m = w/f$

$$\frac{\sqrt[4]{97-x} + \sqrt[4]{x}}{2} \leq \left(\frac{97}{2}\right)^{\frac{1}{4}}$$

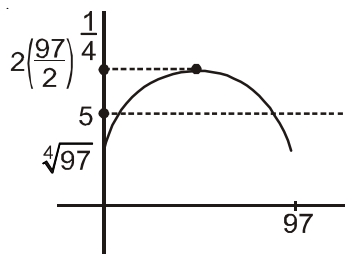
$$\Rightarrow \sqrt[4]{97-x} + \sqrt[4]{x} \leq 2 \times 2.63$$

Which holds for

$$\sqrt[4]{97-x} + \sqrt[4]{x} = 5$$

Now,

Clearly for $\sqrt[4]{97-x} + \sqrt[4]{x} = 5$ has two solution.



23. If $f(x)$ is a polynomial of degree at least two with integral co-efficients then the remainder when it is divided by $(x - a)(x - b)$ is, where $a \neq b$.

$$(1) \quad x \left[\frac{f(a) - f(b)}{b - a} \right] + \frac{af(b) - bf(a)}{a - b}$$

$$(2) \quad x \left[\frac{f(a) - f(b)}{a - b} \right] + \frac{af(b) - bf(a)}{a - b}$$

$$(3) \quad x \left[\frac{f(b) - f(a)}{a - b} \right] + \frac{af(b) - bf(a)}{a - b}$$

$$(4) \quad x \left[\frac{f(b) - f(a)}{a - b} \right] + \frac{bf(a) - af(a)}{a - b}$$

Sol. Answer (2)

$f(x) \rightarrow$ polynomial of degree 'n'

$$= a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

when this polynomial will be divided by $(x - a)(x - b)$, remainder will be of form $Ax + B$

$$\text{Now, } f(a) = A.a + B$$

$$f(b) = A.b + B$$

$$\frac{f(a) - f(b)}{a - b} = A \quad \dots(i)$$

$$\text{Also } B = f(a) - a \frac{f(a) - f(b)}{a - b}$$

$$\frac{af(a) - bf(a) - af(a) + af(b)}{a - b}$$

$$\frac{af(b) - bf(a)}{a - b}$$

$\therefore$ Remainder will be

$$x \left(\frac{f(a) - f(b)}{a - b} \right) + \frac{af(b) - bf(a)}{a - b}$$

24. Let $p = (x_1 - x_2)^2 + (x_1 - x_3)^2 + \dots + (x_1 - x_6)^2 + (x_2 - x_3)^2 + (x_2 - x_4)^2 + \dots + (x_1 - x_6)^2 + \dots +$

$$(x_5 - x_6)^2 = \sum_{1 \leq i < j \leq 6} (x_i - x_j)^2.$$

Then the maximum value of p if each x_i ($i = 1, 2, \dots, 6$) has the value 0 and 1 is

$$(1) \quad 1$$

$$(2) \quad 2$$

$$(3) \quad 4$$

$$(4) \quad 9$$

Sol. Answer (4)

$$\text{Let } p = \sum_{i,j=1}^6 (x_i - x_j)^2$$

If we consider any three of $x_1, x_2, x_3, x_4, x_5, x_6$ is one and another three '0'

$$\Rightarrow p = 9$$

25. If x, y, z are three real numbers such that $x + y + z = 4$ and $x^2 + y^2 + z^2 = 6$, then

- (1) $\frac{2}{3} \leq x, y, z \leq 2$ (2) $0 \leq x, y, z \leq 2$ (3) $1 \leq x, y, z \leq 3$ (4) $2 \leq x, y, z \leq 3$

Sol. Answer (1)

$$x + y = 4 - z$$

$$x^2 + y^2 = 6 - z^2$$

$$\begin{aligned} \therefore 2xy &= (x + y)^2 - (x^2 + y^2) \\ &= (4 - z)^2 - (6 - z^2) \\ &= 2z^2 - 8z + 10 \end{aligned}$$

The quadratic equation whose roots are x and y is

$$t^2 - (x + y)t + xy = 0$$

$$t^2 - (4 - z)t + z^2 - 4z + 5 = 0$$

for real roots,

$$(4 - z)^2 - 4(z^2 - 4z + 5) \geq 0$$

$$3z^2 - 8z + 4 \leq 0 \Leftrightarrow (3z - 2)(z - 2) \leq 0$$

$$\frac{2}{3} \leq z \leq 2$$

$$\text{Similarly } x, y \in \left[\frac{2}{3}, 2 \right]$$

26. If unity is double repeated root of $px^3 + q(x^2 + x) + r = 0$ then

- (1) $pr < 0$ (2) $pq < 0$ (3) $pqr > 0$ (4) $pqr < 0$

Sol. Answer (2)

$$px^3 + qx^2 + qx + r = 0 \quad \begin{matrix} \swarrow \\ 1 \\ \alpha \end{matrix}$$

$$\alpha + 2 = \frac{-q}{p}, \quad 2\alpha + 1 = \frac{q}{p}, \quad \alpha = \frac{-r}{p}$$

$$\begin{array}{l|l} 3\alpha + 3 = 0 & \frac{-q}{p} = 1 \\ \alpha = -1 & \\ pq < 0, pr > 0 & \end{array} \quad \begin{array}{l} r.p > 0 \\ q.p < 0 \end{array}$$

$$\Rightarrow pqr > 0 \text{ or } pqr < 0$$

$$\therefore pq < 0 \text{ is suitable option}$$

27. The number of real solutions of the equation $4x^{99} + 5x^{98} + 4x^{97} + 5x^{96} + \dots + 4x + 5 = 0$ is

- (1) 1 (2) 5 (3) 7 (4) 97

Sol. Answer (1)

$$4x^{99} + 5x^{98} + 4x^{97} + 5x^{96}$$

$$+ x^{98}(4x + 5) + x^{96}(4x + 5) + x^{94}(4x + 5) + \dots + 1(4x + 5)$$

$$(4x + 5)(x^{98} + x^{96} + \dots + 1)$$

$$x = \frac{-5}{4} \text{ as } x^{98} + x^{96} + \dots + 1 \neq 0$$

$$\therefore \text{Only one solution}$$

